Paper 2 — Technical Supplement II: The Foundational Gauge Program

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Companion to Paper 2: "The Structure of Admissible Physics"

Technical Supplement II of two; the classification core is Technical Supplement I

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Abstract

This supplement carries the foundational program of Paper 2: the chain from the APF axioms toward the inputs that the classification core (Technical Supplement I (*The Classification Core*)) consumes, each link stated at its banked grade with its premises named. The chain runs: cost uniqueness (L_{Cauchy}: $F(n) = n$ on the discrete channel domain); the Sep/IJC-routed bridge from superadditivity to a finite noncommutative admissibility algebra, semisimple by the standard finite-dimensional C^* route, derived given the IJC occupancy (which is read off the empirical record); the gauge identification — unconditionally, every continuous physical gauge redundancy is inner, $\mathrm{Gau}_0 \subseteq \mathrm{Inn}(\mathcal{A})$, with equality under a named full-invariance hypothesis; the exact sequence $1 \to Z(\mathrm{U}(\mathcal{A})) \to \mathrm{U}(\mathcal{A}) \to \mathrm{Inn}(\mathcal{A}) \to 1$ with the relative-phase torus as the sole abelian source; the matter action, defined on the lifted group $\widetilde{G}$ with the charge representation a named input (I-typing), the algebra side determining it only up to the central torus; the lift and the global $\mathbb{Z}_6$ quotient; and the $C(G) = \dim(G)$ cost program — the lower bound a theorem on the operational-encoding premise, the upper bound on the disjoint-anchor realizability premise, the equality resting on that operational premise pair pending a cost functor. Enforcement completeness is carried at its honest grade, $[\mathrm{P}_{\mathrm{structural\_reading}} \mid \text{R-EC-inv} + \text{I-typing}]$, with the counter-construction owned in the bank; capacity minimality is A2 over the A1+MD admissible set. The program's newest closures are the two banked EC results: F6 and EC-label-injectivity are logically independent over the declared scan space, all four quadrants inhabited and the F6-free argmin at 42 Weyl degrees of freedom[1], and the one-U(1) necessity leg holds exactly under the kinematic type-inventory reading with the entry-typed inventory[2]. A grade-ledger table near the front states every link, its grade string, its named premises, and its coderef. The open-lane register is headed by the fixed-point program: whether the (template, matter content, single-abelian-grading) triple is the unique simultaneous solution of the full constraint set. The register throughout is a graded program: theorems at their grades, named premises at their consumption sites, opens named as opens.

Grade ledger: every chain link, its grade, its named premises, its coderef

One row per link of the foundational chain. Grade strings are the bank's; a premise named in the third column is consumed at the cited statement, not derived there.

[1] Code reference: `check_L_F6_not_from_EC` in `ec_inventory_reading.py`.

[2] Code reference: `check_L_EC_inventory_reading` in `ec_inventory_reading.py`.

Link (statement)	**Grade**	**Named premises / readings**	**Coderef**
L_{Cauchy}: $F(n) = n$ on $\mathbb{N}$ (§1)	[P]	C1 additivity on separated interfaces (L_{loc}); unit normalization	`check_L_Cauchy_uniqueness` (`L_Cauchy_uniqueness.py`)
Sep/IJC bridge: (IJC) $\Rightarrow$ non-commutative record algebra (§2.1)	[$\mathrm{P}_{\mathrm{math}}$+$\mathrm{P}_{\mathrm{APF}}$]	IJC *occupancy* empirical (read off the record); bridge derived	`check_T_IJC_dichotomy`, `check_T_inseparable_IJC` (`core.py`)
Superadditivity $\rightarrow$ finite semisimple carrier (Prop. 2.2)	[P]	A1, MD, $\Delta > 0$ (IJC), positive B, adjoint closure, finite resolution; standard C^* route	`check_T_alg` (`core.py`)
Gauge identification: $\mathrm{Gau}_0 \subseteq \mathrm{Inn}(\mathcal{A})$ (Thm. 2.12)	[P]	— (unconditional inclusion)	`check_T3` (`core.py`)
Gauge identification, equality leg	[P | full-invariance]	full-invariance hypothesis: anchor set, cost kernel, continuation data Inn-invariant as a family	`check_T3` (`core.py`)
Exact sequence $1 \to Z \to \mathrm{U}(\mathcal{A}) \to \mathrm{Inn}(\mathcal{A}) \to 1$; T_{rel} sole abelian source (Prop. 2.19)	[P]	—	`check_T3` (`core.py`)
Matter action on $\widetilde{G}$; algebra determines it up to T_{rel} (Prop. 2.16)	[P | I-typing]	charge representation = named input of the matter category (I-typing)	`check_L_EC_inventory_reading` (`ec_inventory_reading.py`)
Lift and global $\mathbb{Z}_6$ quotient (Props. 2.25, 2.26)	[P]	maximal-quotient convention (named); T_{rel} observability via charged matter	`check_T_gauge` (`gauge.py`)
EC (enforcement completeness) (Lem. 2.24(i))	[$\mathrm{P}_{\mathrm{structural\_reading}}$ | R-EC-inv + I-typing]	R-EC-inv (kinematic type-inventory reading; A1-motivated, not A1-derived); I-typing; counter-construction owned in-check	`check_L_EC_inventory_reading` (`ec_inventory_reading.py`)
CM (capacity minimality) (Lem. 2.24(ii))	[P]	A2 over the A1+MD admissible set; channel-configuration semantics	`check_L_cost_gauge` (`core.py`)
$C(G) \geq \dim(G) \cdot \varepsilon^*$ (Lem. 2.22)	[P | op. encoding]	operational-encoding premise (invariance of domain applied to the encoding)	`check_L_cost_gauge` (`core.py`)
$C(G) = \dim(G)$, composed operational statement (§2)	[P | op. encoding + disjoint-anchor realizability]	the operational cost premise pair; cost functor = named open lane	`check_L_cost_gauge` (`core.py`)
T_{7B} positive-definite cost kernel (§3.1)	[P]	K2, K3, SP, L_ω, T_{adj}	`check_T7B` (`gravity.py`)
$L_{\mathrm{irr,uniform}}$ (§3.2)	[P] (sufficient-condition)	saturation bound (5)	`check_L_irr_uniform` (`core.py`)
$B1'$ complex carrier (§3.3)	[P]	oriented-composite robustness	`check_B1_prime` (`gauge.py`)
Theorem R prerequisites closed (§3.5)	[P]	as listed per carrier requirement	`check_Theorem_R` (`gauge.py`)
F6 $\perp$ EC independence (all four quadrants inhabited; F6-free argmin = 42)	[P]	pure computation over the declared scan space	`check_L_F6_not_from_EC` (`ec_inventory_reading.py`)
One-U(1) necessity leg (≥ 1 abelian factor)	[$\mathrm{P}_{\mathrm{structural\_reading}}$ | R-EC-inv + I-typing]	consumed by the closure theorems of Technical Supplement I (*The Classification Core*)	`check_L_EC_inventory_reading` (`ec_inventory_reading.py`)

What Is Assumed, What Is Proved, What Is Deferred

Imported from Paper 1 Supplement (proved there; re-proved in the appendix for self-containment). The first four items are the structurally necessary components of PLEC (none derivable from the others; essentiality proofs in Paper 1 Supplement, §1).

1. **A1** (PLEC upper bound): admissibility capacity is finite and positive.
2. **MD** (PLEC positive floor): uniform positive cost per unit of admissibility complexity.

3. **A2** (PLEC argmin selection): the realized configuration is the arg min of total admissibility cost over the admissible set.
4. **BW** (PLEC non-degeneracy): the cost spectrum witnesses a budget-window triple.
5. **SP, K3, FD1–FD6**: constitutive admissibility semantics.
6. T_{form}, T_{embed}, T_{sep}, L_{ε^*}, L_Δ, L_{blk}, L_Π, T_{alg}, T_{GNS}, T2a–T2c, T_{Born}, T_M, L_{loc}, L_{irr}, T3, L_ω, T_{adj}: the finite-carrier chain (each re-proved in Appendix B).

Chain results proved in this supplement.

1. L_{Cauchy}: cost function uniqueness (§1).
2. Local support and commuting algebras from disjoint substrates (Definition 2.5, Lemma 2.7); the generated-algebra block structure (Proposition 2.6).
3. Superadditivity to a finite semisimple carrier by the standard C^* route (Proposition 2.2).
4. Gauge identification: $\mathrm{Gau}_0 \subseteq \mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$ unconditionally; equality under the named full-invariance hypothesis (§2.3).
5. Exact sequence $1 \to Z \to \mathrm{U}(\mathcal{A}) \to \mathrm{Inn}(\mathcal{A}) \to 1$ and the relative-phase torus (Proposition 2.19).
6. The matter category and the action defined on the lifted group $\widetilde{G}$, determined by the algebra only up to central data, closed by the charge representation as named input (Definition 2.14, Proposition 2.16).
7. Lift from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$ and global structure $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ (Propositions 2.25, 2.26).
8. EC and CM at their grades (Lemma 2.24); the generator cost bound and the composed operational statement $C(G) = \dim(G)$ (§2).
9. T_{7B} (§3.1); $L_{\mathrm{irr,uniform}}$ (§3.2); $B1'$ (§3.3); Theorem R consolidation (§3.5).

Consumed downstream by Technical Supplement I (*The Classification Core*). The classification core consumes the outputs above as cited premises; the scan, the waterfall, the one-U(1) closure theorems, hypercharge, and the capacity count are proved there, not here.

Not proved here (honest flags).

1. $L_{\mathrm{irr,uniform}}$ is a *sufficient-condition* theorem: it guarantees gauge-sector irreversibility at interfaces satisfying the saturation bound (5), not at all interfaces.
2. The equality leg of the gauge identification consumes the full-invariance hypothesis; only the inclusion is unconditional.
3. $C(G) = \dim(G)$ rests on the operational cost premise pair pending a cost functor (open lane 2, §5).
4. EC rests on R-EC-inv + I-typing; the counter-construction showing non-derivability at shape level is owned in the bank.
5. The IJC occupancy is empirical by design (named conditioning, §5).

Imported-ingredient hypothesis ledger (verified vs. assumed in the finite-cost environment)

Every off-framework mathematical import, with its hypothesis status *in this document's finite setting.* "Verified" = the theorem's hypotheses are checked here against the finite structures it is applied to; "assumed" = the hypothesis is taken as a premise and flagged where used.

Import	**Applied to**	**Hypothesis status**
Wedderburn–Artin	finite $*$-algebra $\to \bigoplus_k M_{n_k}$	verified (finite-dimensional $*$-algebra with faithful positive state; semisimple by the standard C^* route; T2a)
Field selection $\mathbb{F} = \mathbb{C}$	scalar field of the carrier	derived at T2c from $B1'$ orientation; real/quaternionic branches close at the field gate, not assumed away
Skolem–Noether	per-block automorphisms	verified (applied blockwise to $M_{n_k}(\mathbb{C})$ only; v4.0 removed the central-simple misstatement)
Kadison/Wigner (finite)	symmetry implementation	verified in finite dimension; antiunitary branch excluded by $B1'$, not by assumption
Doplicher–Roberts	framing/comparison only	*not consumed as a theorem*: finite-regime departure from type-III noted in §2.5
Cartan closed-subgroup	classification class	verified (the class is $\prod_k \mathrm{PU}(n_k)$ and closed subgroups; scope rider travels with `check_L_cost_gauge`)
Brouwer invariance of domain	resolution-rank bound	verified (finite-dimensional, local form; Lemma 2.22)

Terminology: four words with one fixed meaning each

faithful of a representation or state: trivial kernel. For states: $\omega(a^*a) = 0 \Rightarrow a = 0$ (GNS null ideal zero). Never used to mean "physically realized."

irreducible of a representation: no proper nonzero invariant subspace. Used only rep-theoretically; "primitive" is used for invariant tensors, never "irreducible."

effective of a group action: only the identity acts trivially (synonym of faithful for actions); used when the group, rather than its representation, is the subject.

physically distinct (multiplet types) distinct isomorphism classes of $\widetilde{G}$-representations (Definition 2.20); copies within a class (generations) are the same type at different multiplicity.

Reader's map: one paper, two supplements

Paper 2's technical material lives in two supplements. This document — Technical Supplement II, *The Foundational Gauge Program* — carries the chain from the APF axioms toward the classification inputs: the admissibility algebra and its gauge identification, the matter category and the lift, the EC/CM grounds at their graded readings, the $C(G) = \dim(G)$ cost program, the type-III/continuum boundary, and the positioning against reconstruction programs, with a grade ledger and an open-lane register. Technical Supplement I (*The Classification Core*) carries the finished classification mathematics on declared inputs — the conditional classification theorem with its C1–C10 ledger, the scan, anomaly and hypercharge closure, the one-U(1) closure theorems, and the capacity count. A reader auditing the classification needs Technical Supplement I; a reader auditing where its premises come from, and at what grade, needs this document.

Axioms-to-SM checklist: minimal load-bearing inputs of the whole program

The map below spans the whole program: the chain rows are proved in this supplement; the template-closure, matter-scan, and hypercharge rows are proved in Technical Supplement I (*The Classification Core*) and appear here so the reader can see the program end to end.

Step	Irreducible input	Output used downstream
PLEC regime	A1 finite capacity; MD positive floor; A2 minimum-cost realization; BW non-degeneracy; constitutive semantics SP/K3/FD1–FD6	Finite admissible search space with a positive admissibility ledger.
Cost uniqueness	Local additivity on independent interfaces; unit normalization	$F(n) = n$ on the discrete channel domain; all cost comparisons use channel counts rather than tunable weights.
Sep/IJC routing	Sep/IJC representation theorem plus the non-separable substrate condition for physical interfaces	Nonzero pool coupling, $\Delta > 0$, and hence a noncommutative admissibility algebra.
Algebraic carrier	Finite-dimensional $*$-algebra; field selection $\mathbb{F} = \mathbb{C}$; faithful GNS state	$\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$ and linear matter modules.
Gauge identification	Internal/kinematic split; Skolem–Noether; cost-equipped block structure	$G_{\text{int}} = \text{Inn}(\mathcal{A}) = \prod_k \text{PU}(n_k)$; outer block permutations are discrete relabelings, not continuous gauge.
Physical action on fields	Matter-category representation (independently specified) and exact sequence $1 \to Z \to U(\mathcal{A}) \to \text{Inn}(\mathcal{A}) \to 1$	Lift from projective algebra action to linear matter action; $\text{PU}(n)$ is represented by its $\text{SU}(n)$ cover where fundamental matter exists.
Abelian/global structure	Relative-phase torus; no-redundancy/minimum-cost condition; anomaly-compatible matter modules	Exactly one $\text{U}(1)_Y$ and global quotient $[\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)_Y]/\mathbb{Z}_6$.
Template closure	Lie classification; oriented composite robustness; Witten anomaly; capacity minimality	$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)_Y$ as the unique minimum-cost gauge template.
Matter scan	Bounded 1,680-template enumeration; F1–F7 exact filters; three-generation input from the capacity ledger	Unique 45-Weyl-fermion SM survivor up to CPT.
Hypercharge	Standard anomaly equations applied to the derived template	Unique rational hypercharge ray and charge quantization.

Necessity is local to each row: relaxing a row does not refute APF, but it opens the indicated alternatives. For example, dropping the non-separable substrate condition leaves the Sep countermodel with $\Delta = 0$; dropping the relative-phase no-redundancy condition allows extra sterile U(1) directions; dropping the finite-regime assumption restores type-III AQFT as an effective continuum limit rather than a finite proof environment.

Framing convention: "forces" and "selects" denote uniqueness, not processes

The supplement's theorems and proofs use the verbs "forces," "selects," and "rejects" as standard mathematical shorthand for uniqueness results, arg min properties, and inadmissibility results respectively. Under the APF ontology, these verbs do not denote processes: nothing is forced by an agent, nothing is selected by a mechanism, nothing is rejected by a filter. Each theorem asserts a property of the set of admissible configurations—that it has a unique member, a minimum member, or no member satisfying some condition. For example, "capacity minimality selects $N_c = 3$" is shorthand for "$N_c = 3$ is the $\arg\min$ of dim(G) over R1–R3-admissible color groups," and the alternative values (SU(4), SU(5), E_6, ...) are *admissible* in the sense that each has cost below the capacity bound, but have higher cost than SU(3) and are therefore not what is. The companion note *What Is, Is Minimum-Cost Admissible* spells this out in full. Readers may substitute the descriptive reading throughout the supplement without any change to the proofs.

Status of A2: argmin realization is a lawlike selection rule, not a derived dynamics

A2 is intentionally not derived in this supplement from a deeper dynamical variational principle. It is one of the four structural inputs of the finite-regime APF package: among admissible continuations satisfying A1, MD, and BW, the realized carrier is the minimum admissibility-cost representative. All selection claims in the supplement have the conditional form

$$\text{A1+MD+BW+A2+listed interface hypotheses} \implies \text{unique minimum-cost admissible structure.}$$

Thus the paper does not claim to prove that nature *must* obey global argmin realization from conventional QFT assumptions. It claims that, if A2 is accepted as the APF lawlike selection rule, the Standard-Model template is the unique admissible minimum under the stated finite ledger. This is why counterfactuals that change A2, make new generators cost-free, or subtract structural gauge generators from the ledger are not small variants of the theorem; they define a different selection problem.

Operational/structural convention

Throughout the supplement, *operations* are tests, preparations, updates, comparisons, and continuations performed at an interface. A *structure* is the equivalence class, ledger, projection, module, or algebra induced by those operations after quotienting observationally indistinguishable continuations. Thus when a structural object is said to "act," the formal meaning is always its represented action on operational records or on the field module generated by those records. This convention prevents an interpretive slippage that would otherwise make properties sound like mechanisms: the algebra is not a physical agent; it is the minimal representation of admissible operational continuations.

Translation table for nonstandard vocabulary

APF term	**Standard-language approximation**
Pool Π	Residual/shared substrate direction not resolved by the current coarse record algebra; comparable to a gluing degree of freedom between contexts.
Anchor M_d	Context-fixed support subspace for a tested distinction; analogous to a local sector or record support.
IJC	Failure of a faithful global Boolean refinement preserving all tested marginals and the defined partial products; finite analogue of a contextual global-section obstruction.
Continuation calculus	Typed partial algebra of operational histories, quotiented by observational equivalence; comparable to a finite process/category-of-contexts presentation.
Admissibility algebra	Finite ∗-algebra generated by record projections and continuation-preserving updates.
Capacity cost	Resource monotone for holding distinctions available; in QFT language, a finite-regime bookkeeping surrogate for the cost of supporting admissible distinctions through independent channels.

Glossary of imported objects. Each entry gives the object's name, one-sentence definition, source in the Paper 1 Supplement (P1S), and where it is first used in this supplement.

Symbol	Definition	Source	Used at
Γ	Enforcement interface (physical boundary)	P1S FD1	Throughout
S_Γ	Substrate: set of configurations at Γ	P1S FD1	Throughout
$\mathcal{D}(\Gamma)$	Admissible distinctions (binary predicates, $\varepsilon > 0$)	P1S FD1	§1
$\varepsilon(d)$	Admissibility cost: $\inf_{\mathcal{P}(d)} \kappa$	P1S FD4	§1
$C(\Gamma)$	Total admissibility budget at Γ	P1S A1	Throughout
ε^*	Uniform cost floor: $\varepsilon(d) \geq \varepsilon^*$ for all d	P1S L_{ε^*}	§1
M_d	Anchor set: minimal perturbation-support subspace	P1S FD3	§2
Π	Pool: $S_\Gamma \ominus \bigoplus_d M_d$	P1S T_{sep}	§2
E_d	Admissibility projection: B-orth. proj. onto M_d	P1S FD5/T_{adj}	§2
$B(\cdot,\cdot)$	Sector-orthogonal bilinear form on V	P1S L_ω	§3.1
$\mathcal{A}$	Admissibility algebra: $\mathrm{Alg}_{\mathbb{R}}(\{E_d, E_{\{d_i,d_j\}}\})$	P1S O3	§2.3
ω	Faithful positive state on $\mathcal{A}$	P1S O4	§2.3
$\mathcal{H}_\omega$	GNS Hilbert space from $(\mathcal{A},\omega)$	P1S T_{GNS}	§2.3
$\Delta(d_1,d_2)$	Superadditive surplus: $\varepsilon(\{d_1,d_2\}) - \varepsilon(d_1) - \varepsilon(d_2)$	P1S L_Δ	§2

APF $\leftrightarrow$ standard terminology.

APF term	Standard / GPT equivalent
Distinction d	Effect / binary test / observable outcome
Admissibility projection E_d	Sharp effect / compression / filter
Pool Π	Shared latent support / coupling sector
Superadditivity surplus Δ	Nonclassicality witness / coupling measure
Sector decomposition (T_{sep})	Face decomposition of state space
Admissibility algebra $\mathcal{A}$	Observable algebra / C^*-algebra of observables
GNS representation	Gelfand–Naimark–Segal construction

Dependency table. Each row gives a result proved in this supplement, its inputs, and the downstream results that depend on it.

Theorem	**Inputs**	**Used by**
L_{Cauchy} (§1)	A1, L_{loc}, L_{ε^*}	cost ranking (Technical Supplement I (*The Classification Core*))
Gauge ident. (§2.3)	T_{alg}, T_{sep}, L_{loc}, T3	Theorem R; closure theorems (Technical Supplement I (*The Classification Core*))
Exact sequence / T_{rel} (§2.3)	gauge ident.	abelian closure (Technical Supplement I (*The Classification Core*))
Matter category + lift (§2.3)	exact sequence; charge rep (I-typing, named input)	T_{field}'s category (Technical Supplement I (*The Classification Core*))
EC / CM (§2)	R-EC-inv + I-typing (EC); A2 over A1+MD (CM)	closure theorems (Technical Supplement I (*The Classification Core*))
$C(G)$ program (§2, §4)	operational premise pair	cost ranking (Technical Supplement I (*The Classification Core*))
T_{7B} (§3.1)	A1, K2, K3, SP, L_ω, T_{adj}	$L_{\text{irr,uniform}}$
$L_{\text{irr,uniform}}$ (§3.2)	L_{loc}, L_{nc}, L_{irr}, T_{7B}, saturation bound	Theorem R (R2 chain)
$B1'$ (§3.3)	A1 (minimality), T_M, rep. theory	Theorem R (R1 chain)

Contents

On operational language. Operational verbs are local readings of an underlying admissibility-space structure under a temporal slicing. They are retained because they are the working vocabulary of physics; the structural reading is always available: “acts,” “forms,” “selects,” and “evolves” mean that the corresponding admissible configuration, quotient, minimum, or continuation relation exists. This convention is inherited from Paper 0 and Paper 1 Supplement v8.24.

Part I

The Foundational Chain

1. Cost function uniqueness (L_{Cauchy})

The admissibility cost function F maps channel count to capacity committed. Paper 2 §4 states the uniqueness result; this section provides the full formal treatment, specifying domain, regularity, and operational meaning of each condition.

1.1 Domain and operational definitions

The admissibility cost function F maps the *number of admissibility channels* consumed by a distinction to the capacity committed. We now specify its domain, its operational meaning, and the three conditions that determine it uniquely.

Definition 1.1 (Enforcement channel count)**.** At an interface Γ with finite capacity $C(\Gamma)$ (A1), each irreducible distinction $d \in \mathcal{D}_{\text{irr}}(\Gamma)$ costs ε^* to enforce (L_{ε^*}, Paper 1 Supplement). A composite distinction occupying n independent irreducible channels has cost $\varepsilon(\{d_1, \ldots, d_n\}) = n \cdot \varepsilon^*$ when the n channels are at *separate interfaces* (L_{loc}), or $n \cdot \varepsilon^* + \Delta$ when co-located (L_Δ): $\Delta \geq 0$ in general, and strictly $\Delta > 0$ exactly when the co-located channels draw on a non-empty shared pool ($\Pi \neq \{0\}$; the surplus is $\Delta = \kappa_\Pi > 0$, L_Δ, and vanishes, $\Delta = 0$, when $\Pi = \{0\}$). The *channel count* is the integer $n(d) := \varepsilon(d)/\varepsilon^*$ evaluated at separate interfaces, where the surplus $\Delta = 0$.

The additive/superadditive determinant is the shared pool. Whether two co-located channels add ($\Delta = 0$) or super-add ($\Delta > 0$) is not a free per-case choice but is fixed by one structural variable: the shared pool Π in the sector decomposition $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ (T_{sep}). The L_Δ surplus is $\Delta = \kappa_\Pi$ with $\kappa_\Pi > 0 \iff \Pi \neq \{0\}$, so $\Pi = \{0\}$ (separated, no common reservoir) gives additivity and $\Pi \neq \{0\}$ (genuine co-location) gives the surplus. The separated-interface additivity used by L_{loc} and C1 and the superadditivity used by L_Δ and the confinement mechanism are therefore one surplus functional evaluated on $\Pi = \{0\}$ versus $\Pi \neq \{0\}$ configurations, not two opposed cost primitives. Accordingly the structural existence ledger L_{count} is additive because its matter and Higgs channels are separated ($T_M + L_{\text{loc}}$, $\Pi = \{0\}$); the gauge-generator contribution is additive at the level of resolution channels (one independent distinction per group-manifold dimension; the generator cost bound, Lemma 2.22), with the cost-exactness $C(G) = \dim G$ carried as the composed operational statement — lower bound on the operational-encoding premise, upper bound on the disjoint-anchor realizability premise — banked as [3] and realized paper-side in Proposition 4.1 of Technical Supplement I (*The Classification Core*).

The domain of F is therefore the positive integers: $F : \mathbb{N} \to \mathbb{R}_{>0}$, where $F(n)$ gives the capacity cost (in units of ε^*) of enforcing n independent channels at separate interfaces.

[3] Code reference: `check_L_cost_gauge` in `core.py`.

Remark 1.2 (Why the domain is $\mathbb{N}$, not $\mathbb{R}^+$)**.** Enforcement channels are discrete. This discreteness is not assumed — it is derived from A1 + MD:

(i) A1 provides finite capacity $C(\Gamma) < \infty$.
(ii) L_{ε^*} (from MD) provides a positive cost floor $\varepsilon^* > 0$ for each irreducible channel.
(iii) Therefore the maximum number of independently admissible channels at any interface is $\lfloor C(\Gamma)/\varepsilon^* \rfloor < \infty$, an integer.

The domain $\mathbb{N}$ is a *consequence*, not a choice. The classical Cauchy problem on $\mathbb{R}^+$ — with its pathological non-measurable solutions — does not arise because the domain is already discrete.

1.2 The three conditions and their derivation

Theorem 1.3 (L_{Cauchy}: Cost Function Uniqueness)**.** *Let $F : \mathbb{N} \to \mathbb{R}_{>0}$ satisfy:*

$$\textbf{\textit{C1 (Additivity):}} \qquad F(a+b) = F(a) + F(b) \quad \textit{for all } a, b \in \mathbb{N}, \tag{1}$$

$$\textbf{\textit{C2 (Monotonicity):}} \qquad a > b \;\Rightarrow\; F(a) > F(b) \quad \textit{for all } a, b \in \mathbb{N}, \tag{2}$$

$$\textbf{\textit{C3 (Unit):}} \qquad F(1) = 1. \tag{3}$$

Then $F(n) = n$ for all $n \in \mathbb{N}$.

Proof. By C1 (induction): $F(n) = F(1 + 1 + \cdots + 1) = n \cdot F(1)$. By C3: $F(1) = 1$. Therefore $F(n) = n$ for all $n \in \mathbb{N}$. C2 is satisfied automatically: $n > m \Rightarrow F(n) = n > m = F(m)$. □

The proof on $\mathbb{N}$ is trivial once the three conditions are established. The substantive content is the *derivation of the conditions from A1*:

Proposition 1.4 (Derivation of C1–C3 from admissibility)**.**

C1 (Additivity). *Let d_1, d_2 be distinctions at* separate *interfaces Γ_1, Γ_2. By L_{loc} (Paper 1 Supplement [2], §4): admissibility at separate interfaces is independent, with independent budgets. The total admissibility cost is $\varepsilon(d_1)+\varepsilon(d_2)$ — costs add exactly when the admissibility regions are disjoint. In channel-count language: $F(n(d_1) + n(d_2)) = F(n(d_1)) + F(n(d_2))$.*
The operational meaning of "additivity" is therefore: independently admissible channels at separate interfaces contribute independently to the total capacity commitment. *This is the content of L_{loc} applied to the cost functional.*
What C1 does not claim: C1 does not assert additivity at a shared *interface. At a shared interface, the superadditive surplus $\Delta > 0$ (L_Δ) produces non-additive costs. C1 applies only to the separated-interface regime, which is precisely the regime relevant to the cost function F (Definition 1.1).*

C2 (Monotonicity). *More admissibility channels require more capacity. This is immediate from A1: each channel costs $\varepsilon^* > 0$ (L_{ε^*}), so adding a channel increases the total cost.*

C3 (Unit). *$F(1) = 1$ is a choice of capacity units: we measure cost in units of the single-channel cost ε^*. This is a normalization convention, analogous to $c = 1$ in natural units. It introduces no physical content.*

Marginal-floor refinement (Paper 10 alignment). The cost-function uniqueness theorem above operates on the separated-interface regime where, by L_{loc}, costs add exactly: $\Delta = 0$. In the broader continuation calculus of Paper 10 (*The Calculus of Finite Continuability*) v1.11, §3, this regime corresponds to a substrate with no shared structural commitments, where the in-isolation cost floor and the marginal floor coincide. In regimes with shared substrate commitments — broken vacuum, confined-phase QCD substrate, topological infrastructure — the in-isolation cost overcounts (the shared piece is paid once by the carrying context, not once per distinction), and the marginal floor

$$\varepsilon^*_\Gamma \;:=\; \inf\{\kappa_\Gamma(S \otimes d) - \kappa_\Gamma(S) : S \text{ admissible},\; d \text{ tested},\; S \otimes d \text{ admissible}\}$$

becomes the operative invariant. Paper 2's L_{Cauchy} and the downstream L_{count} count $C_{\text{total}} = 61$ pass through unchanged in either reading: L_{Cauchy} governs the separated-interface regime where the floors agree; the SM-interface count is read as a count of marginal channel-distinctions on top of the broken-vacuum and confined-phase substrate. Paper 4 Supplement v1.5 §2.2 applies the marginal-floor reading to T_{4F} (the three-generations bound) explicitly. No theorem in this section requires modification.

Connection to resource theories and GPTs. The conditions C1–C3 have close analogues in resource-theoretic frameworks: C1 corresponds to the tensor-product structure of independent resources; C2 to monotonicity under free operations; C3 to the choice of a gold standard (unit resource). The operational stance — deriving algebraic structure from resource constraints — is shared with GPT-style reconstructions (Hardy [11], CDP [12]) and the "accessible fragments" approach. The difference is scope: those programs reconstruct the state space; this program uses the same starting point to derive gauge content and matter spectrum.

1.3 Rational extension (for Paper 4A)

Paper 2's results ($L_{\text{gauge,unique}}$, T_{field}, L_{count}, P_{exhaust}, L_{equip}) use F only on $\mathbb{N}$. The Weinberg angle derivation (Paper 4A) evaluates F at the rational argument $1/d$ (the reciprocal channel count), producing $\gamma = F(d) + F(1/d) = d + 1/d$. We record the rational extension here for completeness.
Proposition 1.5 (Rational extension of F, with operational premise)**.** ***Premise (Sharing-admissibility, S)****: A single admissibility channel evaluated at fractional occupancy $1/q$ ($q \in \mathbb{N}$) contributes $F(1/q)$ to each of q separated-interface sectors that jointly share the channel, and the q shared contributions combine additively under C1. Equivalently, F is extended to $\mathbb{Q}_{>0}$ as the unique additive extension consistent with C1 applied q times across sharing.*

Conclusion. *Under premise S, $F(p/q) = p/q$ for all $p, q \in \mathbb{N}$.*

Proof. For any $q \in \mathbb{N}$: by S, F is defined at p/q; by C1 applied q times, $q \cdot F(p/q) = F(p/q + \cdots + p/q) = F(p) = p$. Therefore $F(p/q) = p/q$. □

Logical status of the sharing premise. Premise S is not derivable from C1 alone: C1 as stated in the integer domain ($a, b \in \mathbb{N}$) is vacuous at fractional arguments, so "C1 extended to $\mathbb{Q}_{>0}$" is definitionally a new domain assignment. S supplies the operational content: it interprets $F(1/q)$ as the per-sector share of a channel decomposed across q separated interfaces. The interpretation is consistent with L_{loc} (locality of capacity) and T_M (interface monogamy)—sharing at separated interfaces does not incur superadditive surplus—but it is a reading beyond C1's native domain. The honest flag is that the Weinberg-angle derivation in Paper 4A rests on C1 + S, not on C1 alone. S is therefore listed alongside MD, BW, and the constitutive admissibility semantics (SP, K3, FD1–FD6) in the assumption inventory when Paper 4A uses the rational extension.

Operational justification for the rational extension. The rational argument $1/d$ arises when the cost functional is evaluated on a *fractional* channel occupancy: a single admissibility channel shared among d sectors contributes $1/d$ of a full channel to each sector. This is the capacity-accounting interpretation of the Weinberg angle formula's "inverse channel count" term. The extension from $\mathbb{N}$

to $\mathbb{Q}_{>0}$ requires no additional regularity assumption (no measurability, no continuity): it follows from C1 applied q times, as the proof above shows.

Exact status of the rational extension in the axiomatics. There are two different statements, and the supplement keeps them separate. First, on the native domain of L_{Cauchy}, $F : \mathbb{N} \to \mathbb{N}$, C1 determines $F(n) = n$; no fractional arguments occur. Second, once one *extends the domain* to shared fractional channel occupancies, premise S supplies the operational meaning of $F(p/q)$ and C1 then forces the unique value $F(p/q) = p/q$. Therefore the rational extension is not a theorem of C1 on $\mathbb{N}$ alone. It is a theorem of *C1 on the enlarged sharing-admissible domain* C1+S. This is the version used in Paper 4A whenever $F(1/d)$ appears. No measurability, continuity, or real-domain Cauchy assumption is added; the only added premise is the operational domain-extension premise S.

No real extension needed. The classical Cauchy problem on $\mathbb{R}^+$ admits pathological (non-measurable, non-monotone, additive) solutions constructible via the Axiom of Choice. These are excluded by C2 (Darboux, 1875): any Lebesgue-measurable (or merely bounded on an interval) additive function $f : \mathbb{R}^+ \to \mathbb{R}$ satisfying $f(x + y) = f(x) + f(y)$ and $f(1) = 1$ is $f(x) = x$. However, no result in this paper or in Paper 4A requires the domain to extend beyond $\mathbb{Q}_{>0}$. The Darboux theorem is available as a safety net but is never load-bearing. [4]

1.4 What $\mathbf{L}_{\text{Cauchy}}$ eliminates

Without L_{Cauchy}, the cost function F is a free parameter. Any monotone function $F : \mathbb{N} \to \mathbb{R}_{>0}$ with $F(1) = 1$ would be compatible with A1. With L_{Cauchy}, the class collapses to a single element:

Candidate	**Condition violated**	**Consequence if allowed**
$F(n) = n^2$	C1 ($9 = F(3) \neq F(1) + F(2) = 5$)	Weinberg angle shifts
$F(n) = \sqrt{n}$	C1 ($\sqrt{3} \neq 1 + \sqrt{2}$)	Capacity non-additive
$F(n) = \log n$	C1 ($\log 3 \neq 0 + \log 2$); C3 ($F(1) = 0$)	No unit cost
$F(n) = c$ (constant)	C2 ($F(2) = F(1)$); C1 ($c \neq 2c$)	Cost insensitive
$F(n) = n^{3/2}$	C1 ($3\sqrt{3} \neq 1 + 2\sqrt{2}$)	Weinberg angle shifts

Each alternative is excluded by C1 alone (additivity). The uniqueness is not a consequence of fine-tuning or accidental cancellation: it is a theorem of algebra (the only additive function on $\mathbb{N}$ with $F(1) = 1$ is the identity). C2 is logically redundant on $\mathbb{N}$ once C1 and C3 are established, but it is stated explicitly because it is the physically motivated condition (more channels cost more) and because it becomes non-redundant on $\mathbb{R}^+$ where it excludes pathological solutions.

[4] Code reference: `check_L_Cauchy_uniqueness` in `L_Cauchy_uniqueness.py`.

2. From superadditivity to gauge structure

This section restates the derivation chain $L_\Delta \to L_{\rm blk} \to L_\Pi \to T_{\rm alg}$ (proved in the Paper 1 Supplement) in condensed form, and then proves the gauge identification theorem that separates internal (gauge) automorphisms from kinematic (diffeomorphism) automorphisms.

The chain is summarized first; the proofs are in the Paper 1 Supplement and are not reproduced. The gauge identification theorem (§2.3) is new to this supplement.

IJC routing. The chain $L_\Delta \to L_{\rm blk} \to L_\Pi \to T_{\rm alg}$ recapitulated below assumes the (IJC) branch of the Sep/IJC Representation Theorem (Paper 1 Supplement v8.9 §"Sep/IJC representation theorem"). Branch (IJC) is the substrate-perturbation joint non-separability condition: at a co-located pair (d_1, d_2), the joint threat class $\mathsf{T}(d_1, d_2)$ strictly exceeds the union $\mathsf{T}(d_1) \cup \mathsf{T}(d_2)$. In the alternative branch (Sep), every joint perturbation reduces to a separable convex combination of single-side perturbations, the pool Π is inert, and Link 1's $\Delta > 0$ degenerates to $\Delta = 0$ (the spectator countermodel $V = M_{d_1} \oplus M_{d_2} \oplus \Pi$ with Π inert; see Paper 1 Supplement v8.41 §"Worked models: Sep, IJC, and coupled-but-classical"). PLEC alone does not force branch (IJC); it must be flagged as a substrate-level premise. The Sep/IJC dichotomy itself is structural (certified by `check_T_IJC_dichotomy` and the inseparability bridge `check_T_inseparable_IJC`, `apf/core.py`): branch (Sep) is the substrate-factorizable case (a commuting hidden-variable extension exists), branch (IJC) the inseparable one for which no such factorization exists. The branch verdict has two parts the framework keeps distinct. The *bridge* — that an inseparable (IJC) interface forces a noncommutative record algebra — is DERIVED internally (Paper 5 Supplement v6.8 Theorem *general-finite-query-noncommutative-bridge*, graded [$\mathrm{P}_{\rm math}$+$\mathrm{P}_{\rm APF}$]): given an interface's declared finite records, branch (IJC) holds iff no faithful Boolean global-section defender exists — an internal Boole / LP feasibility question — and Bell, CHSH, Fine, and Kochen–Specker are RECOVERED as special marginal presentations of this criterion, not imported as theorems. The *occupancy*, by contrast, remains empirical: that a physical quantum-capable interface presents records (its measured correlation table) lying outside the Boole polytope is read off the empirical marginals, not derived from A1 — exactly as the framework reads the Planck parameters and lattice-QCD constants off their records, with Bell/KS serving as the in-house diagnostics for whether those records clear the polytope. (The earlier "substrate-richness forces (IJC)" reading is superseded: substrate-richness permits but does not force inseparability.) Theorems in this section therefore carry the [P+IJC] tag in the sense "proved given the (IJC) branch occupancy at this interface" — the bridge internal and derived, the occupancy empirically certified.

Proposition 2.1 (Worked example: the rotated three-dimensional Sep/IJC witness)**.** *Let* $V = \operatorname{span}\{e_1, e_2, p\}$ *with individual anchor spaces* $M_1 = \mathbb{R}e_1$*,* $M_2 = \mathbb{R}e_2$*, and pool* $\Pi = \mathbb{R}p$*. In the Sep branch the joint defender is* $E_{12}^{\rm Sep} = E_1 + E_2$ *and* $[E_1, E_{12}^{\rm Sep}] = 0$*. In the IJC branch the witness model's joint defender is taken in the rotated two-parameter form*

$$E_{12}^{\rm IJC} = \pi_{\operatorname{span}\{\cos\theta\, e_1 + \sin\theta\, p,\ e_2\}}, \qquad 0 < \theta < \pi/2,$$

— the defining ansatz of this witness model, legitimate because the proposition's role is to exhibit a witness, not to derive a normal form — so that

$$[E_1, E_{12}^{\rm IJC}] = \cos\theta \sin\theta \big(|e_1\rangle\langle p| - |p\rangle\langle e_1|\big) \neq 0.$$

Thus the Sep countermodel is exactly the $\theta = 0$ *case; it is excluded precisely when the substrate geometry forces nonzero pool rotation.*

Proof. In the Sep branch every joint perturbation decomposes into separate single-side perturbations. The least-cost joint support is then the direct sum $M_1 \oplus M_2$, hence $E_{12}^{\mathrm{Sep}} = E_1 + E_2$ and all generated projections commute. This is a genuine countermodel to any attempted derivation of noncommutativity from A1 alone.

In the IJC branch there exists a joint threat not representable as a convex mixture of single-side threats. A direct-sum defender leaves a pool-supported component undefended, so within the ansatz family the least-cost joint defender rotates into Π by an angle $\theta \neq 0$. Writing $w_\theta = \cos\theta\, e_1 + \sin\theta\, p$, the joint projection is $|w_\theta\rangle\langle w_\theta| + |e_2\rangle\langle e_2|$. A direct multiplication gives the displayed commutator. The obstruction is therefore not definitional: Sep and IJC have the same individual anchor spaces and the same finite ledger, but only IJC supplies the gluing obstruction that prevents a faithful all-commuting representation. □

Interpretive consequence. This is a worked example exhibiting the mechanism — it is not a universal derivation of the rotated normal form, and no downstream result consumes it as one. The toy model is the finite-dimensional version of the reviewer-facing "global section" obstruction. In Sep, local defenders glue to a single Boolean refinement. In IJC, the local defenders agree on their individual marginals but the joint defender is rotated into the pool, so no faithful Boolean refinement preserves both the tested marginals and the partial product. The bridge theorem therefore depends on an explicit, checkable substrate condition: nonzero pool rotation, equivalently $\Delta > 0$.

$\Delta = 0$ falsifiability branch

The witness above is not itself a proof that every physical interface lies in the IJC branch. It proves that, once an interface has nonzero pool rotation, the Sep Boolean-refinement defender is unavailable. The physical necessity of $\Delta > 0$ is inherited from the Paper 1 Sep/IJC representation theorem together with the empirical Bell–Kochen–Specker record: interfaces that exhibit Bell/Tsirelson or contextuality signatures cannot be faithfully represented by a single Boolean refinement. Conversely, if an interface is well approximated by $\Delta = 0$, APF predicts the Sep branch: classically composable records, no context-dependent holonomy, no Bell/Tsirelson violation sourced by that interface, and no independent gauge curvature from that interface. Thus the branch condition is falsifiable rather than smuggled.

Proposition 2.2 (Stand-alone proof spine: superadditivity to finite semisimple carrier)**.** *Assume a finite interface Γ satisfying A1, MD, the IJC branch condition $\Delta > 0$, the positive cost kernel B, closure of tested continuations under adjoint, and finite resolution. Then the admissibility records generated by the interface form a finite-dimensional semisimple $*$-algebra whose noncommuting part is nontrivial.*

Proof. A1 and finite resolution bound the number of mutually admissible primitive record distinctions at Γ; hence the algebra generated by tested finite continuations is finite-dimensional after quotienting by observational equivalence. Positivity of B and adjoint closure make that quotient a finite-dimensional complex $*$-algebra carrying a faithful positive state, hence (by GNS) a faithful $*$-representation on a finite-dimensional Hilbert space. A finite-dimensional adjoint-closed complex subalgebra of $\mathcal{B}(\mathcal{H})$ is a finite-dimensional C^*-algebra, and every finite-dimensional C^*-algebra is semisimple (its Jacobson radical is a nil ideal, and a C^*-algebra has no nonzero nil ideals); Wedderburn–Artin then gives a direct sum of matrix algebras over the selected complex field. This is the standard route, and it consumes nothing beyond the stated hypotheses. (Earlier presentations reached semisimplicity through a separately named "radical pricing" premise; the C^* route makes that premise unnecessary, and it is withdrawn — MD's exclusion of zero-cost directions survives only as the physical gloss on what the observational-equivalence quotient removes.) Finally, the banked non-commutativity chain ($L_\Delta \to L_{\text{blk}} \to L_\Pi \to T_{\text{alg}}$, §2.1) shows that $\Delta > 0$ forces a nonzero commutator among tested defenders — the rotated three-dimensional model of Proposition 2.1 exhibits the mechanism — so the semisimple quotient is not wholly commutative. This proposition records the proof boundary explicitly: without IJC one may remain in the Sep/Boolean branch, and without the positivity/finite-resolution hypotheses one does not obtain the finite semisimple carrier used by Skolem–Noether. □

Scope of the proposition. The conclusion is conditional, and two of its steps are scoping clauses rather than closed arguments. First, the argument treats a nonzero pool as cost-active. An interface whose pool carries no positive joint surplus is the Sep branch, and whether a physical interface occupies the IJC branch is an empirical occupancy read off its records — the $\Delta = 0$ falsifiability box above owns exactly this. Second, the step from "not a direct sum of individual defenders" to "mixes sectors" is not established for arbitrary operators: operators that are block-diagonal in some finer decomposition yet lie outside the generated algebra exist, and the proposition's conclusion concerns the algebra generated by the tested defenders, not all of $\text{End}(V)$. The proposition therefore sits inside the hybrid keystone framing of §2.1: the bridge is derived given the IJC occupancy; the occupancy itself is empirical, not derived.

Lemma 2.3 (Boolean refinement blocks the IJC branch)**.** *If all queried finite contexts at an interface admit a faithful common Boolean refinement preserving the tested marginals and the defined partial products, then the Sep/IJC obstruction is absent: the pool rotation angle is $\theta = 0$, the joint defender is the direct-sum defender, and the noncommutative admissibility-algebra branch is not activated. Conversely, a nonzero pool rotation supplies an explicit gluing obstruction to such a faithful Boolean refinement.*

Proof. A faithful Boolean refinement assigns a single joint record algebra whose atoms refine every queried context and whose restrictions reproduce the observed marginals. In the notation of Proposition 2.1, that means the joint defender is represented on the direct sum $M_1 \oplus M_2$ with no pool-supported component. Hence $E_{12} = E_1 + E_2$ and all context projections commute. If instead the minimum joint defender contains the vector $w_\theta = \cos\theta e_1 + \sin\theta p$ with $\theta \neq 0$, the restriction of the joint defender to the individual contexts still reproduces the single-context marginals. Suppose, for contradiction, that a faithful Boolean refinement $\mathcal{B}$ also preserved the defined partial product. Then both E_1 and E_{12} would be sums of atoms in the same Boolean algebra $\mathcal{B}$, hence would commute. But Proposition 2.1 gives $[E_1, E_{12}] = \cos\theta \sin\theta(|e_1\rangle\langle p| - |p\rangle\langle e_1|) \neq 0$. Therefore no such atom assignment exists. This is precisely the finite global-section obstruction familiar from sheaf-theoretic contextuality: the single-context marginals exist, but they do not glue to a faithful global Boolean record preserving the tested product. □

Gauge as continuation-profile-preserving relabeling. In the continuation-calculus vocabulary of Paper 10, gauge transformations are zero-marginal-cost relabelings that preserve all tested continuation profiles. Here this becomes a finite-interface statement: internal automorphisms preserve the admissibility algebra's sector decomposition, while kinematic automorphisms move interface labels and are outer. The Paper 9 gravitational gauge-reduction analogy is methodological only; Remark 2.17 records the parallel without making it load-bearing for Paper 2.

2.1 The non-commutativity chain (summary)

The chain from superadditivity to non-commutativity has four links, with home as noted per link. Link 1 is established in this supplement (Appendix B.3); Links 2–4 are the constructive chain developed in Paper 5 and certified in the bank, whose endpoint (non-commutativity) is the classification theorem of Paper 1 Supplement v8.41 §"Sep/IJC representation theorem" (the *Sep iff commutative record representation* main theorem). We restate the key result of each without re-proof.

Link 1: L_Δ (Superadditivity). For co-located distinctions d_1, d_2 at an interface Γ with non-trivial pool $\Pi \neq \{0\}$: $\Delta(d_1, d_2) := \varepsilon(\{d_1, d_2\}) - \varepsilon(d_1) - \varepsilon(d_2) > 0$. *Source:* Appendix B.3 (this supplement). *Inputs:* A1, K2, K3, T_{sep}, SP.

Link 2: L_{blk} (Direct-Sum Failure). If $\Delta > 0$, the minimum-cost joint defense cannot be a direct sum of individual defenses: $E_{\{d_1,d_2\}} \neq E_{d_1} + E_{d_2}$. *Source:* Paper 5 (non-commutativity chain); bank `check_L_nc` (`core.py`).

Link 3: L_Π (Codespace Rotation). The minimum-cost joint defender π_{W_*} has a codespace W_* rotated into the pool Π by angle $\theta \neq 0$. The off-diagonal component $F_\Pi := \pi_{W_*} - E_{d_1} - E_{d_2} \neq 0$ is a new generator supported on the pool. *Source:* Paper 5; bank `check_L_Pi` (`core.py`).

Link 4: T_{alg} (Non-Commutativity). The commutator $[E_{d_1}, F_\Pi] \neq 0$ witnesses non-commutativity. The algebra $\mathcal{A} = \mathrm{Alg}_{\mathbb{R}}(\{E_d\} \cup \{E_{\{d_i,d_j\}}\})$ admits no faithful commutative $*$-representation. *Source:* classification — Paper 1 Supplement v8.41 §"Sep/IJC representation theorem" (*Sep iff commutative record representation*); construction — Paper 5 + bank `check_T_alg` (`core.py`). *Inputs:* L_Δ, L_{blk}, L_Π.

The chain is pre-Hilbert: no GNS construction, no spectral theory, no Hilbert space is invoked. The non-commutativity is an algebraic fact about $\mathrm{End}(V)$, established before $\mathcal{H}_\omega$ is constructed.

2.2 A two-sector admissibility-algebra example

Worked example: pool, anchors, projections

Let a finite interface have sector space

$$V = M_c \oplus M_w \oplus \Pi, \qquad M_c \cong \mathbb{C}^3, \quad M_w \cong \mathbb{C}^2, \quad \Pi \cong \mathbb{C}.$$

Here M_c and M_w are anchor spaces for two tested distinctions, and Π is the residual pool. The individual admissibility projections are $E_c = \pi_{M_c}$ and $E_w = \pi_{M_w}$. In the Sep branch the joint defender is $E_c + E_w$. In the IJC branch the least-cost joint defender has a graph component into the pool, for example $\pi_{\mathrm{graph}(K)}$ for some nonzero $K : M_c \to \Pi$, so the algebra generated by E_c, E_w, and the joint projection is noncommutative. After complex carrier closure, the corresponding finite admissibility algebra has the canonical block form

$$\mathcal{A} \cong M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}.$$

Its inner automorphism group is $\mathrm{PU}(3) \times \mathrm{PU}(2)$; its block-center supplies a relative-phase torus. The later results of this section show how the field-module lift replaces the projective observable action by $\mathrm{SU}(3) \times \mathrm{SU}(2)$ on matter fields and how the relative torus is pruned to the single hypercharge direction. This toy model is not the Standard Model derivation; it is only the smallest notation map for M_d, E_d, Π, block centers, and inner automorphisms.

Remark 2.4 (Example scope)**.** The example deliberately omits generations, anomaly equations, and the full 1,680-template scan. Its role is definitional: it shows how the nonstandard words "pool," "anchor," and "admissibility projection" are represented by ordinary finite-dimensional subspaces and projections before any classification theorem is invoked.

2.3 Gauge identification theorem

The central question is: how is the gauge group identified with the automorphism group of the admissibility algebra, and how are spatial/kinematic automorphisms (e.g., diffeomorphisms) excluded?

The answer rests on a structural separation: the admissibility algebra $\mathcal{A}$ acts on the *internal* sector decomposition at a single interface Γ, which is finite-dimensional and defined by T_{sep}. Diffeomorphisms are *inter-interface* maps (L_{loc}) and therefore commute with the internal algebra by construction.

Definition 2.5 (Local support: the algebra and the localized unital maps)**.** Two distinct objects are localized at an interface Γ, and they must not be conflated.

(LS1) **The ambient corner algebra.** $\mathcal{A}_{\text{corner}}(\Gamma) := \{A \in \text{End}(V) : A(S_\Gamma) \subseteq S_\Gamma,\ A|_{S_\Gamma^\perp} = 0\} \cong \text{End}(S_\Gamma) \oplus 0$. This is a linear $*$-subalgebra of $\text{End}(V)$ (closed under addition, scalar multiplication, products, and adjoints), with unit the B-orthogonal projection P_Γ onto S_Γ (a *local* unit; $\mathcal{A}_{\text{corner}}(\Gamma)$ is non-unital as a subalgebra of $\text{End}(V)$). The corner is *simple* ($\cong \text{End}(S_\Gamma)$); it is the ambient home for local observables, not the admissibility algebra itself, which is the generated proper subalgebra $\mathcal{A}(\Gamma) \subseteq \mathcal{A}_{\text{corner}}(\Gamma)$ defined below.

(LS2) **The localized unital transformations.** $\mathcal{U}(\Gamma) := \{U \in \text{End}(V) : U(S_\Gamma) \subseteq S_\Gamma,\ U|_{S_\Gamma^\perp} = \text{id}\}$. This set is closed under composition but *not* under addition ($(U + U')|_{S_\Gamma^\perp} = 2\,\text{id}$): it is a monoid, not an algebra, and its invertible B-unitary elements form the *localized unitary group* $U_\Gamma \oplus \text{id}$. Localized symmetries live here; observables live in (LS1).

Earlier versions used a single definition demanding $A|_{S_\Gamma^\perp} = \text{id}$ for all localized operations; that class is not closed under addition and cannot serve as the observable algebra. Where a tensor factorization of V exists, (LS1) plays the role of the conventional $\mathcal{B}(H_\Gamma) \otimes I$; here the substrate decomposition is a direct sum, so the correct algebra is the corner $\text{End}(S_\Gamma) \oplus 0$.

Derivation of local support from FD3. FD3 (anchor-set locality) states that the anchor set M_d of any distinction $d \in \mathcal{D}(\Gamma)$ lies within S_Γ. The admissibility projection E_d has $\text{im}(E_d) = M_d \subseteq S_\Gamma$ and $\ker(E_d) \supseteq S_\Gamma^\perp$ (by T_{adj}: E_d is the B-orthogonal projection onto M_d, hence $E_d|_{M_d^\perp} = 0$, and $S_\Gamma^\perp \subseteq M_d^\perp$). Therefore $E_d(v) = v$ for $v \in M_d$ and $E_d(v) = 0$ for $v \in S_\Gamma^\perp$: each generator annihilates the complement and so satisfies (LS1). The *admissibility algebra* $\mathcal{A}(\Gamma) = \text{Alg}(\{E_d : d \in \mathcal{D}(\Gamma)\})$ is generated inside the corner $\mathcal{A}_{\text{corner}}(\Gamma) \cong \text{End}(S_\Gamma) \oplus 0$, and (LS1) *is* closed under sums, scalar multiples, products, and adjoints. Therefore every $A \in \mathcal{A}(\Gamma)$ has local support in the (LS1) sense. The two algebras are distinct objects carrying distinct symbols: $\mathcal{A}_{\text{corner}}(\Gamma)$ is the simple ambient corner, while $\mathcal{A}(\Gamma)$ is the generated proper subalgebra, and it is the generated algebra — with its non-simple block structure — that every downstream result (the gauge identification, the exact sequence, EC/CM) consumes. (The v3.9 text asserted that the E_d act as the identity on the complement and satisfy the unital form of local support; that was wrong — zero and the identity are different operators — and is repaired by the two-object definition above.)

Proposition 2.6 (Block structure of the generated admissibility algebra)**.** *The generated admissibility algebra $\mathcal{A}(\Gamma) = \text{Alg}(\{E_d\} \cup \{E_{\{d_i,d_j\}}\})$ is a finite-dimensional $*$-subalgebra of the simple corner $\mathcal{A}_{\text{corner}}(\Gamma)$, and it — not the corner — carries the Wedderburn block decomposition consumed downstream: $\mathcal{A}(\Gamma) \cong \bigoplus_k M_{n_k}(\mathbb{C})$, with the Standard-Model specialization $\{n_k\} = \{3, 2, 1\}$, i.e. $M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}$.*

Proof. The generated algebra is a $*$-subalgebra of the corner by the FD3 derivation above (each generator annihilates $S_\Gamma^\perp$). Semisimplicity of the generated algebra is supplied by Proposition 2.2, under that proposition's stated premises, by the standard finite-dimensional C^* route. Wedderburn–Artin (T2a) with field selection (T2c) then gives $\mathcal{A}(\Gamma) \cong \bigoplus_k M_{n_k}(\mathbb{C})$. The specialization $\{n_k\} = \{3, 2, 1\}$ is not proved at this point of the document: it is delivered by Theorem R's carrier requirements together with the fermion scan, and enters downstream as hypothesis H1 of Theorem 7.7 of Technical Supplement I (*The Classification Core*). The corner, being simple, admits no such decomposition; any statement attributing block structure to $\mathcal{A}_{\text{corner}}(\Gamma)$ would be false, which is why the two symbols are kept apart. ☐

Lemma 2.7 (Commutation from disjoint support)**.** *Let Γ_1, Γ_2 be interfaces with B-orthogonal substrates: $B(u, v) = 0$ for all $u \in S_{\Gamma_1}$, $v \in S_{\Gamma_2}$ (from L_ω applied to disjoint interfaces via L_{loc}). Then:*

- *(i) (*Observable algebras.*) For all $A \in \mathcal{A}_{\text{corner}}(\Gamma_1)$ and $B \in \mathcal{A}_{\text{corner}}(\Gamma_2)$ (the (LS1) corners), $AB = BA = 0$; in particular $[A, B] = 0$. The generated admissibility algebras $\mathcal{A}(\Gamma_i) \subseteq \mathcal{A}_{\text{corner}}(\Gamma_i)$ inherit the conclusion as subalgebras.*
- *(ii) (*Localized unital transformations.*) For all $U \in \mathcal{U}(\Gamma_1)$ and $U' \in \mathcal{U}(\Gamma_2)$, $[U, U'] = 0$.*

Proof. Decompose $V = S_{\Gamma_1} \oplus S_{\Gamma_2} \oplus W$, where $W = (S_{\Gamma_1} \oplus S_{\Gamma_2})^\perp$ (exists because V is finite-dimensional and B is positive-definite).

(i) By (LS1), $A = A_1 \oplus 0 \oplus 0$ and $B = 0 \oplus B_2 \oplus 0$ with $A_1 := A|_{S_{\Gamma_1}}$, $B_2 := B|_{S_{\Gamma_2}}$. Then AB annihilates S_{Γ_1} and W (where B vanishes) and annihilates S_{Γ_2} (since B maps it into S_{Γ_2}, which A annihilates); so $AB = 0$, and symmetrically $BA = 0$.

(ii) By (LS2), $U = U_1 \oplus \text{id}_{S_{\Gamma_2}} \oplus \text{id}_W$ and $U' = \text{id}_{S_{\Gamma_1}} \oplus U'_2 \oplus \text{id}_W$. For $v = v_1 + v_2 + w$:

$$\begin{aligned} UU'(v) &= U(v_1 + U'_2 v_2 + w) = U_1 v_1 + U'_2 v_2 + w, \\ U'U(v) &= U'(U_1 v_1 + v_2 + w) = U_1 v_1 + U'_2 v_2 + w. \end{aligned}$$

Therefore $UU' = U'U$. Item (i) is the locality statement for observables (Einstein-causality analogue); item (ii) is the statement consumed by the net-automorphism and gauge arguments, which act by localized unitaries. ☐

Remark 2.8 (Two bilinear structures: B and $g_{\mu\nu}$)**.** There are two bilinear structures in the supplement and they live at separate levels. The form B is the positive-definite admissibility-cost kernel on the finite internal sector space $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ of a single interface. By K2 and SP, $B(v, v) > 0$ for every nonzero internal perturbation. It has no timelike direction and is the form used for local support, sector orthogonality, and Lemma 2.7.

The spacetime metric $g_{\mu\nu}$ is introduced later on the tangent space of the continuum manifold. Its spatial block inherits the positive cost kernel, while the irreversible time direction contributes the negative term:

$$g_{\mu\nu} dx^\mu dx^\nu = -N^2 dt^2 + g_{ij}^{(\text{sp})} dx^i dx^j, \qquad g_{ij}^{(\text{sp})} \sim B_{ij}.$$

Thus Lorentzian signature comes from adjoining the irreversible temporal ordering to the positive spatial cost form, not from changing the sign of B. The finite-interface locality theorem therefore uses B; the later spacetime reading of spacelike separation is a derived bridge, not an input to the gauge-algebra argument.

Relationship to AQFT. In AQFT, Einstein causality ($[\mathcal{A}(\mathcal{O}_1), \mathcal{A}(\mathcal{O}_2)] = 0$ for spacelike-separated regions) is a separate axiom. In the APF, it is a theorem: FD3 (anchor-set locality) forces local support (Definition 2.5), which forces commutativity (Lemma 2.7). FD3 is the APF's analogue of Einstein causality, but it is a constitutive definition of what "admissibility at Γ" means, not an independent physical postulate.

Scope of this result. Lemma 2.7 is a theorem *internal to the finite-capacity interface algebra.* It is not claimed to reproduce full AQFT locality in the continuum/type-III regime. Specifically: it applies to admissibility algebras $\mathcal{A}(\Gamma) = M_n(\mathbb{C})$ at finite n, using the positive-definite cost kernel B for the orthogonal decomposition. The continuum-limit analogue (commutativity of type III local algebras for spacelike-separated regions) is a separate result that requires the full machinery of AQFT (Haag–Kastler axioms, nuclearity estimates, split property). The finite-capacity result is the *precursor* from which the continuum result can be recovered in the $C/\varepsilon^* \to \infty$ limit (§2.5), but the two should not be conflated.

Lemma 2.9 (Internal–kinematic exact sequence)**.** *Let $\{\mathcal{A}(\Gamma)\}_{\Gamma\in\mathcal{I}}$ be a net of finite-dimensional $*$-algebras indexed by a set $\mathcal{I}$, with isotony $\mathcal{A}(\Gamma) \hookrightarrow \mathcal{A}(\Gamma')$ for $\Gamma \subseteq \Gamma'$, and with independence: $\mathcal{A}(\Gamma)$ and $\mathcal{A}(\Gamma')$ commute when $\Gamma \cap \Gamma' = \emptyset$ (Lemma 2.7). Every net automorphism α induces a permutation $\pi(\alpha)$ of the index set $\mathcal{I}$ (preserving the isotony order), and π is a group homomorphism. Let $\mathrm{Aut}_{\mathrm{int}} := \ker(\pi)$ (automorphisms fixing $\mathcal{I}$ pointwise, mapping each $\mathcal{A}(\Gamma)$ to itself) and let $K := \mathrm{im}(\pi)$ be the induced group of admissible index transformations. Then the sequence*

$$1 \longrightarrow \mathrm{Aut}_{\mathrm{int}} \longrightarrow \mathrm{Aut}(net) \xrightarrow{\pi} K \longrightarrow 1$$

is exact: $\mathrm{Aut}_{\mathrm{int}}$ *is a normal subgroup and* $\mathrm{Aut}(net)/\mathrm{Aut}_{\mathrm{int}} \cong K$*.*

Proof. π is a homomorphism because index actions compose; its kernel is by definition the fibrewise automorphisms, and kernels are normal. Exactness at the ends is the definition of K and injectivity of the inclusion. □

Remark 2.10 (No semidirect splitting is claimed)**.** Earlier versions asserted $\mathrm{Aut(net)} = \mathrm{Aut}_{\mathrm{int}} \rtimes \mathrm{Aut}_{\mathrm{kin}}$ with a *unique* factorization of every automorphism into internal and kinematic parts. Both claims exceed what the construction delivers: a semidirect decomposition requires a group homomorphism $K \to \mathrm{Aut(net)}$ splitting π, and no such section is constructed here; and a putative "kinematic part" built from fiber isomorphisms extracted from α itself is not determined by the index action alone, since different fiber isomorphisms cover the same permutation — the ambiguity is exactly a coset of $\mathrm{Aut}_{\mathrm{int}}$. Nothing downstream needs the splitting: the gauge identification consumes only the kernel $\mathrm{Aut}_{\mathrm{int}}$, which the exact sequence supplies.

Theorem 2.11 (Gauge Identification)**.** *Let $\mathcal{A}$ be the admissibility algebra at interface Γ (from T_{alg}), and let $\mathrm{Aut}(\mathcal{A})$ denote its $*$-automorphism group. Then:*

(i) The net automorphisms sit in the exact sequence of Lemma 2.9: $1 \to \mathrm{Aut}_{\mathrm{int}} \to \mathrm{Aut}(net) \xrightarrow{\pi} K \to 1$, where $\mathrm{Aut}_{\mathrm{int}}$ (the kernel of the index action) consists of the automorphisms preserving the interface label Γ and K is the induced group of index transformations. No splitting of this sequence is claimed or needed (Remark 2.10).

(ii) $\mathrm{Aut}_{\mathrm{int}}(\mathcal{A})$ contains the inner automorphism group $\mathrm{Inn}(\mathcal{A})$, and equals it blockwise. By T2c ($\mathbb{F} = \mathbb{C}$) and T2a (Wedderburn–Artin), $\mathcal{A} \cong \bigoplus_{k=1}^{K} M_{n_k}(\mathbb{C})$ is semisimple *(central simple only when $K = 1$); Skolem–Noether applies per block, giving $\mathrm{Inn}(\mathcal{A}) \cong \prod_{k=1}^{K} \mathrm{PU}(n_k)$.*

(iii) The physical gauge group G is identified with $\mathrm{Inn}(\mathcal{A})$. Kinematic automorphisms (diffeomorphisms) project non-trivially to K and are excluded from G.

(iv) G is non-abelian: $\mathrm{Inn}(\mathcal{A})$ is non-trivial and non-commutative because $\mathcal{A}$ is non-commutative (T_{alg}) with at least one block of size $n_k \geq 2$.

Proof. Step 1: Internal vs. kinematic automorphisms. By Lemma 2.9 applied to the admissibility net $\{\mathcal{A}(\Gamma)\}_{\Gamma\in\mathcal{I}}$ (with independence from L_{loc}), the index-action homomorphism π places every automorphism in the exact sequence $1 \to \mathrm{Aut}_{\mathrm{int}} \to \mathrm{Aut}(\mathrm{net}) \to K \to 1$. The internal automorphisms are canonically the kernel — no choice of complement or splitting is involved. Diffeomorphisms act non-trivially on $\mathcal{I}$ and hence project non-trivially to K; the gauge group is the kernel. This gives (i).

Step 2: Wedderburn–Artin and Skolem–Noether. After field selection (T2c: $\mathbb{F} = \mathbb{C}$), the admissibility algebra is a finite-dimensional $*$-algebra over $\mathbb{C}$. Wedderburn–Artin (T2a) gives $\mathcal{A} \cong \bigoplus_{k=1}^{K} M_{n_k}(\mathbb{C})$. By Skolem–Noether, every automorphism of $M_{n_k}(\mathbb{C})$ is inner: $\alpha(a) = UaU^*$ for some unitary U. The inner automorphism group of each block is $\mathrm{Inn}(M_{n_k}) \cong \mathrm{PU}(n_k) = \mathrm{U}(n_k)/\mathrm{U}(1)$. For the full semisimple algebra: $\mathrm{Inn}(\mathcal{A}) = \prod_{k=1}^{K} \mathrm{PU}(n_k)$. This gives (ii).

Step 2b: Block permutations. For $K \geq 2$, the full automorphism group of a finite-dimensional semisimple algebra is:

$$\mathrm{Aut}\Big(\bigoplus_{k=1}^{K} M_{n_k}(\mathbb{C})\Big) \;=\; \Big(\prod_{k=1}^{K} \mathrm{PU}(n_k)\Big) \;\rtimes\; S_{\mathrm{iso}},$$

where $S_{\mathrm{iso}} \subseteq S_K$ is the group of permutations among isomorphic blocks (blocks with the same n_k). The inner automorphisms are $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$; the block permutations S_{iso} are outer.

The physical status of S_{iso}:

(a) Each Wedderburn block corresponds to a distinct admissibility sector under T_{sep}. Sectors are distinguished by their anchor sets M_{d_k}, which carry distinct admissibility costs $\varepsilon(d_k)$.

(b) A block permutation $\sigma \in S_{\mathrm{iso}}$ exchanges sectors. This is an automorphism of the *abstract* algebra but not of the *cost-equipped* algebra $(\mathcal{A}, \varepsilon)$: it maps $\varepsilon(d_k) \mapsto \varepsilon(d_{\sigma(k)})$, which differs from $\varepsilon(d_k)$ unless the permuted sectors are cost-degenerate.

(c) Cost-degenerate sectors (identical admissibility costs at every state), if any occurred, would make S_{iso} a nontrivial discrete relabeling of repeated summands. For the derived algebra $M_3 \oplus M_2 \oplus \mathbb{C}$ no two summands are isomorphic, so S_{iso} is trivial and no such relabeling exists. In either case S_{iso} is discrete, carries no connection one-form, and does not contribute to the gauge group G. Generations are not carried here: they are multiplicities within one representation type of the matter category (Definition 2.14; the copy clause of Definition 2.20), and the generation count is the imported capacity–beta derivation (§8 of Technical Supplement I (*The Classification Core*)), not a count of algebra blocks.

(d) Non-degenerate blocks: $\varepsilon(d_k) \neq \varepsilon(d_j)$ for $k \neq j$. The permutation σ is not an automorphism of $(\mathcal{A}, \varepsilon)$, so S_{iso} is trivial on the cost-equipped algebra.

Conclusion. The continuous gauge group is $G = \mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$, regardless of whether S_{iso} is trivial (non-degenerate costs) or nontrivial (degenerate costs). The reason is that S_{iso} is *discrete and outer*: it permutes Wedderburn blocks but does not generate any continuous family of automorphisms. The continuous gauge group sees only the connected component of $\mathrm{Aut}(\mathcal{A})$, which is $\mathrm{Inn}(\mathcal{A})$.

What survives of S_{iso}. S_{iso} is discrete and outer, generates no continuous family of automorphisms, and contributes no gauge bosons, in any regime. For the derived algebra it is trivial outright: $M_3 \oplus M_2 \oplus \mathbb{C}$ has no repeated summands, so there are no cost-degenerate blocks to permute. Flavor structure lives elsewhere. The observed generations are copies within one representation type of the matter category — multiplicity spaces, not algebra summands (Definition 2.20's copy clause) — and their splitting by Yukawa couplings is Paper 4B's territory.

Step 3: Identification. The physical gauge group acts on the internal degrees of freedom at a single interface, preserving the cost-equipped algebraic structure. By Steps 1–2, this is precisely $\mathrm{Inn}(\mathcal{A})$. Kinematic automorphisms are outer (they change the interface label) and are not part of the gauge group. Block permutations among isomorphic summands are outer and discrete, and carry no generation content (the copy clause of Definition 2.20; §8 of Technical Supplement I (*The Classification Core*)). This gives (ii) and (iii).

Step 4: Non-abelian. $\mathcal{A}$ is non-commutative (T_{alg}). A commutative algebra has $\mathrm{Inn}(\mathcal{A}) = \{1\}$ (every inner automorphism is trivial). Since $\mathrm{Inn}(\mathcal{A})$ is non-trivial, G is non-trivial. For $M_n(\mathbb{C})$ with $n \geq 2$, $\mathrm{PU}(n)$ is non-abelian (e.g., $\mathrm{PU}(2) \cong \mathrm{SO}(3)$). This gives (iv). □

Theorem 2.12 (Continuous APF gauge redundancies are inner; equality under full invariance)**.** *Let $\mathcal{A}(\Gamma)$ be a finite admissibility algebra equipped with its interface support data, cost kernel B, anchor assignment M, and admissible continuation profile E. Write $\mathrm{Gau}_0(\Gamma)$ for the identity component of the group of physical gauge redundancies preserving these data (the identity component of $\mathrm{Aut}_0(\mathcal{A}(\Gamma), B, M, E)$). Then, unconditionally,*

$$\mathrm{Gau}_0(\Gamma) \;\subseteq\; \mathrm{Inn}(\mathcal{A}(\Gamma)) \;=\; \prod_k \mathrm{PU}(n_k),$$

so every continuous physical gauge direction is generated by an inner derivation of the finite admissibility algebra. Equality, $\mathrm{Gau}_0(\Gamma) = \mathrm{Inn}(\mathcal{A}(\Gamma))$*, holds under an explicit* full-invariance hypothesis*: the anchor set, cost kernel, and continuation data are* Inn*-invariant as a family — conjugation by any unitary of $\mathcal{A}(\Gamma)$ maps the preserved data onto itself. The hypothesis is substantive, not bookkeeping: a fixed anchor projector onto a proper sub-summand subspace M_d is preserved only by its stabilizer, which reduces the group to a proper subgroup of that block's* $\mathrm{PU}(n_k)$*. Outer block permutations, including accidental permutations among isomorphic blocks, are kinematic relabelings and cannot contribute gauge bosons.*

Proof. Write the Wedderburn decomposition as $\mathcal{A} = \bigoplus_k M_{n_k}(\mathbb{C})$. The automorphism group of the abstract algebra is

$$\mathrm{Aut}(\mathcal{A}) \cong \Big(\prod_k \mathrm{PU}(n_k)\Big) \rtimes S_{\mathrm{iso}},$$

where S_{iso} permutes isomorphic simple summands. The first factor is the identity component and has Lie algebra $\bigoplus_k \mathfrak{su}(n_k)$. The second factor is finite and discrete, so it has no infinitesimal generators.

The inclusion (unconditional). A continuous physical gauge redundancy is in particular a $*$-automorphism of $\mathcal{A}(\Gamma)$ in the identity component of $\mathrm{Aut}(\mathcal{A})$; the discrete factor S_{iso} is separated from the identity, so the identity component is $\prod_k \mathrm{PU}(n_k) = \mathrm{Inn}(\mathcal{A}(\Gamma))$ (Skolem–Noether per block). Hence $\mathrm{Gau}_0(\Gamma) \subseteq \mathrm{Inn}(\mathcal{A}(\Gamma))$, with no hypothesis beyond the finite complex semisimple carrier.

The reverse inclusion (conditional). More is needed than summand preservation. Inner automorphisms rotate states within a fixed simple summand and preserve the *summand's* support projector and cost profile; but the preserved data list includes the anchor assignment, and anchors are sub-summand objects. For a generic Ad_U acting inside a block, a proper anchor subspace M_d is not mapped to itself — it is preserved only by the stabilizer of M_d. The full-invariance hypothesis is exactly what licenses the reverse inclusion: if the anchor set, cost kernel, and continuation data are Inn-invariant as a family, then every inner automorphism preserves the data, so $\mathrm{Inn}(\mathcal{A}(\Gamma)) \subseteq \mathrm{Gau}_0(\Gamma)$ and equality holds.

Block permutations. A nontrivial element of S_{iso} exchanges summand labels. If the exchanged summands are not identical as cost-equipped local codespaces, support or cost preservation fails. If they are identical, the permutation remains a discrete symmetry of repeated copies; it is separated from the identity component and has no connection one-form or curvature. It therefore behaves as a kinematic relabeling rather than a continuous local gauge redundancy.

What downstream consumes. The gauge-identification chain consumes the unconditional inclusion together with the exact sequence (Lemma 2.9, Proposition 2.19): the abelian-source and closure arguments need every continuous gauge direction to be inner (the inclusion) and the central structure of $\mathrm{U}(\mathcal{A})$ (the exact sequence). No downstream result consumes the unconditional equality. Where the identification register $G = \mathrm{Inn}(\mathcal{A})$ is used (Theorem 2.11), the gauge-preserved data are read at block resolution — summand granularity — which is the full-invariance hypothesis in the form the physical criterion intends; the hypothesis boundary is stated here so that the reader can see exactly where it enters. If an additional continuous symmetry is postulated that acts outside $\mathcal{A}$, it is not derived by the APF gauge identification theorem and must pay for a new admissibility record. □

Remark 2.13 (Essential hypotheses for $G_{\mathrm{int}} = \mathrm{Inn}(\mathcal{A})$)**.** The conclusion uses five finite-algebra hypotheses: finite resolution, a faithful positive state/GNS representation, unitary implementation of zero-marginal-cost relabelings, complex-sector selection, and — for the equality direction only — the full-invariance hypothesis (anchor set, cost kernel, and continuation data Inn-invariant as a family; without it only the inclusion $\mathrm{Gau}_0 \subseteq \mathrm{Inn}(\mathcal{A})$ is available). Relaxing them changes the theorem. Extra commutative summands contribute central phases but no nonabelian inner automorphisms; real or quaternionic blocks move the problem to the field-selection gate rather than the gauge-identification theorem; nonfaithful states quotient out invisible summands and must be replaced by the faithful quotient before Skolem–Noether applies. Thus the theorem is robust under small perturbations preserving the faithful finite complex $*$-algebra, but not under changing the algebraic category.

Definition 2.14 (The matter category)**.** A *matter multiplet* is a finite-dimensional irreducible unitary representation of the lifted gauge group $\widetilde{G} = \prod_k \mathrm{SU}(n_k) \times \mathrm{U}(1)$ — equivalently, a choice of one irreducible representation per factor together with a U(1) charge, including tensor products across factors (bifundamentals such as $(\mathbf{3}, \mathbf{2})_{1/6}$). The matter category is an *independent structural input* at this point of the derivation: it is the category of representations of the group that the algebra determines, not a category read off from the algebra's own modules.

Remark 2.15 (Why left $\mathcal{A}$-modules cannot carry the matter)**.** Earlier versions took a matter multiplet to be an irreducible left $\mathcal{A}$-module. That category is too small: for $\mathcal{A} = \bigoplus_k M_{n_k}(\mathbb{C})$ the central idempotents act on any irreducible left module as scalar idempotents summing to 1, so exactly one acts as the identity (Schur), and every irreducible left module factors through a *single* block — the irreducibles are $\mathbb{C}^{n_1}, \ldots, \mathbb{C}^{n_K}$, never a simultaneous bifundamental. A quark doublet $(\mathbf{3}, \mathbf{2})$ therefore cannot arise as an irreducible left module of $M_3 \oplus M_2 \oplus \mathbb{C}$. Carrying bifundamentals requires commuting representations of the several factors on one space — equivalently, representations of the product group (the category of Definition 2.14), a bimodule structure, or an equivalent device. The fermion scan of Part II of Technical Supplement I (*The Classification Core*) searches Definition 2.14's category; the algebra fixes the *group*, and the group's representation theory — not the algebra's module theory — carries the matter.

Proposition 2.16 (Gauge automorphisms act on fields by the lifted matter representation)**.** *Let $(\mathcal{A}, \omega)$ be the finite admissibility algebra with faithful GNS representation $\rho_\omega : \mathcal{A} \to \mathcal{B}(\mathcal{H}_\omega)$, and let Ψ denote a matter multiplet carrying an irreducible unitary representation of the lifted gauge group $\widetilde{G} = \prod_k \mathrm{SU}(n_k) \times \mathrm{U}(1)$ (Proposition 2.25; the matter category is specified in Definition 2.14 below — it is* not *the category of irreducible left $\mathcal{A}$-modules). A zero-marginal-cost internal relabeling* *preserving all tested expectations is represented by a unitary normalizer $U \in U(\mathcal{A})$. The matter action itself is* defined on $\widetilde{G}$*: for* $\tilde{g} \in \widetilde{G}$,

$$a \mapsto \pi(\tilde{g})\, a\, \pi(\tilde{g})^*, \qquad \psi \mapsto \rho_\Psi(\tilde{g})\, \psi,$$

where $\pi : \widetilde{G} \to \mathrm{Inn}(\mathcal{A})$ *is the composite of the blockwise covering maps* $\mathrm{SU}(n_k) \twoheadrightarrow \mathrm{PU}(n_k)$ *(the abelian coordinate acts trivially on observables), and* ρ_Ψ *— the symbol is only ever applied to elements of* $\widetilde{G}$ *— is* Ψ*'s representation from the matter category. Three clauses fix what the algebra side does and does not determine.*

- *(i)* Observable side. *A zero-marginal-cost relabeling acts on observables by conjugation* $a \mapsto UaU^*$*; after quotienting the central kernel this is an element of* $\mathrm{Inn}(\mathcal{A})$ *(Proposition 2.19).*
- *(ii)* Determination up to the central torus. *The algebra side determines the corresponding matter transformation only up to the central data: conjugation data fix an element of* $\mathrm{Inn}(\mathcal{A})$ *and leave the action of* $Z(\widetilde{G})$ *— in particular of the relative-phase torus* T_{rel} *— undetermined. No global homomorphic splitting* $U(\mathcal{A}) \to \widetilde{G}$ *is claimed: the blockwise phase–*SU *factorizations are multivalued (each determined only up to* $\mathbb{Z}_{n_k}$*), and no continuous global section exists.*
- *(iii)* The charge representation closes the gap. *Which one-parameter subgroup of* T_{rel} *acts on* Ψ*, and with what charge, is supplied by the charge representation* ρ_Ψ *itself — a named input of the matter category, per the entry-typed inventory I-typing (Definition 2.14; Definition 2.20).*

Thus the algebraic identification $G_{\mathrm{alg}} = \mathrm{Inn}(\mathcal{A})$ *and the physical gauge action on matter fields are two presentations of the same internal symmetry — adjoint/projective on observables, linear on fields — with the central (abelian) data carried by the matter category's charge representation, not derived from the algebra side.*

Proof. (i): Finite dimensionality and faithfulness imply that every tested expectation has the form $\omega(b^*ab)$ for $a, b \in \mathcal{A}$. A relabeling that preserves all such expectations and does not move the interface label is, by the finite-dimensional Kadison/Wigner form, implemented by a unitary or antiunitary normalizer of the represented algebra; the oriented complex carrier $B1'$ excludes the antiunitary branch for continuous gauge transformations. Conjugation by U is the map of Proposition 2.19; scalars in $Z(U(\mathcal{A}))$ act trivially by conjugation, so the induced automorphism group is $\mathrm{Inn}(\mathcal{A})$.

(ii): Exactness of $1 \to Z(U(\mathcal{A})) \to U(\mathcal{A}) \to \mathrm{Inn}(\mathcal{A}) \to 1$ says precisely that conjugation data determine U only up to $Z(U(\mathcal{A}))$; passing to the lifted group, the corresponding matter transformation is determined only up to the central torus's action. A global continuous homomorphic section of the blockwise phase–SU factorization would invert the n_k-fold covers $\mathrm{U}(1) \times \mathrm{SU}(n_k) \to \mathrm{U}(n_k)$, which admit no continuous global sections; the up-to-central-data statement is therefore the strongest algebra-side conclusion available, and (ii) is sharp.

(iii): The matter category is an independent structural input (Definition 2.14); ρ_Ψ assigns each multiplet its $\widetilde{G}$-representation, including its abelian charge. The action displayed in the statement is therefore well defined on $\widetilde{G}$ by construction, and it descends to the faithful quotient identified in Proposition 2.26 because the derived template's charges satisfy the $\mathbb{Z}_6$ condition (Corollary 2.27). This is the R-EC-inv + I-typing conditionality in its lift form: the physicality of the abelian direction is carried by the charged matter that sees it, not by the algebra's conjugation action, and the surviving relative phase is fixed downstream by Proposition 7.1 of Technical Supplement I (*The Classification Core*). □

Remark 2.17 (Methodological parallel: Paper 9 v1.2 gauge-reduction)**.** The gauge-identification theorem follows a reusable APF audit template: start with a larger automorphism/response group, quotient continuation- profile-preserving relabelings, and name the residual obstruction that must be closed. Paper 9 applies the same template in a weak-field comparison sector, reducing a local $GL^+(2,\mathbb{R})$ clock-ruler response by gauge equivalence and isolating the trace obstruction. The analogy is useful for wayfinding only. Paper 2's gauge derivation depends on the finite Wedderburn/Skolem–Noether chain above, not on the Paper 9 gravitational normal-form program.

Remark 2.18 (Emergence of abelian U(1) factors)**.** The inner automorphism group $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$ contains no abelian factors: $\mathrm{PU}(n)$ is semisimple for $n \geq 2$, and trivial for $n = 1$. This raises the question: where does the $\mathrm{U}(1)_Y$ hypercharge factor of the Standard Model originate?

The answer involves the relative-phase torus of the multi-block unitary group, together with a no-redundancy theorem.

The algebraic relationship between the groups acting on the algebra (by conjugation) and on the Hilbert space (linearly) is captured by an exact sequence.

Proposition 2.19 (Exact sequence for gauge representations)**.** *Let* $\mathcal{A} = \bigoplus_{k=1}^{K} M_{n_k}(\mathbb{C})$ *with* $\mathrm{U}(\mathcal{A}) = \prod_k \mathrm{U}(n_k)$*. Define the conjugation map* $\pi : \mathrm{U}(\mathcal{A}) \to \mathrm{Aut}(\mathcal{A})$ *by* $\pi(U)(a) = UaU^*$*. Then:*

(i) The image of π *is* $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$*.*

(ii) The kernel of π *is the center* $Z(\mathrm{U}(\mathcal{A})) = \prod_k \mathrm{U}(1)_k$*, consisting of block-diagonal phase matrices* $(e^{i\theta_1}\mathbf{1}_{n_1}, \ldots, e^{i\theta_K}\mathbf{1}_{n_K})$*.*

(iii) The sequence

$$1 \longrightarrow Z(\mathrm{U}(\mathcal{A})) \longrightarrow \mathrm{U}(\mathcal{A}) \xrightarrow{\pi} \mathrm{Inn}(\mathcal{A}) \longrightarrow 1$$

is exact.

(iv) The overall diagonal phase $\mathrm{U}(1)_{\mathrm{diag}} = \{(e^{i\theta}\mathbf{1}_{n_1}, \ldots, e^{i\theta}\mathbf{1}_{n_K}) : \theta \in \mathbb{R}\}$ *acts trivially on both* $\mathcal{A}$ *(by conjugation) and on* $\mathcal{H}_\omega$ *(as a global phase). The quotient* $Z/\mathrm{U}(1)_{\mathrm{diag}}$ *is a torus of rank* $K-1$*, the relative-phase torus* T_{rel}*.*

(v) The algebra's GNS carrier $\mathcal{H}_\omega$ *transforms under* $\mathrm{U}(\mathcal{A})$ *linearly (each block entering with its left-regular multiplicity:* n_k *copies of the fundamental* $\mathbb{C}^{n_k}$*), not under* $\mathrm{Inn}(\mathcal{A})$ *(which acts on* $\mathcal{A}$ *by conjugation, i.e., the adjoint representation). Matter fields do* not *live in* $\mathcal{H}_\omega$*: they carry the independently specified matter category (Definition 2.14); the GNS space is the algebra-side bookkeeping object in which the fundamental of each block is exhibited as a subrepresentation. The two actions agree on the Lie algebra (*$\mathfrak{u}(\mathcal{A}) \to \mathfrak{inn}(\mathcal{A})$ *is surjective with kernel* $\mathfrak{z}$*), but differ globally:* $\mathrm{U}(\mathcal{A})$ *admits representations that* $\mathrm{Inn}(\mathcal{A})$ *does not.*

Proof. (i) By Skolem–Noether, every automorphism of $M_n(\mathbb{C})$ has the form $a \mapsto UaU^*$ for some $U \in \mathrm{U}(n)$. For the semisimple algebra, each block is mapped to itself by an inner automorphism (outer block permutations are handled separately in Step 2b of Theorem 2.11), so $\mathrm{Im}(\pi) = \prod_k \mathrm{Inn}(M_{n_k}) = \prod_k \mathrm{PU}(n_k)$. (ii) $\pi(U) = \mathrm{id}$ iff $UaU^* = a$ for all $a \in \mathcal{A}$, iff U is central. For $M_n(\mathbb{C})$, the center is $\mathrm{U}(1) \cdot \mathbf{1}_n$; for the direct sum, $Z = \prod_k \mathrm{U}(1)_k$. (iii) Surjectivity of π onto $\mathrm{Inn}(\mathcal{A})$ is (i); the kernel statement is (ii); the sequence is exact at $\mathrm{U}(\mathcal{A})$ by construction. (iv) The diagonal $\mathrm{U}(1)_{\mathrm{diag}}$ is a subgroup of Z with K-dimensional Z quotient of dimension $K-1$. Triviality on $\mathcal{H}_\omega$ is because states are defined up to global phase: $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same physical state. (v) The GNS representation maps each $M_{n_k}(\mathbb{C})$ block to operators on $\mathbb{C}^{n_k} \subset \mathcal{H}_\omega$; the natural group action on $\mathbb{C}^{n_k}$ is $v \mapsto Uv$ (fundamental), not $a \mapsto UaU^*$ (adjoint). The fundamental is a linear representation of $\mathrm{U}(n_k)$ but only a projective representation of $\mathrm{PU}(n_k)$. □

Definition 2.20 (Admissible gauge structure)**.** Let G be a compact Lie group acting faithfully on a finite-dimensional Hilbert space $\mathcal{H}$, and let $\mathcal{M} = \{r_1, \ldots, r_N\}$ be the set of physically distinct irreducible multiplet types occurring in $\mathcal{H}$. *Type* means an isomorphism class of $\widetilde{G}$-representations, not an individual copy: replicated copies of one representation (the observed generations are the physical example) are one type with multiplicity, and EC does *not* demand that gauge labels separate copies within a type. That generation identity is carried outside the gauge labels (by mass and mixing structure) is a scoping *premise* of this definition, stated here explicitly rather than derived. The copy clause is scoped to the *full realized* $\widetilde{G}$: types are isomorphism classes of $\widetilde{G}$-representations, and it is this inventory — with u^c and d^c antecedently distinct entry-types (the entry-typed inventory I-typing, a named input consumed at Theorem 7.7 of Technical Supplement I (*The Classification Core*) H3; see Lemma 2.24) — that CM's restriction tests act on. Applied instead to a proper subgroup $H \subsetneq \widetilde{G}$, the clause would recount u^c/d^c as one type at multiplicity two and void the restriction test; the scoping forbids that reading.

Enforcement completeness (EC). G is *admissibility-complete* if the map $r_i \mapsto [r_i]_G$ from multiplet types to G-representation labels is injective: distinct physical types carry distinct gauge quantum numbers.

Capacity minimality (CM). G is *capacity-minimal* if no generator channel of its enforcing configuration can be removed while preserving admissibility completeness. Removal is an operation on channel configurations, not on groups; the group-level comparison space is the closed proper subgroups of G (with their restricted faithful representations). Formally: for every closed connected subgroup $H \subsetneq G$ enforced by a configuration with at least one generator channel removed, the label map $r_i \mapsto [r_i]_H$ is *not* injective.

Admissibility. G is *admissible* if it is both admissibility-complete and capacity-minimal.

Before deriving EC and CM, we establish the bridge from Lie-algebra generators to admissibility channels, which CM requires. The bridge is not as simple as "one generator = one channel": the relationship between Lie-algebra generators and admissibility channels involves incompatibility (non-commuting generators cannot be simultaneously monitored), coarse-graining (degenerate eigenvalues reduce the information content of a single generator), and the distinction between Cartan (commuting) and root (non-commuting) generators. We treat these systematically.

Definition 2.21 (Enforcement monitoring of a generator)**.** Let $g_a \in \mathfrak{g}$ be a Hermitian generator of a compact Lie algebra acting on a finite-dimensional Hilbert space $\mathcal{H}$ of dimension n. The *admissibility monitoring* of g_a is the projection-valued measure (PVM) $\{P_{a,\lambda}\}_\lambda$ on the spectrum $\sigma(g_a) \subseteq \mathbb{R}$, where $P_{a,\lambda}$ projects onto the λ-eigenspace of g_a. The *distinction count* of g_a is $d(g_a) := |\sigma(g_a)| - 1$ — one less than the number of distinct eigenvalues. This is the number of binary questions needed to determine which eigenspace a state occupies (using a binary search on the ordered eigenvalue list).

Lemma 2.22 (Generator cost bound by resolution rank)**.** *Let G be a compact Lie group of the class fixed by the gauge identification —* $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$ *and its closed subgroups (closed subgroups of a Lie group are Lie, by Cartan's closed-subgroup theorem) — acting faithfully and non-trivially on the admissibility structure at Γ, with* $\dim(G) = p$*. Then the admissibility cost of* enforcing *G-equivariance is at least $p \cdot \varepsilon^*$:*

$$C(G) \;\geq\; \dim(G) \cdot \varepsilon^*.$$

This is the paper-side statement of the banked gauge clause $n(G) = \dim(G)$, $C(G) = \dim(G) \cdot \varepsilon$ [5]*; the two independent proof routes below are the bank's.*

[5] Code reference: `check_L_cost_gauge` in `core.py`.

Proof. Route A (resolution rank; primary). Enforcing G-equivariance requires the admissibility structure to distinguish automorphisms that act differently on observables: if $\alpha_{g_1}(A) \neq \alpha_{g_2}(A)$ for some observable A and the structure cannot separate g_1 from g_2, then only a quotient of G is enforced, not G (orbit separation). G is a p-dimensional manifold: every group element in a neighborhood of the identity is specified by p real coordinates, and by invariance of domain (Brouwer, local form: a continuous injection from an open $U \subseteq \mathbb{R}^p$ into $\mathbb{R}^k$ forces $k \geq p$), no continuous injective resolution of that neighborhood uses fewer than p independent distinctions. This step consumes a named premise, *operational encoding*: the enforced group action is registered by a continuous *injective* encoding of an identity neighborhood into the admissibility structure's distinction coordinates — invariance of domain applies to that encoding, not to the abstract group manifold. The premise is motivated by orbit separation (a relabeling the structure cannot separate from a neighbor is not enforced as part of G), and it is a premise of this lemma, not a theorem of the framework. Resolving the enforced group therefore requires at least p independent admissibility distinctions, each an irreducible channel costing at least ε^* (L_{ε^*}). The count is a property of the group manifold — it is invariant under change of Lie-algebra basis, and it does not assert that any particular generator corresponds to a separate physical measurement.

Route B (bracket non-closure; confirmatory). The bracket $[T_a, T_b]$ is realized by composition (four exponentials), and composition is non-free (L_{nc}: interaction cost ≥ 0, generically positive). Each bracket-generated direction therefore costs at least ε^* (L_{ε^*}); after closure all $\dim(G)$ directions are populated, each costing at least ε^*. Total $\geq \dim(G) \cdot \varepsilon^*$.

Both routes give a resolution count of at least $\dim(G)$, hence the lower bound of the statement: $C(G) \geq \dim(G) \cdot \varepsilon^*$ — in L_{Cauchy}'s unit normalization, $C(G) \geq \dim(G)$ in ε^* units. Equality holds under the disjoint-anchor realizability premise; the composed operational statement below is where the equality lives. □

Remark 2.23 (The withdrawn measurement-channel argument)**.** Earlier versions argued this bound by assigning one *operational measurement channel* per Lie-algebra basis generator, claiming that commuting linearly independent generators require separate spectral interfaces because no basis element can be a function of another. That argument is withdrawn: it is false. For $g_1 = \mathrm{diag}(1, 0, -1)$ and $g_2 = \mathrm{diag}(0, 1, -1)$, the operators commute and are linearly independent, yet g_2 is a polynomial in the nondegenerate g_1, and a single rank-one projective measurement in the common eigenbasis determines both eigenvalues — linear independence in the Lie algebra does not imply separate measurement channels, and "one generator, one channel" is not invariant under change of generator basis. The corrected route above counts *resolution rank of the enforced group manifold*, not measurements on states: an informationally complete measurement may characterize many observables, but it cannot reduce the number of independent distinctions needed to resolve a p-dimensional group of enforced relabelings below p. The bank's statement (`check_L_cost_gauge`) has carried the resolution-rank route since v24.3.404; the withdrawn presentation was this document's, not the bank's.

Scope and caveats. The lemma establishes a *lower bound* on the admissibility cost, not an exact cost: co-located non-abelian channels may incur superadditive surplus (L_Δ), making the actual cost strictly greater. The lower bound suffices for CM: a redundant generator wastes *at least* ε^* capacity, which A2 then eliminates (Lemma 2.24). The scope rider travels with the bank statement: the argument holds for $\mathrm{Inn}(\mathcal{A})$-class groups and their closed subgroups, which is exactly the class the classification of Part II of Technical Supplement I (*The Classification Core*) ranges over.

The composed operational statement $C(G) = \dim(G)$. The equality the cost ranking consumes is a composition of two named operational premises, stated once here. The *lower* bound is the lemma above, a theorem on the operational-encoding premise (invariance of domain applied to a continuous injective encoding of an identity neighborhood). The *upper* bound rests on the disjoint-anchor realizability premise: the $\dim(G)$ resolution distinctions can be anchored on disjoint supports, so C1-additivity applies and no superadditive surplus accrues across generator channels. Under the pair, $C(G) = \dim(G) \cdot \varepsilon^*$ exactly. The pair is the operational cost premise of the gauge-cost program: a cost functor deriving both premises from the admissibility semantics is the named open lane, and until it closes, every consumer of the equality — the Lie-algebra cost ranking and the classification theorem's cost convention — carries the pair by name. The paper-side realization of the composed statement, with the same two premises named, is Proposition 4.1 of Technical Supplement I (*The Classification Core*)[6].

Lemma 2.24 (EC and CM from the admissibility semantics)**.** *EC and CM are derived principles of the admissibility framework. They are not consequences of A1 (finite capacity) alone; their dependencies on the two axioms (A1, A2) and the regularity input MD are stated cleanly under the dual-axiom framing of Paper 1, §2:*

(i) ***EC*** *requires: A1 (finite capacity) +* T_{sep} *(sector decomposition) +* L_{loc} *(locality) + FD3 (anchor-set locality) + the gauge identification (Theorem 2.11). EC is the statement that the internal gauge labels at a single interface must resolve all physically distinct internal types. It is a* completeness principle for the interface-local admissibility semantics*. Its structural-separation ingredient is A1-derived, but EC itself rests on two named, non-derived elements: the* kinematic type-inventory reading *R-EC-inv (at a single interface, kinematic type-distinctness is exhausted by the interface's counted labels; label-identical species are copies, not types — A1-motivated via the admissibility referent, not A1-derived) and, where EC is consumed against a fixed template, the* entry-typed inventory *I-typing (the template's entries are antecedently distinct types:* $u^c \neq d^c$ *prior to any abelian label). A2 plays no role in EC.*

(ii) ***CM*** *requires: A2 (minimum cost) applied to the admissible set defined by A1 + MD (via* L_{ε^*}*) + the generator–channel bridge (Lemma 2.22). A configuration carrying a redundant generator is* not *inadmissible under A1 — it still satisfies* $\Sigma\varepsilon \leq C$*. But A1 + MD (via* L_{ε^*}*) guarantees that removing the redundant generator yields a strictly cheaper configuration in the same admissible set, by* $\varepsilon^* > 0$*. A2 then selects the cheaper one as the realised gauge structure. CM is therefore the statement that the realised admissible structure carries no redundant generators — a direct consequence of A2.*

Proof. EC from the structural separation. The argument for EC requires a premise that is not A1 alone but is *derived from* A1: the internal–external structural separation. By T_{sep} (Paper 1 Supplement), the admissibility structure at each interface Γ decomposes into sectors. By Definition 2.5 and Lemma 2.7 (both derived from FD3 and L_{loc}, which are themselves derived from A1): at a single interface, the admissibility algebra $\mathcal{A}(\Gamma)$ acts on the *internal* sector decomposition of S_Γ, while inter-interface (spacetime) distinctions — spatial separation, momentum, mass — are properties of the *external* structure (the inter-interface map L_{loc}). At a single interface, the only admissibility operations available are those in $\mathcal{A}(\Gamma)$, whose automorphism group is the gauge group G (Theorem 2.11).

Now suppose two physically distinct multiplet types $r_i \neq r_j$ carry identical gauge quantum numbers: $[r_i]_G = [r_j]_G$. Since $G = \mathrm{Aut}(\mathcal{A}(\Gamma))$ and the G-representation labels exhaust the distinctions available within $\mathcal{A}(\Gamma)$, no admissibility operation at Γ distinguishes r_i from r_j. (The exhaustion

[6] Code reference: `check_L_cost_gauge` in `core.py`.

step just used is, within the algebra-module vocabulary, a triviality — isomorphic modules are internally indistinguishable; the *physical* claim, that the algebra's counted module structure exhausts kinematic type-distinctness at the interface, is the reading R-EC-inv, consumed exactly here.) But A1 requires that all physically distinct states be admissible (this is the operational content of "admissibility capacity": the budget $C(\Gamma)$ is the capacity to enforce distinctions at Γ). If two types are indistinguishable at every interface at which they co-occur, they are admissibility-equivalent and therefore physically identical — contradicting $r_i \neq r_j$.

The objection "perhaps they are distinguished by mass or momentum rather than gauge quantum numbers" is addressed by the structural separation: mass and momentum are spacetime (external) properties resolved by the inter-interface structure, not by the admissibility algebra at a single interface. EC is the statement that the *internal* sector must resolve all internal types; it does not claim that gauge quantum numbers are the only physically meaningful labels, only that they must be complete within the internal sector.

CM from A2 over the admissible set. CM requires three ingredients in addition to A2:

(a) L_{ε^*}: each irreducible distinction costs at least $\varepsilon^* > 0$. This is derived from A1 + MD (admissibility isotropy) in the Paper 1 Supplement. It is *not* derivable from A1 alone: without the isotropy assumption MD, one cannot exclude the possibility that some channels have arbitrarily small cost.

(b) Lemma 2.22: resolving the enforced group costs at least one channel per group-manifold dimension ($n(G) = \dim(G)$, the resolution-rank route; the banked gauge clause `check_L_cost_gauge`). This is derived from faithfulness and non-triviality of the G-action (which follow from Theorem 2.11, itself derived from A1) and invariance of domain on the group manifold — not from any one-generator-one-measurement claim (Remark 2.23).

(c) A1 + MD jointly define the admissible set $\mathcal{A}(\rho, R)$ of *channel configurations* satisfying $\Sigma\varepsilon \leq C$ with $\varepsilon(d) \geq \mu^* n(d)$. A channel configuration is a finite set $\mathcal{C}$ of costed generator-channels; the group it enforces is the closed subgroup $G = \langle \mathcal{C} \rangle$ generated by the configuration's channels, with its faithful representation on the interface carrier.

Given (a)–(c): CM is stated at the channel-configuration level (the semantics of the admissible set above). Consider any admissible configuration $\mathcal{C}$ containing a channel X whose removal preserves injectivity of the label map $r_i \mapsto [r_i]_{\langle \mathcal{C} \setminus \{X\} \rangle}$ — a channel that does not refine any multiplet distinction. The configuration $\mathcal{C}' := \mathcal{C} \setminus \{X\}$ is a well-defined set-theoretic removal of one costed channel. Removal is defined on configurations, never on groups: where a group-level comparison is needed, the comparison space is closed subgroups, quotients, and central-factor removals with specified faithful representations, and the group enforced by $\mathcal{C}'$ is $\langle \mathcal{C} \setminus \{X\} \rangle$, a closed subgroup of $\langle \mathcal{C} \rangle$ (Cartan). $\mathcal{C}'$ is admissible (it satisfies the same capacity bound and isotropy condition; removing a channel only relaxes them). By (a) and (b), $\Sigma_{d \in \mathcal{C}} \varepsilon(d) - \Sigma_{d \in \mathcal{C}'} \varepsilon(d) = \varepsilon(X) \geq \varepsilon^* > 0$, so $\mathcal{C}'$ is strictly cheaper than $\mathcal{C}$ inside the same admissible set. By A2 (minimum cost), the realised configuration $\mathcal{C}_{\text{realized}}(\rho, R) = \arg\min_{\mathcal{C} \in \mathcal{A}(\rho, R)} \Sigma\varepsilon(d)$ cannot contain X. Iterating over all redundant channels: the realised admissible structure contains no redundant generators. □

Status of the premises: derived, not additional. A careful reading of Lemma 2.24 might suggest that EC and CM import "extra" assumptions beyond A1. This would be a genuine vulnerability if the required premises were independently postulated. In fact, every premise in the dependency chain is derived:

Premise	**Content**	**Derived from**	**Reference**
T_{sep}	Sector decomposition of S_Γ	A1 + T_{form}	P1S §2
L_{loc}	Independent interfaces	A1 + FD1–FD2	P1S §4
FD3	Anchor-set locality	Foundational definition	P1S §1
Gauge ident.	$G = \mathrm{Inn}(\mathcal{A})$	T_{alg} + T2a + Skolem–Noether	This supplement
L_{ε^*}	Cost floor $\varepsilon^* > 0$	A1 + MD	P1S §3
Gen.-channel bridge	$\dim(\mathfrak{g})$ channels	Faithfulness + T_{form}	This supplement
A2	Min-cost selection over $\mathcal{A}(\rho, R)$	Second axiom (P1, §2)	P1, §2; P1S §2

The only external inputs beyond the two named axioms (A1, A2) are the regularity condition MD (admissibility isotropy), which enters through L_{ε^*}, and FD3 (anchor-set locality), a foundational definition rather than an empirical assumption. The remaining premises are mathematical consequences of A1 (or A1 + MD) and the upstream chain. The dependency for CM in the table now reads A1 + MD + L_{ε^*} + generator–channel bridge (which together define the admissible set with cost floor) + A2 (which selects the min-cost member).

The correct characterization is therefore: *the structural separation that EC rests on is A1-derived, but EC itself is not a consequence of A1 alone. Two of its elements are named rather than derived: the inventory-exhaustiveness step — the claim that the counted labels exhaust the kinematic distinctions available at the interface, i.e. the reading R-EC-inv — and the type/copy scoping premise of Definition 2.20 (with its entry-typed inventory I-typing where a fixed template is consumed). CM follows from A2 applied to the admissible set defined by A1 + MD: A1 bounds total cost and admits every configuration satisfying that bound; MD gives the positive cost floor (via L_{ε^*}) that makes a redundant generator strictly costly; A2 then selects the minimum-cost member of the admissible set, which by construction contains no redundant generators.* That the named elements are genuinely non-derived is witnessed by a finite counter-construction: two interfaces with independent budgets on which the species-to-label map is non-injective while every distinction is enforced and priced within the shapes of the banked premises — A1 + MD + T_{sep} + L_{loc} do not entail EC-injectivity over species[7]. EC's honest grade is therefore $[\mathrm{P}_{\mathrm{structural\_reading}} \mid \text{R-EC-inv + I-typing}]$; CM's is unconditional given A2 over the A1 + MD admissible set. The earlier-version claim "EC and CM follow from A1" was an overclaim on both counts: EC consumes the named reading and input above, and CM is properly an A2 result (operating on the A1 + MD-defined admissible set). Neither correction imports a modeling assumption beyond the framework's stated axioms and the two named elements: A2 is one of the two axioms named in the Paper 1, §2 box, and MD is the regularity input also listed in the Paper 1 axiom inventory.

CM is not a selection principle (descriptive reading). Capacity minimality is sometimes described as "Occam's razor for gauge structure," but this risks making it sound like an aesthetic preference or, worse, a dynamical mechanism that ranges over candidate structures and rejects the non-minimal ones. Neither reading is correct. Lemma 2.24 establishes CM as a *consequence of A2* applied to the admissibility set defined by A1 + MD: a configuration with a redundant generator is admissible (it satisfies the capacity bound and isotropy condition), but it sits at strictly higher total cost than the same configuration with the redundant generator removed (also admissible, by $\varepsilon^* > 0$). A2 selects the cheaper one as the realised structure. The generator–channel bridge (Lemma 2.22) makes the

[7] Code reference: `check_L_EC_inventory_reading` in `ec_inventory_reading.py`.

gap quantitative: each redundant generator costs exactly ε^*. Read correctly, CM is therefore a *descriptive label* on the realised admissible structure, not an algorithm: the realised configuration is the $\arg\min_G \dim G$ subject to EC, and that argmin is exactly what A2 fixes. No dynamical search is performed and none is needed; the framework reads off admissibility space's argmin (A1 admits, A2 selects).

Physical gauge group. By the gauge identification theorem (Theorem 2.11) and Proposition 7.1 of Technical Supplement I (*The Classification Core*), the Lie algebra of the physical gauge group is $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Y$. At the global (topological) level:

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y}{\mathbb{Z}_6},$$

where the $\mathbb{Z}_6$ quotient is derived in Proposition 2.26 below, $\mathrm{SU}(n_\alpha)$ is the universal cover of $\mathrm{PU}(n_\alpha)$, and $\mathrm{U}(1)_Y$ is the unique relative-phase factor from Proposition 7.1 of Technical Supplement I (*The Classification Core*).

The derivation chain produces $\mathrm{Inn}(\mathcal{A}) = \prod_\alpha \mathrm{PU}(n_\alpha)$ as the automorphism group of the algebra, but the physical gauge group that acts faithfully on matter fields is $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$. The passage between these two descriptions involves (a) a lift from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$ required by the representation theory of matter, and (b) a question about global structure (center quotients) that determines the topology of the gauge group. We address both formally.

Proposition 2.25 (Lift from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$)**.** *The physical gauge group acting on matter fields in $\mathcal{H}_\omega$ is $\mathrm{SU}(n)$, not $\mathrm{PU}(n)$, for each Wedderburn block $M_n(\mathbb{C})$ with $n \geq 2$.*

Proof. Step 1 (Projective obstruction). $\mathrm{Inn}(\mathcal{A}) = \prod_\alpha \mathrm{PU}(n_\alpha)$ acts on $\mathcal{A}$ by conjugation: $\alpha_U(a) = UaU^*$. This is the *adjoint* action. Matter multiplets, however, live in the independently specified matter category (Definition 2.14), which requires the *fundamental* representation of each block; the GNS Hilbert space $\mathcal{H}_\omega$ is the algebra's carrier (of dimension $\sum_k n_k^2$, containing n_k copies of each fundamental, Proposition 2.19(v)) and serves here only to exhibit that the fundamental exists as a subrepresentation: $v \mapsto Uv$ for $v \in \mathbb{C}^n \subset \mathcal{H}_\omega$. The fundamental is a linear representation of $\mathrm{U}(n)$ but only a *projective* representation of $\mathrm{PU}(n)$: the lift $\mathrm{PU}(n) \to \mathrm{GL}(n, \mathbb{C})$ requires choosing a phase for each element, and there is no continuous choice that is globally consistent (the obstruction is the second group cohomology $H^2(\mathrm{PU}(n), \mathrm{U}(1)) \cong \mathbb{Z}_n$).

Step 2 (Linearization by lift to universal cover). The universal covering group of $\mathrm{PU}(n)$ is $\mathrm{SU}(n)$, with covering map $\mathrm{SU}(n) \twoheadrightarrow \mathrm{PU}(n)$ and kernel $Z(\mathrm{SU}(n)) \cong \mathbb{Z}_n$ (the center, generated by $e^{2\pi i/n}\,\mathbf{1}_n$). Every projective representation of $\mathrm{PU}(n)$ lifts to a genuine (linear) representation of $\mathrm{SU}(n)$. The Lie algebras are isomorphic: $\mathfrak{su}(n) = \mathfrak{pu}(n)$, so the gauge field content (connection 1-form A_μ, curvature $F_{\mu\nu}$, covariant derivative D_μ) is identical regardless of whether one uses $\mathrm{SU}(n)$ or $\mathrm{PU}(n)$. The distinction is purely global: $\mathrm{SU}(n)$ admits the fundamental representation while $\mathrm{PU}(n)$ does not.

Step 3 (Carrier requirement forces the fundamental). The carrier requirement R1 (Theorem R, §3) demands a faithful complex carrier of dimension n (from $B1'$). R1 is a requirement on the matter category (Definition 2.14), and this forcing argument runs there — $\mathcal{H}_\omega$ entered only in Step 1, to exhibit that the fundamental exists. The fundamental representation of $\mathrm{SU}(n)$ on $\mathbb{C}^n$ is the unique faithful irreducible representation of dimension n; the adjoint ($\mathrm{PU}(n)$ on $\mathfrak{su}(n) \cong \mathbb{C}^{n^2-1}$) has dimension $n^2 - 1$ and is not faithful on $\mathbb{C}^n$. R1 therefore forces the fundamental, which in turn forces $\mathrm{SU}(n)$ as the gauge group acting on matter. □

Proposition 2.26 (Global structure of the gauge group)**.** *The gauge group acting on the matter content of the APF is:*

$$G_{\text{phys}} \;=\; \frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y}{\mathbb{Z}_6}\,,$$

where $\mathbb{Z}_6 \cong Z(\mathrm{SU}(3))\times Z(\mathrm{SU}(2))\cap\ker(Y)$ *is the subgroup of the center that acts trivially on all matter representations.*

Proof. The group that acts faithfully on matter is not the direct product $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y$ but its quotient by the subgroup of center elements acting trivially on all derived multiplets.

Step 1 (Center identification). The center of $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y$ is $\mathbb{Z}_3\times\mathbb{Z}_2\times\mathrm{U}(1)_Y$. A central element $(e^{2\pi ik/3}\,\mathbf{1}_3,\ (-1)^m\,\mathbf{1}_2,\ e^{iY\theta})$ acts on a multiplet (r_3, r_2, Y_f) by the phase $e^{2\pi ik\,c(r_3)/3}\,(-1)^{m\,t(r_2)}\,e^{iY_f\theta}$, where $c(r_3)$ is the $\mathbb{Z}_3$ triality of the SU(3) representation and $t(r_2)$ is the $\mathbb{Z}_2$ character of the SU(2) representation.

Step 2 (Trivially acting subgroup — explicit verification). A central element acts trivially on *all* derived multiplets iff $e^{2\pi ik\,c(r_3)/3}\,(-1)^{m\,t(r_2)}\,e^{iY_f\theta}=1$ for every multiplet in the SM template. The derived multiplet list (from the fermion scan, T_{field}) assigns each multiplet definite quantum numbers.

Normalization. We work with the integer-valued charge $Y_{\text{int}} := 6Y$ (so that $Y_Q = 1/6$ becomes $Y_{\text{int}} = 1$, etc.). The $\mathrm{U}(1)_Y$ element $e^{i\theta}$ acts on a multiplet with charge Y_{int} by the phase $e^{iY_{\text{int}}\theta}$. A center element $(e^{2\pi ik/3}\,\mathbf{1}_3,\ (-1)^m\,\mathbf{1}_2,\ e^{i\theta})$ acts on multiplet (c, t, Y_{int}) by:

$$\phi(k,m,\theta) \;=\; e^{2\pi ik\,c/3}\;e^{\pi im\,t}\;e^{iY_{\text{int}}\,\theta}. \tag{4}$$

Setting $\theta = \pi/3$ (the candidate $\mathbb{Z}_6$ generator) and $(k,m) = (1,1)$:

$$\phi(1,1,\tfrac{\pi}{3}) = \exp\Big[\,i\pi\,\frac{2c+3t+Y_{\text{int}}}{3}\Big].$$

This equals 1 iff $2c+3t+Y_{\text{int}} \equiv 0 \pmod 6$. We verify this for every multiplet type in one generation of the derived fermion template:

Multiplet	SU(3)	SU(2)	Y	c	t	Y_{int}	$2c{+}3t{+}Y_{\text{int}}$
Q_L	$\mathbf{3}$	$\mathbf{2}$	$1/6$	1	1	1	$6\equiv 0$
u_R	$\mathbf{3}$	$\mathbf{1}$	$2/3$	1	0	4	$6\equiv 0$
d_R	$\mathbf{3}$	$\mathbf{1}$	$-1/3$	1	0	-2	$0\equiv 0$
L_L	$\mathbf{1}$	$\mathbf{2}$	$-1/2$	0	1	-3	$0\equiv 0$
e_R	$\mathbf{1}$	$\mathbf{1}$	-1	0	0	-6	$-6\equiv 0$
H (Higgs)	$\mathbf{1}$	$\mathbf{2}$	$1/2$	0	1	3	$6\equiv 0$

All entries satisfy $2c+3t+Y_{\text{int}} \equiv 0 \pmod 6$. The Higgs doublet $(1, 2, 1/2)$ is included because it participates in gauge interactions; its compatibility is necessary for consistency of the Yukawa sector.

The $\mathbb{Z}_6$ generator is therefore $g := (\omega_3\,\mathbf{1}_3,\ -\mathbf{1}_2,\ e^{i\pi/3})$, and the six elements $\{g^0, g^1, \ldots, g^5\}$ exhaust the kernel. To confirm that the kernel is exactly $\mathbb{Z}_6$ and not a proper subgroup, note that $\mathbb{Z}_3\times\mathbb{Z}_2 \cong \mathbb{Z}_6$ (since $\gcd(2,3) = 1$), so $\mathbb{Z}_6$ is the largest discrete subgroup of $Z(\mathrm{SU}(3))\times Z(\mathrm{SU}(2))$. It remains to verify that elements of the center with *trivial* U(1) phase do not act trivially: $(\omega_3,\ \mathbf{1}_2,\ 1)$ acts on Q_L as $\omega_3 \neq 1$, and $(\mathbf{1}_3, -\mathbf{1}_2, 1)$ acts on Q_L as $-1 \neq 1$. The nontrivial U(1) phase $e^{i\pi/3}$ in the generator g is essential: the kernel is a *diagonal* $\mathbb{Z}_6$ inside $Z(\mathrm{SU}(3)) \times Z(\mathrm{SU}(2)) \times \mathrm{U}(1)_Y$, not the direct product $\mathbb{Z}_3 \times \mathbb{Z}_2 \times \{1\}$.

Step 3 (Quotient). The faithful gauge group is therefore $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$. This is the global form of the Standard Model gauge group, consistent with the analysis of Aharony, Seiberg, and Tachikawa (2013) and Tong (2017). $\square$

Corollary 2.27 (Fermion template compatible with quotient)**.** *Every multiplet type in the derived fermion template (T_{field}, §5 of Technical Supplement I (*The Classification Core*)) defines a well-defined (linear, not merely projective) representation of $G_{\text{phys}} = [\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$.*

Proof. A representation of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$ descends to a representation of the quotient G_{phys} if and only if the $\mathbb{Z}_6$ kernel acts trivially on that representation. The table above verifies this condition ($2c + 3t + Y_{\text{int}} \equiv 0 \pmod 6$) for each of the five fermion multiplet types in one generation and for the Higgs doublet. Since the three generations carry identical quantum numbers, the check extends to all 45 Weyl fermions (15 types × 3 generations). No exotic representation outside the $\mathbb{Z}_6$-compatible sublattice appears in the derived matter content.

Conversely, the $\mathbb{Z}_6$ compatibility is not an accident: it follows from the anomaly cancellation conditions (§6 of Technical Supplement I (*The Classification Core*)), which relate the $\mathrm{U}(1)_Y$ charges to the SU(3) and SU(2) representations. Specifically, the anomaly equation $z^2 - 2z - (N_c^2 - 1) = 0$ with $N_c = 3$ yields $z \in \{4, -2\}$; the resulting charge assignments automatically satisfy the $\mathbb{Z}_6$ condition because the anomaly system forces $Y_{\text{int}} \equiv -(2c + 3t) \pmod 6$ for every multiplet. This is a derived consistency, not a coincidence. $\square$

Why the PSU(n) issue is not a loophole. The reviewer's concern can be sharpened: since $\mathrm{Inn}(\mathcal{A}) = \mathrm{PU}(3) \times \mathrm{PU}(2) \cong \mathrm{PSU}(3) \times \mathrm{SO}(3)$, and neither PSU(3) nor SO(3) admits the fundamental representation, the matter content might appear incompatible with the group that the algebra naturally produces. The resolution is that the algebra and the Hilbert space carry *different* representations of the same underlying structure:

(a) The algebra $\mathcal{A} = M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}$ is acted on by $\mathrm{Inn}(\mathcal{A})$ via conjugation (adjoint representation). This action is well-defined on $\mathrm{PU}(n)$ because the phase ambiguity cancels: $e^{i\theta} U a U^* e^{-i\theta} = U a U^*$.
(b) The GNS Hilbert space $\mathcal{H}_\omega$ carries a linear action of $\mathrm{U}(\mathcal{A}) = \mathrm{U}(3) \times \mathrm{U}(2) \times \mathrm{U}(1)$ (Proposition 2.19). The fundamental representation $v \mapsto Uv$ requires the full unitary group, not the quotient. $\mathcal{H}_\omega$ is the algebra-side bookkeeping object (of dimension $\sum_k n_k^2$, with n_k copies of each fundamental); it exhibits the fundamental of each block as a subrepresentation, but it is not the matter space.
(c) The physical gauge group G_{phys} is the image of the lifted group in the general linear group of the *matter* representation space — the carrier of the derived multiplet content (Definition 2.14) — modulo the center acting trivially there; the GNS space plays no role in this definition beyond the bookkeeping of item (b). The exact sequence (Proposition 2.19) shows that this image is $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$: larger than $\mathrm{Inn}(\mathcal{A})$ (because it includes the $\mathrm{U}(1)_Y$ grading from the center) but smaller than $\mathrm{U}(\mathcal{A})$ (because the diagonal U(1) and the $\mathbb{Z}_6$ kernel are removed).

One premise rides this list and is named here (it is consumed wherever the $\mathrm{U}(1)_Y$ grading is called physical): central unitaries act trivially by conjugation on $\mathcal{A}$, so the relative phases of T_{rel} are physical exactly because the independently specified charged matter representations (Definition 2.14) see them; absent charged matter, the grading would be unobservable. The passage $\mathrm{Inn}(\mathcal{A}) \to G_{\mathrm{phys}}$ is therefore not a "choice" to work with $\mathrm{SU}(n)$ instead of $\mathrm{PU}(n)$; it is a derived consequence of the map $\mathrm{U}(\mathcal{A}) \twoheadrightarrow G_{\mathrm{phys}}$ (Propositions 2.19–2.26), forced by the requirement that matter fields transform linearly and that the kernel of the representation is exactly $\mathbb{Z}_6$. One convention rides this conclusion and is named here: the computation identifies the kernel of the *chosen matter representation* and hence the maximal faithful quotient *for that representation*. Calling this quotient "the" global gauge group is the convention of taking the maximal quotient compatible with the derived matter content — the standard choice, adopted explicitly, not an additional theorem.

Physical significance of the global structure. The distinction between $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$ and $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ is invisible at the level of the Lie algebra (perturbative physics), but has observable consequences in the non-perturbative sector:

(a) *Allowed monopole charges.* The Dirac quantization condition $eg = n/2$ depends on the fundamental group $\pi_1(G_{\mathrm{phys}})$. For the quotient group, $\pi_1([\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)]/\mathbb{Z}_6) \cong \mathbb{Z}$, and the minimal monopole charge is 6 times the naive Dirac quantum.
(b) *Allowed line operators.* In the path-integral formulation, the global form determines which Wilson and 't Hooft line operators are allowed. The $\mathbb{Z}_6$ quotient restricts the allowed electric and magnetic charges to a sublattice of the weight lattice.
(c) *Topological sectors.* The distinct homotopy groups of the quotient group vs. the product group affect the classification of instantons and the vacuum structure.

The APF predicts the $\mathbb{Z}_6$ quotient because the derivation proceeds through $\mathrm{Inn}(\mathcal{A})$ — which is $\mathrm{PU}(3)\times\mathrm{PU}(2)$, already a quotient — and then lifts to $\mathrm{SU}(n)$ only as far as the matter representations require. The lift introduces exactly the covering needed for the fundamental representation (Proposition 2.25), but the center elements that act trivially on all matter multiplets remain identified. The global form is therefore a prediction, not an assumption: it follows from the algebraic derivation chain $\mathrm{Inn}(\mathcal{A}) \to \mathrm{PU}(n) \to \mathrm{SU}(n)/\Gamma$ combined with the specific matter content from the fermion scan.

Reconciliation summary. The three levels of description are:

Group	**Acts on**	**Representation type**
$\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$	$\mathcal{A}$ (algebra)	Adjoint (conjugation)
$\mathrm{U}(\mathcal{A}) = \prod_k \mathrm{U}(n_k)$	$\mathcal{H}_\omega$ (Hilbert space)	Fundamental (linear)
$G_{\mathrm{phys}} = [\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y]/\mathbb{Z}_6$	Matter multiplets	Fundamental, faithful

The passage from the first row to the third involves two steps: lifting from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$ (forced by the fundamental representation, Proposition 2.25), then quotienting by the trivially acting center $\mathbb{Z}_6$ (forced by faithfulness on the derived matter content, Proposition 2.26). The Lie algebras are identical at all three levels; the distinctions are global (topological).

Status of $\mathrm{U}(1)_Y$*.* The abelian factor is determined by a two-layer argument:

Layer 1: How many abelian directions? Proposition 7.1 of Technical Supplement I (*The Classification Core*) (this section): exactly one $\mathrm{U}(1)$ from the relative-phase torus T_{rel} is admissible. Enforcement completeness (EC) forces at least one; capacity minimality (CM) excludes all but one.

Layer 2: Which direction? Theorem 6.1 of Technical Supplement I (*The Classification Core*) (§6 of Technical Supplement I (*The Classification Core*)): among all one-dimensional subgroups of

$T_{\rm rel}$, the unique one consistent with all four anomaly-cancellation conditions for the surviving fermion template is hypercharge Y, with assignments $Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$, up to $u \leftrightarrow d$ relabeling and overall normalization (the ratios are derived; the unit is the compact-U(1) charge-lattice convention, front-matter ledger).

Neither layer alone suffices. Layer 1 without Layer 2 gives "one abelian factor exists" without identifying it. Layer 2 without Layer 1 gives "these hypercharges cancel anomalies" without explaining why a second independent U(1) is excluded. Together they uniquely determine the Lie algebra of $G_{\rm phys}$ as $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Y$; the global form $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ is determined by Proposition 2.26.

$\mathrm{U}(1)_Y$ is not an inner automorphism of $\mathcal{A}$; it is a central element of the unitary group that acts non-trivially on matter representations. The non-abelian gauge factors (SU(3), SU(2)) are structural (from $\mathrm{Inn}(\mathcal{A})$); the abelian factor is a grading channel (from $T_{\rm rel}$, selected by Proposition 7.1 of Technical Supplement I (*The Classification Core*) and fixed by Theorem 6.1 of Technical Supplement I (*The Classification Core*)), physical because the charged matter representations see it (Definition 2.14). The non-abelian factors are derived from A1 via non-closure; the abelian factor follows from admissibility completeness under its named reading (R-EC-inv + I-typing, Theorem 7.7 of Technical Supplement I (*The Classification Core*) H3–H4), minimality, and anomaly cancellation. Neither is postulated, and the conditionality of the abelian leg is stated where it is proved.

Connection to NCG. This mechanism parallels Connes' "unimodular" condition in NCG, where the same $\mathrm{U}(1)_Y$ emerges from the determinant constraint on the automorphism group of the spectral triple [14]. In the APF, the selection principle is admissibility completeness (Proposition 7.1 of Technical Supplement I (*The Classification Core*)) rather than unimodularity, but the algebraic mechanism is identical: the abelian factor comes from the *center* of the multi-block unitary group, not from inner automorphisms.

Connection to T3. T3 (Paper 1 Supplement) derives the gauge structure from the Doplicher–Roberts reconstruction theorem, which recovers a compact group from its representation category. The gauge identification theorem above provides the complementary result: it identifies *which* automorphisms constitute the gauge group, and separates them from kinematic automorphisms. Together, T3 and Theorem 2.11 establish that the admissibility algebra produces a non-abelian compact Lie gauge group acting on internal degrees of freedom at each interface.

Connection to noncommutative geometry. The identification $G = \mathrm{Inn}(\mathcal{A})$ parallels a central theme of Connes' NCG program, where the gauge group of the Standard Model arises as the inner automorphism group of an almost-commutative spectral triple [14]. The present framework arrives at the same structural identification from different premises: the algebra $\mathcal{A}$ is derived from A1 ($T_{\rm alg}$), not postulated as a spectral triple; the inner/outer separation follows from interface locality ($L_{\rm loc}$) rather than from the order-one condition; and the carrier constraints (Theorem R) replace the KO-dimension and reality conditions. A comparison of the constraint structures is in Appendix A of Technical Supplement I (*The Classification Core*).

State independence. The gauge group $G = \mathrm{Inn}(\mathcal{A})$ depends only on the abstract $*$-algebra $\mathcal{A}$, not on the choice of faithful state ω used in the GNS construction. This is because $\mathrm{Inn}(\mathcal{A})$ is defined purely algebraically: it is the group of automorphisms $\alpha_U : a \mapsto UaU^*$ for U unitary in $\mathcal{A}$. This definition references only the algebra's multiplication and involution, not any representation or state.

For the same reason, the Wedderburn block structure $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$, the block sizes $\{n_k\}$, and the product $\mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$ are all ω-independent. For any *faithful* state the GNS null ideal is zero, so $\mathcal{H}_\omega \cong \mathcal{A}$ as a vector space and $\dim \mathcal{H}_\omega = \sum_k n_k^2$ — the same dimension for every faithful ω, with left-regular multiplicities (n_k copies of $\mathbb{C}^{n_k}$). What varies with ω is the inner product's weighting across blocks, not the dimension. (Earlier versions claimed different faithful states could produce different GNS dimensions; that is false in this finite-dimensional setting.) The gauge group is the same in every realization.

Which AQFT algebra does $\mathcal{A}$ correspond to? In the Doplicher–Haag–Roberts (DHR) framework of AQFT, two algebras play distinct roles: the *observable algebra* $\mathcal{A}_{\mathrm{obs}}$ (gauge-invariant quantities accessible to local measurements) and the *field algebra* $\mathcal{F}$ (which includes charged fields that transform non-trivially under the gauge group). The gauge group G acts on $\mathcal{F}$ by automorphisms that fix $\mathcal{A}_{\mathrm{obs}}$ pointwise: $\mathcal{A}_{\mathrm{obs}} = \mathcal{F}^G$, the fixed-point subalgebra.

The APF admissibility algebra $\mathcal{A}$ corresponds to the *field algebra* $\mathcal{F}$, not the observable algebra. This identification is forced by the construction:

(a) $\mathcal{A}$ is generated by all admissibility projections $\{E_d\}$ and joint defenders $\{E_{\{d_i,d_j\}}\}$, including those that act on individual charged sectors (M_{d_k} in the Wedderburn decomposition). An operator that projects onto a single color-triplet subspace is not gauge-invariant: it distinguishes between colors, and a gauge rotation maps it to a different projection. Such operators belong to $\mathcal{F}$, not to $\mathcal{A}_{\mathrm{obs}}$.

(b) Gauge transformations are inner automorphisms of $\mathcal{A}$ (Theorem 2.11): $\alpha_U(a) = UaU^*$ for $U \in \mathcal{U}(\mathcal{A})$. This is precisely the AQFT statement that gauge transformations are inner on the field algebra. In AQFT, gauge transformations are *not* inner automorphisms of $\mathcal{A}_{\mathrm{obs}}$ — they act trivially there (by definition of gauge invariance). No conflict arises because the APF identifies $\mathcal{A} = \mathcal{F}$, not $\mathcal{A} = \mathcal{A}_{\mathrm{obs}}$.

(c) The observable algebra in the APF is the gauge-invariant subalgebra: $\mathcal{A}_{\mathrm{obs}} := \mathcal{A}^G = \{a \in \mathcal{A} : \alpha_U(a) = a \ \forall U \in G\}$. This is derived, not postulated: once $G = \mathrm{Inn}(\mathcal{A})$ is identified (Theorem 2.11), $\mathcal{A}^G$ is determined. The Doplicher–Roberts reconstruction (T3, Paper 1 Supplement) then recovers G from the superselection structure of $\mathcal{A}_{\mathrm{obs}}$, closing the circle.

In summary: the APF's $\mathcal{A}$ is the field algebra $\mathcal{F}$; gauge $= \mathrm{Inn}(\mathcal{A})$ is the standard DHR statement that gauge transformations act innerly on $\mathcal{F}$; and the observable algebra $\mathcal{A}_{\mathrm{obs}} = \mathcal{A}^G$ is the gauge-invariant fixed-point subalgebra, consistent with the standard AQFT framework. [8]

2.4 Why the gauge group is a compact Lie group

The gauge group $G = \mathrm{Inn}(\mathcal{A})$ is a compact Lie group by direct construction, not by an appeal to general topological-group theory:

Step 1 (Matrix Lie group). $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$ (Wedderburn–Artin, T2a). By the gauge identification theorem (Theorem 2.11), $G = \prod_k \mathrm{PU}(n_k)$. Each $\mathrm{PU}(n_k) = \mathrm{U}(n_k)/\mathrm{U}(1)$ is a matrix Lie group: it is a closed subgroup of $\mathrm{GL}(n_k^2, \mathbb{R})$ (via the adjoint representation), hence a Lie group by the closed-subgroup theorem (Cartan, 1930).

Step 2 (Compactness). A1 (finite capacity) bounds $\dim(\mathcal{H}_\omega) < \infty$ (T_{form}). $\mathrm{U}(n)$ is compact (closed and bounded in $M_n(\mathbb{C})$); $\mathrm{PU}(n)$ is the continuous image of a compact group, hence compact. A finite product of compact groups is compact. Therefore $G = \prod_k \mathrm{PU}(n_k)$ is compact.

[8] Code reference: `check_T3` in `core.py`.

Consequence. The gauge group G is a non-abelian compact Lie group acting faithfully on a finite-dimensional Hilbert space. The classification of such groups reduces to the classification of compact simple Lie algebras (Section 4 of Technical Supplement I (*The Classification Core*)).

2.5 Finite capacity vs. type III local algebras

Finite dimensionality is not a convenience in this supplement; it is the mathematical expression of A1 at a finite interface. It enters the substrate theorem, Wedderburn–Artin decomposition, compactness of $\prod_k \mathrm{PU}(n_k)$, finite Lie-algebra classification, Theorem R, the fermion scan, and the integer capacity count. The comparison with AQFT should therefore be read as a regime comparison, not as an attempted finite-dimensional truncation of a type III theory.

In AQFT, local algebras for continuum regions are generically type III factors. APF instead treats such behavior as a continuum-refinement shadow of finite-capacity interface algebras. The finite statement is deliberately stronger and simpler:

Proposition 2.28 (Type III incompatibility)**.** *Let $\mathcal{A}(\Gamma) = M_n(\mathbb{C})$ with $n = \lfloor C(\Gamma)/\varepsilon^* \rfloor < \infty$. Then $\mathcal{A}(\Gamma)$ has a faithful normal trace, contains minimal projections, and satisfies the split property for disjoint finite interfaces. Hence it is type I_n, not type III.*

Proof. All three statements are standard for a finite matrix algebra: the normalized trace is faithful and tracial; rank-one projections are minimal; and commuting finite subalgebras have an algebraic tensor product already complete in norm. Type III factors have no faithful normal semifinite trace and no nonzero finite projections, so the regimes are incompatible at finite n. □

Proposition 2.29 (Controlled limit to type III)**.** *For a refinement sequence $\mathcal{A}_n = M_n(\mathbb{C})$ equipped with compatible KMS states, the inductive/GNS limit can generate a Powers–Araki–Woods type III_λ factor; in the saturation limit $\lambda \to 1$ this approaches type III_1. The finite trace then fails to extend to a normal faithful trace on the limit factor.*

Proof. This is the standard UHF/CAR-to-Powers-factor mechanism: finite matrix algebras embed into an inductive system, the product KMS state defines the GNS representation, and the modular spectrum fixes the Connes parameter $\lambda = e^{-\beta\varepsilon^*}$. Taking the infinite-refinement limit before asking local-algebra questions destroys the finite trace. □

Proposition 2.30 (Continuum-limit audit plan)**.** *A finite APF net is compatible with type-III AQFT only under an explicit three-part limiting plan: (i) nested finite subalgebras embed by inclusion/refinement maps; (ii) the limiting state satisfies the usual regularity hypotheses used in split/nuclearity arguments; and (iii) the sector labels locked at finite capacity persist rather than opening new independent low-cost sectors. Under these hypotheses, strict tensor factorization is a finite-regime tool, while the continuum limit may have the standard type-III obstruction to traces and exact local subsystems.*

Proof. The first condition supplies the inductive net. The second prevents a pathological limit in which the GNS closure loses the local regularity needed for AQFT comparison. The third is the APF-specific stability condition: if refinement introduced a new independent low-cost sector, the finite gauge/matter classification would have been performed on an incomplete carrier. When all three hold, the type-III limit affects the idealized local-algebra type but not the already locked finite block labels used by the gauge and matter derivations. □

Robustness audit under the limit. The continuum passage is not allowed to re-open the finite classification. The structures below are robust because they are locked at finite capacity before the limiting operation; the non-robust entries are explicitly bridge or regulator questions:

Structure	**Status under type-III limit**	**Reason**
Block labels $M_n(\mathbb{C})$ and $\prod \mathrm{PU}(n)$	Robust, provided sector-stability hypothesis (iii) holds.	They are finite-capacity superselection data before refinement.
Matter multiplet template and hypercharge ray	Robust under stable embedding.	The anomaly equations and center kernel are algebraic on the finite modules.
Strict tensor factorization	Not robust.	Replaced by split/nuclearity approximations and Definition 2.31.
Trace on local algebra	Not robust.	The finite trace need not extend to the type-III GNS closure.
Regulator-level beta coefficients	Limit-dependent bridge.	Paper 4A must supply the RG regulator/renormalization step.

2.5.1 Gravitational dressing and the scope of locality

The finite locality lemma is pre-gravitational and gauge-fixed at the level of interface algebras. It says: if two finite substrates are B-orthogonal and disjoint, their admissibility operations commute. This is compatible with gravitational dressing results, where diffeomorphism- invariant dressed observables necessarily carry long-range metric data and therefore fail to factorize strictly. In APF language, the dressing field is part of the shared interface substrate; once it is included, the interfaces are not independent in the sense required by Lemma 2.7. Thus gravitational nonlocality is not a counterexample to the finite lemma; it identifies the regime where the lemma's disjoint-support hypothesis is false.

Scope of the finite locality lemma

Lemma 2.7 is a finite-interface theorem with an explicit hypothesis: disjoint B-orthogonal substrates. It is not a theorem about fully diffeomorphism-invariant gravitational observables. Once a dressing tail is required to define an observable, the tail is part of the substrate and supplies shared support. The correct APF replacement for exact subsystem independence in that regime is finite-capacity interface independence, not strict tensor factorization.

Definition 2.31 (Finite-capacity interface independence)**.** Two finite interface algebras $\mathcal{A}(\Gamma_1)$ and $\mathcal{A}(\Gamma_2)$ are finite-capacity independent at tolerance η when every joint continuation cost decomposes as

$$C_{12}(x, y) = C_1(x) + C_2(y) + I_{12}(x, y), \qquad |I_{12}(x, y)| \leq \eta$$

for all tested records x, y, and when the interface term I_{12} is accounted for as shared substrate capacity rather than as a tensor-product factor. Exact tensor factorization is the special case $\eta = 0$. A gravitational dressing tail gives $\eta > 0$: it is not a violation of the finite locality lemma but evidence that the disjoint-support hypothesis has failed.

Proposition 2.32 (A1-bounded Reeh–Schlieder mechanism)**.** *At finite capacity, cyclicity/separating behavior is limited by the number of admissible channels. Reeh–Schlieder-type entanglement can be approximated on accessible subalgebras, but exact continuum cyclicity for arbitrarily fine local excitations requires the $C/\varepsilon^* \to \infty$ limit.*

Proof. A finite algebra has finite rank and a finite-dimensional state space, so a local algebra cannot generate an infinite independent set of excitations from the vacuum vector. Refinement increases the accessible rank; the usual AQFT behavior is recovered only after the unbounded limit. □

2.5.2 Modular theory and the Unruh effect as a limit

Finite matrix algebras have modular automorphism groups for faithful states, but the full continuum Unruh/KMS structure belongs to the limiting factor. The APF reading is therefore: finite interfaces carry regulated modular flow; the familiar thermal wedge behavior is the large-refinement limit of that flow.

2.5.3 Haag duality and locality

For disjoint finite interfaces, Haag-duality-like factorization is trivial: commuting finite algebras split as tensor factors once the substrate orthogonality assumptions hold. In the continuum, Haag duality and the split property require the usual AQFT nuclearity/regularity hypotheses. APF claims only that the finite theorem is the regulated precursor.

2.5.4 Independent evidence for finite local capacity

The physical motivation for A1 is the convergence of entropy bounds rather than a preference for finite-dimensional algebra. Bekenstein–Hawking entropy, Bousso's covariant bound, holography, and black-hole information unitarity all point toward finitely many distinguishable states associated with a finite-area interface. APF takes this as fundamental and treats continuum type III behavior as idealized emergent behavior.

Theorem 2.33 (Stability of structural predictions under $n \to \infty$)**.** *The finite structural predictions used in Paper 2—the block dimensions, compact gauge factors, anomaly equations, and template-scan survivor—are stable under refinement provided the finite stage has already locked the same separated sectors and no new independent low-cost sector opens in the limit.*

Proof. Refinement adds resolution within already separated sectors. It does not change the finite block labels, anomaly equations, or template admissibility unless a new sector with independent admissibility support appears. The latter case is exactly excluded by the persistent-pooling/no-new-sector hypothesis used in the refinement statement below. □

Definition 2.34 (Refinement Hypothesis (RH))**.** Interfaces admit arbitrarily fine operational refinements whose finite subalgebras embed compatibly into later refinements.

Definition 2.35 (Capacity Density (CD))**.** The number of available channels scales with interface area according to a finite density bound, so the channel count diverges only in the continuum limit.

Definition 2.36 (Persistent Pooling (PP))**.** The shared-pool/nonseparability structure persists under refinement; no refinement splits the locked interface into independent cost-free sectors.

Proposition 2.37 (Effective type-III behavior in the refinement limit)**.** *Under RH, CD, and PP, a sequence of finite APF interface algebras can have continuum observables whose GNS closure is effectively type III while the finite-stage gauge and matter predictions remain unchanged.*

Proof. RH supplies the inductive system, CD supplies unbounded channel growth in the limit, and PP preserves the finite locked sector structure. The limit therefore has the usual type III obstructions to a faithful trace, while the finite-stage algebraic invariants used for the Standard-Model template are stable by Theorem 2.33. □

Regime: finite-capacity admissibility algebras

Paper 2 derives the gauge/matter structure in finite APF interface algebras. AQFT type III locality is treated as a controlled continuum limit, not as an input. Gravitational dressing is accommodated by denying the disjoint-support hypothesis whenever the dressing field supplies a shared long-range substrate.

3. Carrier prerequisites: metric, sector-uniform irreversibility, and complex type

Theorem R (Paper 2, §3.5) derives three carrier requirements (R1–R3) from admissibility. The derivation chains for R1 and R2 depend on three lemmas originally developed in companion papers. This section provides self-contained proofs of all three, eliminating external dependencies.

The dependency order is: $T_{7B} \to L_{\text{irr,uniform}} \to B1'$. T_{7B} is used by $L_{\text{irr,uniform}}$ (Step 2); $L_{\text{irr,uniform}}$ closes the R2 chain (chiral carrier); $B1'$ closes the R1 chain (complex ternary carrier). $B1'$ is logically independent of $L_{\text{irr,uniform}}$; both appear in this section because both close formerly external dependencies.

All upstream requirements (A1, K2, K3, SP, L_{nc}, L_{loc}, L_{irr}, T_{sep}, T_{GNS}, T3) are proved in the Paper 1 Supplement. No result in this section imports from Papers 3–7.

3.1 Cost kernel from shared-interface non-factorization (T_{7B})

At separate interfaces, admissibility costs are additive by L_{loc} (Paper 1 Supplement [2], §4). At a shared interface Γ, co-located distinctions produce a superadditive surplus $\Delta > 0$ (L_Δ, Paper 1 Supplement [2]). The question is: what mathematical object tracks this non-factorization when admissibility operates across *multiple* directions at a single interface?

The answer is already implicit in the Paper 1 Supplement's derivation chain: the sector-orthogonal bilinear form B (constructed by L_ω from K3) and the characterization of admissibility costs as B-norms (from T_{adj}) together force the external feasibility cost to be an *exact* quadratic form. T_{7B} makes this explicit.

Theorem 3.1 (T_{7B}: Positive-Definite Cost Kernel from Shared-Interface Non-Factorization)**.** *Let* Γ *be an interface with substrate* $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ *(*T_{sep}*) and sector-orthogonal bilinear form* B *(*L_ω*, Paper 1 Supplement [2]). Let* $\varepsilon_{\text{ext}} : S_\Gamma \to \mathbb{R}_{\geq 0}$ *denote the external feasibility cost. Then:*

(i) $\varepsilon_{\text{ext}}(v) = B(v, v)$ *for all* $v \in S_\Gamma$*: the external feasibility cost is an exact quadratic form.*
(ii) The associated bilinear form $g := B$ *is positive-definite.*
(iii) g *is unique up to one positive scalar* per sector *of the* T_{sep} *decomposition (the within-sector gauge freedom of* B*); it is unique up to a single overall scalar exactly when an inter-sector normalization is imposed, which this theorem does not supply. (v3.9 stated "an overall scalar" while the proof delivers the per-sector freedom; the statement now matches the proof.)*

No additional regularity assumption is needed.

Proof. Step 1: Defense costs are B*-norms.* By T_{adj} (Paper 1 Supplement [2], §5.7), the admissibility projection for distinction d is the B-orthogonal projection $E_d = \pi^{(B)}_{M_d}$ onto its anchor set M_d. The cost of defending d against a perturbation v is the B-norm of the component of v in M_d:

$$\kappa_d(v) = B(E_d\, v,\, E_d\, v) = \|E_d\, v\|_B^2.$$

This is the minimum-cost defense: T_{adj} proves that the B-orthogonal projection minimizes the defense cost among all idempotents with image M_d.

Step 2: K3 gives exact additivity over disjoint supports. By the sector decomposition $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ and K3 (additivity on disjoint supports), the total defense cost decomposes as:

$$\varepsilon_{\text{ext}}(v) = \sum_d \|E_d\, v\|_B^2 + \|\pi_\Pi\, v\|_B^2 = \sum_d B(v_d, v_d) + B(v_\Pi, v_\Pi),$$

where $v = \sum_d v_d + v_\Pi$ is the sector decomposition of v, and π_Π is the B-orthogonal projection onto Π.

Step 3: Sector orthogonality collapses the sum. By L_ω ($B(v_d, v_{d'}) = 0$ for $d \neq d'$, and $B(v_d, v_\Pi) = 0$ for all d):

$$\varepsilon_{\mathrm{ext}}(v) = \sum_d B(v_d, v_d) + B(v_\Pi, v_\Pi) = B\Big(\sum_d v_d + v_\Pi,\ \sum_d v_d + v_\Pi\Big) = B(v, v).$$

The cross-terms vanish by sector orthogonality. Therefore $\varepsilon_{\mathrm{ext}}(v) = B(v, v)$ *exactly*: the quadratic form is not a leading-order approximation but an identity. This gives (i).

Step 4: Positive-definiteness. For $v \neq 0$: $B(v, v) > 0$ because B is positive-definite (L_ω, Paper 1 Supplement). Therefore $g = B$ is positive-definite, giving (ii).

Step 5: Uniqueness. L_ω constructs B with forced inter-sector orthogonality and free within-sector gauge ($B_d \mapsto \alpha_d B_d$, any $\alpha_d > 0$). The choice of capacity units (A1) fixes an overall scale; the remaining gauge is the within-sector normalization convention, which does not affect the geometric structure. This gives (iii). □

Regularity conditions used: A1 (finite capacity), K2 (positive perturbation cost), K3 (disjoint-support additivity), SP (no null directions), T_{sep} (sector decomposition), L_ω (sector-orthogonal bilinear form), T_{adj} (admissibility projections are B-orthogonal).

Status: [P]. No additional regularity assumption (R7B, K3-bilinear, or differentiability) is needed. The quadraticity is *derived* from $L_\omega + T_{\mathrm{adj}}$, not assumed.

Lemma 3.2 (No metric assumed upstream of T_{7B})**.** *The bilinear form B used in T_{7B} is constructed from A1, K3, and FD3 alone. No norm, inner product, quadratic cost law, or metric structure is assumed at any point in the construction chain.*

Proof. The chain (each step uses only preceding outputs):

(1) A1 provides real-valued costs $\varepsilon(d) > 0$.
(2) T_{embed} maps the convex state space into $\mathbb{R}^{N+1}$ using cost coordinates (convexity from OR1; no inner product invoked).
(3) T_{sep} decomposes $S_\Gamma = \bigoplus_d M_d \oplus \Pi$ from K3 + FD3. This is a *pre-metric algebraic direct sum*: no inner product is used (Appendix B.2).
(4) L_ω constructs B as a direct sum of free within-sector forms with *forced* zero cross-terms. The vanishing $B(u_d, v_{d'}) = 0$ for $d \neq d'$ is a *theorem* of K3 (Appendix B.4), not a definition. The within-sector forms B_d are free (normalization gauge).
(5) T_{adj} identifies admissibility maps as B-orthogonal projections (Appendix B.5). Uses only L_ω's inter-sector orthogonality.

T_{7B} then shows $\varepsilon_{\mathrm{ext}}(v) = B(v, v)$ by combining (4) and (5) with K3 (Step 2 of T_{7B}'s proof). The quadratic form is the *output* of this chain, not an input at any stage. □

Why the cost kernel is positive-definite, not Lorentzian. The cost kernel $g = B$ measures admissibility cost at a single interface Γ. Admissibility cost is positive for every nonzero perturbation (K2 + SP), so g is positive-definite. Lorentzian signature $(-, +, +, +)$ arises at a different level: it is the signature of the *inter-interface* causal structure, where the timelike direction is determined by L_{irr} (irreversibility provides the arrow of time) and the spatial directions are determined by

L_{loc} (independent interfaces provide spatial separation). The positive-definite g at each interface is the internal (Riemannian) metric; the Lorentzian spacetime metric is constructed in the exited spacetime bridge (Paper 6; boundary in §3 of Technical Supplement I (*The Classification Core*)) by combining g with the causal ordering of interfaces. The disambiguation of these two bilinear structures is detailed in Remark 2.8. This is the same structural split as in standard GR, where the spatial metric is positive-definite and the Lorentzian signature comes from including the timelike direction.

Roadmap. The full derivation of Lorentzian spacetime from the cost kernel g is carried by Paper 6 (boundary statement in §3 of Technical Supplement I (*The Classification Core*)). The chain runs: $L_{\text{irr}} \to$ causal partial order $\to$ smooth manifold (Kolmogorov) $\to$ Lorentzian conformal class (Hawking–King–McCarthy) $\to d = 4$ (Lovelock uniqueness + DOF minimality) $\to$ Einstein equations. [9]

3.2 Sector-uniform irreversibility ($L_{\text{irr,uniform}}$)

L_{irr} (Paper 1 Supplement [2], §5) derives irreversibility from A1 + L_{nc} + L_{loc}: cross-interface correlations commit capacity that no local observer can recover. That proof establishes irreversibility *generically* — it guarantees that irreversible channels exist at interfaces where cross-interface correlations are superadditive. The question for Theorem R's R2 requirement is whether this irreversibility extends to the *gauge sector specifically*: does the sector of internal (non-gravitational) admissibility contain its own irreversible channels, or could it remain fully reversible while inheriting apparent irreversibility from gravity alone?

Theorem 3.3 ($L_{\text{irr,uniform}}$: Gauge-Sector Irreversibility at Saturated Interfaces)**.** *Let Γ be a shared interface with non-trivial pool ($\Pi \neq \{0\}$), let G denote the set of gauge-sector distinctions at Γ, and let R_Γ denote the gravitational record distinctions at Γ. Suppose:*

(a) The gauge sector is non-trivially coupled to gravity at Γ (i.e., the cost kernel g from T_{7B} has nonzero cross-terms between G and R_Γ).

(b) The interface saturation exceeds the bound

$$s := \frac{\varepsilon(G) + \varepsilon(R_\Gamma)}{C_\Gamma} > 1 - \frac{\Delta(G, R_\Gamma)}{C_\Gamma - \varepsilon(R_\Gamma)}, \tag{5}$$

where $\Delta(G, R_\Gamma) \geq \varepsilon^* > 0$ is the superadditive surplus from L_Δ.

(c) A gauge-sector transition $G_1 \to G_2$ commits positive shared surplus with the record sector: $\Delta(G_2, R_\Gamma) > 0$.

Then the gauge sector contains an irreversible channel at Γ: there exist gauge-sector histories H, H' differing only in gauge distinctions such that the reversal $H' \to H$ is inadmissible.

Proof (three steps). Step 1: Records are interface-local. By L_{loc} (Paper 1 Supplement [2], §4), admissibility is distributed over independent interfaces, each with its own budget. Irreversibility arises because cross-interface correlations commit capacity that no *single-interface* observer can recover (L_{irr}, Step 3). In particular, irreversible records at Γ are distinction sets *at* Γ, not global objects.

Step 2: Gauge–gravity coupling at shared interfaces. By hypothesis (a) and T_{7B} (Theorem 3.1), gauge distinctions G contribute to the cross-terms of the cost kernel g at Γ. Therefore gauge

[9] Code reference: `check_T7B` in `gravity.py`.

and gravitational admissibility share interface Γ. This sharing implies: there exist admissible histories H, H' that differ by gauge-sector distinctions and yield different gravitational records $R_\Gamma(H) \neq R_\Gamma(H')$. If no such histories existed, gauge distinctions would have no recordable consequences at Γ, contradicting hypothesis (a).

Step 3: Budget overrun from non-closure. Since G and R_Γ coexist at Γ with $\Pi \neq \{0\}$, L_{nc} gives:

$$\varepsilon_\Gamma(G \cup R_\Gamma) \;>\; \varepsilon_\Gamma(G) + \varepsilon_\Gamma(R_\Gamma).$$

Consider the reversal of a gauge transition $G_1 \to G_2$. The forward transition commits cross-interface surplus $\Delta(G_2, R_\Gamma) > 0$ (hypothesis (c)). The reversal $G_2 \to G_1$ while R_Γ persists requires recovering this surplus. By L_{irr}, the surplus includes cross-interface terms that no single-interface operation can free. The reversal cost satisfies:

$$\varepsilon_\Gamma(\text{reverse } G_2 \to G_1 \mid R_\Gamma) \;\geq\; \varepsilon_\Gamma(G_1) + \Delta(G_1, R_\Gamma).$$

The remaining budget after the forward transition is $C_\Gamma - \varepsilon(R_\Gamma) - \varepsilon(G_2)$. By hypothesis (b) (saturation bound (5)), this remaining budget is strictly less than the reversal cost. The reversal is inadmissible. □

Corollary 3.4 (Near-capacity interfaces)**.** *At any interface within one admissibility channel of full capacity ($s > 1 - \varepsilon^*/C_\Gamma$), condition (b) is automatically satisfied because $\Delta \geq \varepsilon^* > 0$ (L_Δ + L_{ε^*}). Conditions (a) and (c) hold at any interface where the gauge sector is non-trivially coupled and undergoes transitions. Therefore gauge-sector irreversibility is generic at near-capacity shared interfaces.*

Regularity conditions used: L_{loc} (locality), L_{nc} (non-closure), L_{irr} (irreversibility from cross-interface correlations), T_{7B} (cost kernel from shared-interface non-factorization), A1 (finite budget), SP (non-trivial gauge coupling).

Scope of the sufficient condition. $L_{\text{irr,uniform}}$ is a sufficient-condition theorem: it guarantees gauge-sector irreversibility at interfaces satisfying hypotheses (a)–(c), not at all interfaces. This is deliberate. The downstream use (R2: chiral carrier requirement) needs only the *existence* of irreversible gauge-sector channels at physically relevant interfaces — not their universality. The corollary shows that the hypotheses are automatically satisfied at near-capacity shared interfaces, which is the regime where the gauge architecture is determined. A universal irreversibility theorem (holding at arbitrarily low saturation) would be false: at very low saturation ($s \ll 1$), the budget is loose enough that all transitions are reversible, and the interface is effectively classical.

Key structural point. The argument does *not* assume that the gauge sector has any specific algebraic structure beyond being a non-trivially coupled set of distinctions at a shared interface. The irreversibility is inherited from the *interface geometry* (T_{7B}) and the *non-closure mechanism* (L_Δ), not from any property specific to non-abelian gauge theories. Chirality (R2) then follows because a *vector-like* gauge sector, despite sharing the interface, produces zero *intrinsic* gauge irreversibility (all gauge-level processes are CPT-reversible; see Paper 2, §3.4). The existence of gauge-sector irreversible channels (this theorem) combined with the impossibility of intrinsic irreversibility in a vector-like theory (Paper 2, §3.5) forces chirality. [10]

[10] Code reference: `check_L_irr_uniform` in `core.py`.

3.3 Complex type from oriented composite robustness ($B1'$)

The R1 carrier requirement (Theorem R) demands a complex carrier: $V \ncong V^*$ as G-modules. The physical reason is that composites must be *oriented* — robustly distinguishable from their anti-composites — to provide independent admissibility channels under T_M. This subsection proves that real and pseudoreal carriers cannot satisfy this requirement: the equivariant antilinear map J collapses composite orientation at zero capacity cost.

Definition 3.5 (Oriented composite)**.** Let V be a faithful irreducible carrier for a gauge group G, and let $B \in (V^{\otimes k})^G$ be a gauge-singlet composite (a G-invariant tensor in $V^{\otimes k}$). The conjugate composite is $\bar{B} \in (V^{*\otimes k})^G$. The composite B is *oriented* if B and $\bar{B}$ are robustly distinguishable: no admissibility-preserving relabeling internal to the carrier maps $B \mapsto \bar{B}$.

Theorem 3.6 ($B1'$: Complex Carrier from Oriented Composite Robustness)**.** *Let V be a faithful irreducible carrier for a compact group G. Suppose:*

- *(B1) L_{irr}-stable gauge-singlet composites B and $\bar{B}$ exist whose oriented distinction is robust under admissible refinements.*
- *(B2) This distinction is not enforced by an independent external grading channel (minimality clause from A1).*

Then V must be of complex type*: $V \ncong V^*$ as G-modules.*

Proof (by contradiction). Suppose $V \cong V^*$ as G-modules. Then V is either *real type* (admits a G-equivariant $\mathbb{R}$-linear isomorphism $\phi : V \to V^*$ with $\phi = \bar{\phi}$) or *pseudoreal type* (admits a G-equivariant antilinear map $J : V \to V$ with $J^2 = -\mathrm{id}$, but no real structure). In either case, there exists a G-equivariant antilinear map $J : V \to V$.

Step 1: Tensorial extension. Extend J to $V^{\otimes k}$ by

$$J^{\otimes k}(v_1 \otimes v_2 \otimes \cdots \otimes v_k) \;=\; J(v_1) \otimes J(v_2) \otimes \cdots \otimes J(v_k).$$

Since J is G-equivariant ($J \circ g = g \circ J$ for all $g \in G$), $J^{\otimes k}$ is G-equivariant on $V^{\otimes k}$. In particular, $J^{\otimes k}$ maps gauge-invariant subspaces to gauge-invariant subspaces: if $B \in (V^{\otimes k})^G$, then $J^{\otimes k}(B) \in (V^{\otimes k})^G$.

Step 2: Identification of composite with conjugate. We split cases.

Real-type case. If V is real-type, $V \cong V^*$ via a G-equivariant real structure ϕ with $\phi = \bar{\phi}$. The associated $J : V \to V$ satisfies $J^2 = +\mathrm{id}$ (a genuine involution), so $J^{\otimes k}$ is an involution on $V^{\otimes k}$ for every k and provides a direct G-equivariant identification $(V^{\otimes k})^G \leftrightarrow (V^{*\otimes k})^G$. The map $B \mapsto \bar{B}$ is well-defined as a square-trivial involution, and the identification is orientation-collapsing in the sense required below.

Pseudoreal-type case. If V is pseudoreal-type, then by the standard classification (Humphreys 1972; Fulton–Harris 1991, §IV.23) $V \cong V^*$ as G-modules *by definition of pseudoreal type.* The carrier-complexity hypothesis $V \ncong V^*$ of the theorem statement is therefore already false for pseudoreal V, and the proof does not need to invoke any $J^{\otimes k}$ argument in this case. The J-extension machinery used below is only necessary in the real-type case. The earlier formulation, which attempted to handle both types under a single $J^{\otimes k}$ identification and raised a potential signed-tensor subtlety for odd k, is therefore unnecessary: pseudoreal carriers are ruled out by the structural hypothesis $V \ncong V^*$ directly.

Accordingly, the remainder of the proof treats only the real-type case, where $J^{\otimes k}$ is a genuine involution and the identification $(V^{\otimes k})^G \leftrightarrow (V^{*\otimes k})^G$ is unambiguous.

Concretely: let $B = \sum_i c_i \, v_{i_1} \otimes \cdots \otimes v_{i_k}$ be a gauge singlet. Then

$$J^{\otimes k}(B) \;=\; \sum_i \bar{c}_i \, J(v_{i_1}) \otimes \cdots \otimes J(v_{i_k})$$

is also a gauge singlet (by equivariance), and it is the image of B under conjugation composed with the equivariant isomorphism $V \cong V^*$. Therefore $J^{\otimes k}(B)$ is identified with $\bar{B}$.

Step 3: The identification is admissibility-preserving. The map J is internal to the carrier — it is a relabeling of the carrier basis vectors, not a change of physical state. It costs zero admissibility capacity: the admissibility structure at the interface is unchanged because J commutes with all G-actions (equivariance), so the admissibility projections E_d and the cost functional ε are invariant under J. Therefore $J^{\otimes k} : B \mapsto \bar{B}$ is an admissibility-preserving relabeling.

Step 4: Oriented distinction collapses. By Steps 2–3, there exists an admissibility-preserving relabeling (internal to the carrier, at zero capacity cost) that maps $B \mapsto \bar{B}$. By Definition 3.5, the composite B is *not* oriented: the "forward" and "backward" versions collapse into the same equivalence class under admissible relabelings.

Step 5: Contradiction with hypotheses. Hypothesis (B1) requires that the oriented distinction $B \neq \bar{B}$ be robust. Step 4 shows it is not robust if $V \cong V^*$. The only escape is to introduce an independent external grading channel that "protects" the $B/\bar{B}$ distinction from the J-identification. But this violates hypothesis (B2) (minimality under A1: no additional admissibility channel may be introduced solely to protect a distinction that the carrier itself should sustain).

Therefore $V \not\cong V^*$: the carrier must be of complex type. □

Counterexample (pseudoreal confinement). $G = \mathrm{SU}(2)$ with fundamental representation $V = \mathbb{C}^2$ (pseudoreal: $V \cong V^*$ via the ε-tensor, $J = i\sigma_2 K$ where K is complex conjugation). The "baryon" $B = \varepsilon_{ij} \, v^i w^j$ is a bilinear singlet. $J^{\otimes 2}(B) = \bar{B}$, and the identification is G-equivariant: there is no robust oriented distinction. "Baryon" and "antibaryon" in an SU(2) confining theory are the same singlet reached by different paths.

Positive example. $G = \mathrm{SU}(3)$ with fundamental representation $V = \mathbb{C}^3$ (complex: $V \not\cong V^*$, since the Dynkin labels $[1, 0]$ and $[0, 1]$ are distinct). No G-equivariant antilinear $J : V \to V$ exists. The baryon $B = \varepsilon_{ijk} \, v^i w^j x^k$ and the antibaryon $\bar{B} = \varepsilon^{ijk} \, \bar{v}_i \bar{w}_j \bar{x}_k$ are robustly distinct: no internal relabeling maps one to the other.

Regularity conditions used: A1 (minimality clause, hypothesis B2), T_M (admissibility independence, which motivates hypothesis B1), L_{irr} (stability of composites, hypothesis B1). The proof itself is pure G-module theory — the only non-algebraic input is the admissibility-preservation criterion (Step 3), which invokes equivariance and zero-cost relabeling.

Consequence for Theorem R. $B1'$ closes the R1 derivation chain. The ternary carrier ($k = 3$, from the trilinear invariant requirement) must be complex type to sustain oriented composites. Combined with faithfulness and minimality ($\dim V = 3$ is the minimum for an irreducible trilinear invariant), R1 is fully derived: a faithful complex 3-dimensional carrier. [11]

[11] Code reference: `check_B1_prime` in `gauge.py`.

3.4 Worked example: three-dimensional interface

We illustrate T_{7B} and $L_{\text{irr,uniform}}$ with the simplest nontrivial interface: $\dim(S_\Gamma) = 3$ with two one-dimensional sectors and a one-dimensional pool.

Setup. $S_\Gamma = \mathbb{R}^3$ with standard basis $\{e_1, e_2, e_3\}$. Sector decomposition (T_{sep}): $M_{d_1} = \text{span}(e_1)$, $M_{d_2} = \text{span}(e_2)$, $\Pi = \text{span}(e_3)$. Admissibility projections: $E_{d_1} = \text{diag}(1,0,0)$, $E_{d_2} = \text{diag}(0,1,0)$. Individual admissibility costs: $\varepsilon(d_1) = \varepsilon(d_2) = \varepsilon^* = 1$.

Computing Δ. The pool $\Pi = \text{span}(e_3) \neq \{0\}$, so L_Δ applies. A pool-supported perturbation $p = \alpha\, e_3$ with $|\alpha| > 0$ threatens joint admissibility (SP: physically meaningful DOF; K2: positive cost). Defense costs $\kappa_\Pi = \varepsilon^* = 1$ (one pool channel). By K3: $\varepsilon(\{d_1, d_2\}) = 1 + 1 + 1 = 3$. Therefore $\Delta = 1 > 0$.

Computing ε_{ext} *and verifying* T_{7B}. The external feasibility cost for a general perturbation $v = (v_1, v_2, v_3)^T$ is specified by the cost matrix:

$$G \;=\; \begin{pmatrix} 1 & \lambda & 0 \\ \lambda & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \varepsilon_{\text{ext}}(v) = v^T G\, v = v_1^2 + v_2^2 + v_3^2 + 2\lambda\, v_1 v_2,$$

where $\lambda \in [0,1)$ is the pool-mediated coupling between sectors M_{d_1} and M_{d_2}. The diagonal entries are the individual sector costs; the off-diagonal λ is the cross-term from L_Δ (pool-mediated defense cost); the e_3-diagonal entry is the direct pool cost.

- $\lambda = 0$: sectors uncoupled; $\Delta = 0$ (no cross-term); G is diagonal; the interface is classical.
- $\lambda > 0$: sectors coupled through Π; $\Delta > 0$; G has off-diagonal structure. The pool mediates an admissibility interaction that is invisible to individual-sector accounting.

G is symmetric and positive-definite for $|\lambda| < 1$ (eigenvalues $1 \pm \lambda$ and 1, all positive). $\varepsilon_{\text{ext}}(v) = v^T G\, v$ is manifestly a quadratic form. The polarization identity gives $g(u, v) = u^T G\, v$, which is the bilinear form. All three conclusions of T_{7B} are verified:

(i) ε_{ext} is a quadratic form with kernel G.
(ii) G is positive-definite $\Rightarrow$ non-degenerate.
(iii) the diagonal part B of G is unique up to one positive scalar *per sector*, matching T_{7B}(iii); the off-diagonal λ is fixed by the pool coupling, not by the normalization gauge.

The connection to Δ: the superadditive surplus is $\Delta = 2\lambda \cdot |v_1 v_2|$, which vanishes iff $\lambda = 0$ (no pool coupling). The relation between G and T_{7B}'s objects must be stated carefully, because sector-orthogonality is exactly the vanishing of cross-sector terms. The sector-orthogonal bilinear form B (L_ω) is the *diagonal part* of G: its cross-sector values are zero by construction ($B(v_1, v_2) = 0$ for v_1, v_2 in distinct sectors), its diagonal entries are B-norms of sector projections (T_{adj}), and K3 gives exact additivity over disjoint supports. The off-diagonal λ is *not* B's cross-sector value (which is zero); it is the L_Δ pool-mediated coupling, a distinct object added to B: at $\lambda = 0$ the cost matrix is B itself (the Sep baseline), and $\lambda > 0$ is the IJC deformation carried by Π.

Illustrating $L_{\text{irr,uniform}}$. Set $\lambda = 1/2$ and $C_\Gamma = 3$. Let $G = \{d_1\}$ (gauge sector) and $R_\Gamma = \{d_2\}$ (gravitational record), with costs $\varepsilon(G) = \varepsilon(R_\Gamma) = 1$. The superadditive surplus is $\Delta(G, R_\Gamma) = 2\lambda = 1$. Interface saturation: $s = (1+1)/3 = 2/3$.

Check hypothesis (b) of $L_{\text{irr,uniform}}$: $s = 2/3 > 1 - \Delta/(C_\Gamma - \varepsilon(R_\Gamma)) = 1 - 1/(3-1) = 1/2$. Satisfied.

After a gauge transition $G_1 \to G_2$, the reversal cost is: $\varepsilon(\text{reverse}) \geq \varepsilon(G_1) + \Delta = 1 + 1 = 2$. The remaining budget is $C_\Gamma - \varepsilon(R_\Gamma) - \varepsilon(G_2) = 3 - 1 - 1 = 1 < 2$. The reversal is inadmissible.

Now set $C_\Gamma = 5$ (loose budget, $s = 2/5$): $s = 2/5 < 1/2$. Hypothesis (b) fails. Remaining budget $= 5 - 1 - 1 = 3 > 2$. The reversal succeeds. The gauge sector is reversible at this low saturation, as the theorem predicts.

3.5 Consolidation: Theorem R prerequisites closed

With T_{7B}, $L_{\text{irr,uniform}}$, and $B1'$ proved in this supplement, the three carrier requirements of Theorem R (Paper 2, §3.5) are now derived entirely within Paper 1, Paper 2, and their respective supplements:

Requirement	**Derivation chain**	**Key new lemma**	**Status**
R1 (complex ternary)	$L_{\text{nc}} \to T_{\text{conf}} \to T_M \to B1'$	$B1'$ (§3.3)	[P]
R2 (pseudoreal binary)	$L_{\text{irr,uniform}} \to T_M \to$ chirality	$L_{\text{irr,uniform}}$ (§3.2)	[P]
R3 (abelian grading)	admissibility completeness (R-EC-inv + I-typing) + A1 minimality	self-contained in Paper 2	[P | R-EC-inv + I-typing]

No result in this section depends on Papers 3–7. The formerly external dependency on “Paper 7 ($B1'$)” cited in Paper 2’s abstract and §3.4 is now resolved.

4. The gauge-cost program

The composed operational statement $C(G) = \dim(G)$ — lower bound a theorem on the operational-encoding premise, upper bound the disjoint-anchor realizability premise (§2) — is this program's current state. The named open is a cost functor: a derivation of both premises from the admissibility semantics, which would upgrade the equality from premise-conditioned to derived. The subsection below records what capacity arguments alone currently deliver on the asymptotic-freedom side of the same cost story.

4.1 Can the sign be established from capacity alone?

The imported-coefficient check (§2 of Technical Supplement I (*The Classification Core*)) verifies $b_0 > 0$ using the imported one-loop formula. The reviewer asks whether any part of the sign can be established purely from capacity or positivity arguments, without importing the formula from perturbative QFT. The answer is partially affirmative: the *dominance of the gauge self-interaction term over the matter screening term* can be given a capacity argument, even though the precise coefficient 11/3 cannot.

Proposition 4.1 (Capacity sign pressure with imported group weights)**.** *Let G be a non-abelian gauge group. Capacity accounting fixes the sign of the two competing contributions—adjoint self-correlation is anti-screening, matter channels are screening—but it does not by itself fix the numerical weights multiplying those contributions. The comparison used here imports the standard adjoint-Casimir and Dynkin-index weights (C_A and T_R). With those imported weights and the derived matter content, the non-abelian factors have positive one-loop coefficient.*

Argument. Step 1 (Self-interaction as capacity self-cost). Each gauge generator occupies one admissibility channel (Lemma 2.22). Non-abelian gauge fields self-interact: the structure constants f^{abc} couple all p channels to each other. In the admissibility language, this self-interaction is the cross-channel cost of maintaining p mutually correlated admissibility channels at a single interface. The superadditive surplus $\Delta > 0$ (L_Δ) ensures that this self-cost is *positive*: co-located non-abelian channels cost more than the sum of their individual costs. This positive self-cost makes the effective coupling *decrease* at short distances (where more channels are simultaneously active), because the admissibility budget is finite and the self-cost consumes capacity that would otherwise be available for matter interactions.

Step 2 (Matter as screening). Charged matter fields screen the gauge interaction: a virtual fermion–antifermion pair partially cancels the gauge field, reducing the effective coupling at long distances. In the admissibility language, each charged matter DOF provides an additional channel through which admissibility can operate, partially offsetting the capacity consumed by the gauge self-interaction. The screening contribution is proportional to N_{matter}.

Step 3 (Counting argument). For the derived matter content: $N_{\text{matter}}^{\text{SU}(3)} = 12$ charged Weyl DOF per generation $\times$ 3 generations $= 36$, while $p = \dim(\mathfrak{su}(3)) = 8$. Naively, $N_{\text{matter}} > p$, so the counting argument alone does not settle the sign. However, the self-interaction term carries a group-theoretic enhancement: each generator couples to all $p-1$ others through the adjoint representation, giving an effective self-interaction count of $p \times C_A = p^2$ (where C_A is the quadratic Casimir of the adjoint). For SU(3): $p \times C_A = 8 \times 3 = 24$. The matter screening, by contrast, is weighted by T_R (the index of the matter representation, which is 1/2 for the fundamental), giving an effective screening count of $T_R \times N_f = (1/2) \times 12 = 6$ per generation, or 18 for three generations. Since $24 > 18$, the self-interaction dominates. □

What this establishes. The argument above does not derive the one-loop formula or the precise coefficient 11/3. What it establishes is the *physical mechanism*: non-abelian self-interaction produces antiscreening (capacity self-cost from L_Δ), while matter produces screening, and the self-interaction dominates for the derived matter content because the adjoint Casimir C_A enhances the self-interaction count. The dominance can be stated as a *capacity inequality*:

$$p \cdot C_A \ > \ T_R \cdot N_f^{\text{total}} \qquad \Longrightarrow \qquad b_0 > 0. \tag{6}$$

For SU(3): $8 \times 3 = 24 > 6 \times 3 = 18$. For SU(2): $3 \times 2 = 6 > (1/2) \times 12 = 6$; the inequality is *saturated*, and the precise coefficient 11/3 (rather than some other number) is needed to establish positivity. This is the boundary where the capacity argument alone becomes insufficient and the perturbative formula is required.

What remains for Paper 4A. Paper 4A ($L_{\text{beta,capacity}}$) derives the one-loop β-function coefficient, including the precise 11/3 and 4/3 factors, from the capacity structure: the 11/3 arises from the number of independent polarization states of a spin-1 field in $d = 4$ (from T_8) combined with the self-interaction structure of the adjoint representation; the 4/3 arises from the spin-1/2 screening of a Weyl fermion. The capacity-based derivation replaces the standard Feynman-diagram calculation with an admissibility-cost calculation, but arrives at the same coefficients. Here we can only establish the mechanism (antiscreening from self-interaction) and the dominance condition (6); the coefficients are imported.

5. Open-lane register

The program's named opens, each stated as a question with its current grade boundary. Heading the register:

1. The fixed-point program. Is the (template, matter content, single-abelian-grading) triple the unique simultaneous solution of the full constraint set? The classification core proves minimality among the enumerated candidates given the template; the closure theorems verify abelian consistency given the matter content. A fixed-point theorem would close the loop: uniqueness of the triple under the joint constraints, with no consumption order. Nothing banked asserts it today.

2. The cost functor. Derive the operational cost premise pair (operational encoding; disjoint-anchor realizability) from the admissibility semantics, upgrading $C(G) = \dim(G)$ from premise-conditioned to derived (§4).

3. The EC reading. Derive R-EC-inv from the axioms, or prove it underivable in a stronger sense than the current counter-construction (which shows A1 + MD + T_{sep} + L_{loc} do not entail EC-injectivity over species at shape level)[12].

4. A ground for F6. Electromagnetic completeness is an independent declared assumption of the scan, and its independence from EC is a computed theorem[13]. The candidate lane is a record-carrier route: whether the unbroken electromagnetic charge's separating role can be grounded in the record semantics rather than declared.

[12]Code reference: `check_L_EC_inventory_reading` in `ec_inventory_reading.py`.
[13]Code reference: `check_L_F6_not_from_EC` in `ec_inventory_reading.py`.

5. Cap completeness. The scan's enumeration caps are declared inputs; the proof obligations that no admissible sub-45-DOF template exists outside them are owned in the classification core's reproducibility contract and remain open.

6. The full-invariance hypothesis. The gauge identification's equality leg consumes it (Theorem 2.12). Derive it — anchor data entering at block granularity for gauge purposes — or restate the downstream chain so that only the unconditional inclusion is consumed everywhere (the present text already consumes only the inclusion plus the exact sequence).

One conditioning is named here so it is not mistaken for an open lane: the IJC *occupancy* is empirical by design — whether a physical interface presents branch-(IJC) records is read off its correlation record, exactly as coupling constants are read off theirs. It is a named conditioning of the chain, not a gap the program owes a proof for.

A. Comparison with reconstruction programs

The comparisons kept in this document are the ones that bear on the foundational chain — operational reconstructions, algebraic QFT, and assumption inventories; the "SM from principles" comparisons that bear on the classification claim live in Technical Supplement I (*The Classification Core*)'s comparison appendix.

A.1 Operational and categorical reconstructions

Several programs reconstruct quantum theory from operational or information-theoretic axioms. We summarize each and note where the APF diverges.

Hardy (2001) [11]. Five "reasonable axioms" (probabilities, simplicity, subspaces, composite systems, continuous reversibility) derive finite-dimensional quantum theory over $\mathbb{C}$. The axioms are stated at the kinematics level: they constrain the state space and measurement structure but do not determine gauge content, matter spectrum, or dynamics. APF shares the "finite capacity" starting point (Hardy's axiom 1 is a form of finite dimensionality) but pushes through to gauge group and matter content via non-closure, which Hardy does not address.

Chiribella–D'Ariano–Perinotti (2011) [12]. Six axioms (causality, perfect distinguishability, ideal compression, local distinguishability, pure conditioning, purification) derive QT. The purification axiom is the key differentiator from classical probability. The output is a general probabilistic theory isomorphic to finite-dimensional quantum mechanics. Like Hardy, CDP stops at kinematics. The APF's constitutive principle SP (substrate faithfulness) plays a role analogous to CDP's local distinguishability, and K3 (forced additivity) parallels their ideal compression, but the APF does not assume purification — instead, it derives the Born rule from the admissibility algebra via Busch's generalized Gleason theorem.

Remark A.1 (Complex structure and reconstruction axioms)**.** The APF's $B1'$ result should be compared with local-tomography and continuous-reversibility reconstructions of complex quantum theory. Those programs typically isolate $\mathbb{C}$ by assuming how composites behave (e.g., local tomography, purification, or continuous reversible dynamics). APF uses a different but related composite axiom: oriented composite robustness. Real carriers fail to preserve the needed oriented charged distinction, while quaternionic carriers add an anti-linear/reality structure not forced by the finite record ledger. Thus $B1'$ is not a replacement proof of all complex quantum theory; it is the APF-specific gate selecting complex charged modules for the gauge/matter theorem.

Höhn (2017) [13]. Reconstructs qubit quantum theory from "questions" (binary tests) with finite information capacity. The closest cousin to APF's starting point: finite capacity $\to$ quantum structure. Höhn's framework is carefully delimited to qubits and does not attempt gauge content. APF can be viewed as an extension of Höhn's program that asks: what happens when you push the "finite information" idea beyond qubit reconstruction to the full field content?

Effectus theory (Westerbaan, van de Wetering). Categorical framework using "effectuses" to axiomatize quantum-like theories. Sequential product axioms force operator-algebraic structure. The APF's admissibility algebra (§2) can be compared to the sequential product in effectus theory: both derive non-commutativity from the structure of sequential measurements. The effectus approach is more general (it classifies all effectuses, not just the quantum one); the APF is more specific (it selects the quantum case via A1 and derives gauge content).

Tull (2019); Spekkens (2007). Tull derives operator algebras from categorical CPM (completely positive maps) structure. Spekkens constructs a toy model from epistemic restrictions. Both illuminate the boundary between classical and quantum but do not derive field content or gauge structure.

Finite-algebra structures in quantum error correction. A structurally close neighbor from a different tradition: in operator-algebra quantum error correction and categorical code frameworks, the correctable/logical structure of a finite code is organized by a finite $*$-algebra, and the *inner* automorphisms of the code algebra are exactly the locally implementable ("gauge-like") transformations, while outer automorphisms act as logical symmetries. That is the same inner/outer split this supplement derives operationally — gauge = inner = implementable-at-no-observable-cost, kinematic/logical = outer — arrived at from error-correction requirements rather than capacity minimization. The convergence is evidence that identifying the physical gauge group with $\mathrm{Inn}(\mathcal{A})$ of a finite operational algebra is natural in finite settings generally, not an artifact of the admissibility axioms. The APF adds what the QEC tradition does not attempt: a selection principle (PLEC) over the algebras themselves, from which the *particular* finite algebra $M_3 \oplus M_2 \oplus \mathbb{C}$ is argued to be the minimum-cost choice.

A.2 Locally covariant QFT and the General Boundary Formulation

Two programs are particularly relevant to the APF because they share structural elements (local algebras, causality, boundaries) while differing fundamentally on infinite-dimensionality.

Locally covariant QFT (Brunetti–Fredenhagen–Verch, 2003). LCQFT axiomatizes QFT as a functor from the category of globally hyperbolic Lorentzian manifolds (with isometric embeddings as morphisms) to the category of $*$-algebras (with injective $*$-homomorphisms). The key axioms are:

- *Locality*: spacelike-separated regions have commuting algebras.
- *Covariance*: the assignment $\mathcal{O} \mapsto \mathcal{A}(\mathcal{O})$ is functorial under isometric embeddings.
- *Time-slice*: if $\mathcal{O}_1 \hookrightarrow \mathcal{O}_2$ contains a Cauchy surface of $\mathcal{O}_2$, then $\mathcal{A}(\mathcal{O}_1) \cong \mathcal{A}(\mathcal{O}_2)$.

The LCQFT framework works with *infinite-dimensional* algebras (typically type III factors for local regions) and *presupposes* a background Lorentzian manifold.

The APF differs from LCQFT on three structural points:

(a) *Finite vs. infinite dimensions.* LCQFT's local algebras are infinite-dimensional by construction (they are obtained via the GNS representation of a state on a C^*-algebra of smeared fields). The APF's local algebras are finite-dimensional (A1: $\dim(\mathcal{H}_\omega) < \infty$). The reconciliation is via the controlled limit (Proposition 2.29): the APF's type I algebras approximate LCQFT's type III algebras in the $C/\varepsilon^* \to \infty$ limit, with the structural predictions (gauge group, matter content) stable under the limit (Theorem 2.33).

(b) *Spacetime derived vs. presupposed.* LCQFT takes a Lorentzian manifold as input; the APF derives it (exited spacetime bridge, Paper 6: $L_{\mathrm{irr}} \to \prec \to M \to g_{\mu\nu}$). The APF can therefore be viewed as providing a *foundation* for LCQFT: it derives the background manifold that LCQFT assumes, and the local algebras that LCQFT assigns to regions emerge as the admissibility algebras at interfaces.

(c) *Causality.* Both frameworks implement causality via commutativity of spacelike-separated algebras. In LCQFT, this is an axiom; in the APF, it is derived (Lemma 2.7). The LCQFT time-slice axiom has an APF analogue: if $\Gamma_1 \subseteq \Gamma_2$ and Γ_1 "contains a Cauchy surface" (i.e., the admissibility structure at Γ_1 determines the admissibility structure at Γ_2), then $\mathcal{A}(\Gamma_1) \cong \mathcal{A}(\Gamma_2)$. This is a consequence of L_{loc} and the deterministic evolution of the admissibility structure.

Oeckl's Positive Formalism / General Boundary Formulation (2003–2016). Oeckl's program assigns quantum amplitudes to *boundaries* of spacetime regions, rather than to states on Cauchy surfaces. The formalism uses:

- *Boundary Hilbert spaces*: each boundary Σ (not necessarily a Cauchy surface) carries a Hilbert space $\mathcal{H}_\Sigma$.
- *Amplitude maps*: each spacetime region M bounded by Σ defines a linear map $\rho_M : \mathcal{H}_\Sigma \to \mathbb{C}$ (or more generally a positive map).
- *Composition*: gluing regions along shared boundaries composes amplitude maps.

The General Boundary Formulation (GBF) generalizes this to non-globally-hyperbolic spacetimes and is compatible with quantum gravity scenarios where the background spacetime is not fixed.

The APF's interface structure has a striking structural parallel to Oeckl's boundary formalism:

(a) *Interfaces as boundaries.* An APF interface Γ is the boundary between two admissibility regions. The admissibility algebra $\mathcal{A}(\Gamma)$ plays the role of Oeckl's boundary Hilbert space $\mathcal{H}_\Sigma$ (in the finite-dimensional case, the algebra and its GNS Hilbert space are equivalent objects).
(b) *Cost kernel as amplitude.* The admissibility cost $\varepsilon_{\text{ext}}(v) = B(v, v)$ (T_{7B}) is a positive functional on the interface sector space, analogous to Oeckl's positive amplitude ρ_M. The positivity of B corresponds to Oeckl's positivity condition, which ensures physical probabilities are non-negative.
(c) *Composition via pool coupling.* When two interfaces share a pool Π (L_Δ), the admissibility structure composes: the combined cost is superadditive ($\varepsilon(v_1 + v_2) \geq \varepsilon(v_1) + \varepsilon(v_2)$). This parallels Oeckl's composition of amplitude maps via gluing.

The key differences:

- Oeckl works with infinite-dimensional Hilbert spaces and does not impose a capacity bound. The APF's A1 restricts to finite dimensions.
- Oeckl's formalism is *kinematic* (it assigns amplitudes but does not derive gauge content or matter spectrum). The APF pushes through to field content via the admissibility algebra's Wedderburn decomposition.
- The GBF is compatible with indefinite causal structure (useful for quantum gravity); the APF derives a definite causal structure from L_{irr} (exited causal-order construction; Paper 6, BA1).

Masanes–Müller (2011). Derives finite-dimensional quantum theory from four postulates: continuous reversibility, tomographic locality (joint states determined by local measurements), existence of an information unit, and the subspace axiom. The output is quantum theory over $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$ (with $\mathbb{C}$ selected by tomographic locality + bit symmetry). The APF shares the finite-dimensionality starting point and the field selection ($\mathbb{F} = \mathbb{C}$ from T2b), but arrives at $\mathbb{C}$ via a different mechanism (Skolem–Noether applied to the admissibility algebra, not tomographic locality). Like Hardy and CDP, Masanes–Müller stops at kinematics and does not address gauge content.

A.3 Systematic comparison: infinite-dimensionality and causality

The reviewer asks for a systematic contrast on two axes — infinite-dimensionality and causality — across the relevant programs. The following table provides this.

Program	**Local algebra type**	**Causality treatment**	**APF relationship**
AQFT (Haag–Kastler)	Type III_1 factors (Reeh–Schlieder)	Axiom: $[\mathcal{A}(\mathcal{O}_1), \mathcal{A}(\mathcal{O}_2)] = 0$ for spacelike $\mathcal{O}_1, \mathcal{O}_2$	APF recovers type III in $n \to \infty$ limit; derives commutativity from L_{loc}
LCQFT (Brunetti–Fredenhagen–Verch)	Infinite-dim. $*$-algebras; functorial on Lorentzian manifolds	Presupposes globally hyperbolic manifold; time-slice axiom	APF derives the manifold; provides a finite-dim. foundation
Oeckl (GBF)	Infinite-dim. boundary Hilbert spaces	Compatible with indefinite causal structure	Structural parallel (interfaces $\sim$ boundaries); APF adds A1 finiteness and derives causal order
Hardy (2001)	Finite-dim. (axiom 1)	Not addressed (kinematics only)	Shared starting point; APF extends to gauge content
CDP (2011)	Finite-dim. (output)	Causality axiom (no signaling)	APF derives no-signaling from L_{loc}; pushes beyond kinematics
Masanes–Müller	Finite-dim. (postulate)	Not directly addressed	Shared $\mathbb{F} = \mathbb{C}$ result; different mechanism
Connes NCG	Finite-dim. internal algebra; ∞-dim. external	Lorentzian structure from spectral triple	APF derives the algebra that NCG postulates
DR reconstruction	Infinite-dim. (type III in AQFT context)	Inherited from AQFT axioms	APF uses DR at finite n; structural parallel for abelian factors
APF	Type I_n ($M_n(\mathbb{C})$); type III recovered as $n \to \infty$	Derived from L_{irr} (exited; Paper 6, BA1); HKM/Malament for signature	—

Three patterns emerge from the comparison:

(i) Finite vs. infinite dimensions is a foundational choice, not a technical detail. Programs that start with infinite dimensions (AQFT, LCQFT, Oeckl, DR) must contend with type III algebras, ultraviolet divergences, and the absence of a faithful trace. Programs that start with finite dimensions (Hardy, CDP, Masanes–Müller, APF) avoid these issues but must explain how the infinite-dimensional continuum is recovered. The APF's controlled limit (Proposition 2.29) and structural stability theorem (Theorem 2.33) provide this explanation; the kinematic reconstruction programs do not address the question (they stop at finite-dimensional QT and do not discuss the continuum limit).

(ii) Causality is either axiomatized or derived, never both. AQFT and CDP axiomatize causality (as commutativity of spacelike algebras or no-signaling); the APF derives it from L_{irr} (irreversibility $\to$ causal order) and L_{loc} (independent interfaces $\to$ commutativity). Oeckl's GBF is compatible with indefinite causality, making it the most flexible — but at the cost of not determining the causal structure. The APF's derivation of causality from a non-causal starting point (finite admissibility capacity has no intrinsic temporal or causal content) is the closest parallel to the causal-set program (Sorkin, 1991), though the APF derives a smooth manifold rather than working with a discrete causal set.

(iii) Only the APF and Connes NCG derive gauge content. All other programs in the table stop at the kinematics of quantum theory (state space, measurement structure, composition rules). The passage from kinematics to dynamics and gauge content requires additional structure: in NCG, the

spectral triple; in the APF, the admissibility algebra's Wedderburn decomposition + non-closure + anomaly cancellation. The APF's advantage over NCG is that it *derives* the algebra, whereas NCG postulates it; NCG's advantage is that it produces the full SM Lagrangian (including Yukawa couplings and the Higgs potential), which the APF defers to Paper 4A.

A.4 Assumption inventory comparison

Program	**Inputs**	**Derives**	**Stops at**
Hardy (2001)	5 axioms	QT kinematics	State space
CDP (2011)	6 axioms	QT kinematics	State space
Masanes–Müller (2011)	4 postulates	QT over $\mathbb{C}$	State space
Höhn (2017)	Finite info	Qubit QT	Qubit
LCQFT (BFV 2003)	Functorial axioms	QFT on curved spacetime	Assumes manifold
Oeckl (GBF 2003)	Boundary axioms	QT amplitudes	Kinematics
Connes NCG	Spectral triple	SM Lagrangian	Assumes algebra
APF	A1+MD+A2+BW+SP+K3	QT + gauge chain at banked grades	Classification inputs (Techn

The APF's named assumption count (the four PLEC components A1, MD, A2, BW; plus constitutive semantics SP, K3, FD1–FD6) is numerically comparable to Hardy's 5 or CDP's 6, though a direct comparison of "assumption weight" is imprecise: the APF's constitutive definitions (FD1–FD6) under PLEC's four constitutive features (A1+MD+A2+BW) carry more internal structure than any single-axiom presentation suggests. The significant difference is scope: the APF's downstream chain extends toward gauge content and the matter spectrum — as a graded program with named premises, closed by the classification core of Technical Supplement I (*The Classification Core*) — and, at bridge grade only, toward cosmological readings; the other programs do not attempt this scope.

DHR/DR categorical boundary. The finite APF admissibility net is not asserted to be a type-III Haag–Kastler net. The functorial comparison is: a finite interface category of admissible localized endomorphisms maps to a tensor subcategory of transportable sectors; the finite field algebra generated by those sectors has compact internal automorphism group $\mathrm{Inn}(\mathcal{A})$; the continuum DHR/DR theorem is recovered only after adding the split/nuclearity/scaling assumptions stated in Proposition 2.30. Thus APF uses DR as a structural analogue and limiting target, not as a hidden premise inside the finite derivation.

B. Self-contained proofs of imported results

This appendix provides complete proofs of every result imported from the Paper 1 Supplement that is used in this document. With this appendix, no external document is needed to verify any theorem in this supplement.

Input assumptions. All proofs below use only: A1 (finite admissibility capacity $0 < C(\Gamma) < \infty$), MD (uniform per-test cost floor $\varepsilon \geq \mu^* > 0$), K3 (additivity on disjoint supports), SP (no null directions in the physical quotient), and the formal definitions FD1–FD6. No result from Papers 2–7 is used.

B.1 L_{ε^*}: Uniform cost floor

Lemma B.1 (L_{ε^*}). *There exists $\varepsilon^* > 0$ such that $\varepsilon(d) \geq \varepsilon^*$ for all $d \in \mathcal{D}$.*

Proof. By MD: $\varepsilon(d) \geq \mu^* \cdot n(d) \geq \mu^* > 0$ for every d (since $n(d) \geq 1$). Set $\varepsilon^* := \mu^*$. □

B.2 T_{sep}: Sector decomposition

Theorem B.2 (T_{sep}). *For distinctions $d_1, d_2 \in \mathcal{D}(\Gamma)$ with anchor sets $M_{d_1}, M_{d_2} \subseteq S_\Gamma$:*

$$M_{d_1} \cap M_{d_2} = \{0\} \quad \Longleftrightarrow \quad \varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2).$$

The biconditional is stated at empty residual pool, $\Pi = \{0\}$; when $\Pi \neq \{0\}$ the joint cost carries, in addition, the superadditive surplus κ_Π of L_Δ (Appendix B.3). For any maximal antichain $\mathcal{B}$ of pairwise independently admissible distinctions, the substrate decomposes as $S_\Gamma = \bigoplus_{d \in \mathcal{B}} M_d \oplus \Pi$, where Π is the residual pool.

Proof. ($\Rightarrow$) If $M_{d_1} \cap M_{d_2} = \{0\}$, perturbations threatening d_1 and d_2 have disjoint supports (FD3: anchor-set locality). The joint defense decomposes into independent defenses on M_{d_1} and M_{d_2}. By K3 (additivity on disjoint supports): $\varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2)$.

($\Leftarrow$, contrapositive) If $v \in M_{d_1} \cap M_{d_2}$ with $v \neq 0$, then defending d_1 and d_2 separately each commits capacity to $\mathrm{span}\{v\}$; defending jointly commits once. By SP and K2, $\kappa(\mathrm{span}\{v\}) > 0$. Write $M_{d_i} = \mathrm{span}\{v\} \oplus M'_{d_i}$. By K3 on the three disjoint subspaces:

$$\varepsilon(\{d_1, d_2\}) \leq \kappa(\mathrm{span}\{v\}) + \kappa(M'_{d_1}) + \kappa(M'_{d_2}) < \varepsilon(d_1) + \varepsilon(d_2).$$

The direct-sum decomposition follows by iterating the biconditional over a maximal antichain. □

Uses: A1, K3, FD3, SP. No inner product or linear map structure.

B.3 L_Δ: Superadditivity

Theorem B.3 (L_Δ). *For co-located distinctions d_1, d_2 at Γ with $M_{d_1} \cap M_{d_2} = \{0\}$ and $\Pi \neq \{0\}$: $\Delta(d_1, d_2) := \varepsilon(\{d_1, d_2\}) - \varepsilon(d_1) - \varepsilon(d_2) > 0$.*

Proof. Step 1. The joint perturbation class $\mathcal{P}(\{d_1, d_2\})$ strictly contains $\mathcal{P}(d_1) \cup \mathcal{P}(d_2)$: perturbations p with $\mathrm{supp}(p) \subseteq \Pi$ are in $\mathcal{P}(\{d_1, d_2\})$ but not in either individual class (their anchors are disjoint from Π by T_{sep}). These perturbations change physically meaningful DOF (SP) at positive cost (K2).

Step 2. A defense confined to $M_{d_1} \oplus M_{d_2}$ acts as the identity on Π and cannot resist Π-supported perturbations.

Step 3. Any admissible joint defense must commit positive capacity $\kappa_\Pi > 0$ in Π (by K2: resisting a non-null perturbation requires positive capacity).

Step 4. M_{d_1}, M_{d_2}, Π are pairwise disjoint (T_{sep}). By K3: $\varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2) + \kappa_\Pi$.

Step 5. $\Delta = \kappa_\Pi > 0$. □

Uses: A1, K2, K3, T_{sep}, SP. Pre-algebraic.

B.4 L_ω: Sector-orthogonal bilinear form

Lemma B.4 (L_ω)**.** *Given T_{sep} and K3, there exists a positive-definite symmetric bilinear form $B : V \times V \to \mathbb{R}$ with $B(u_d, v_{d'}) = 0$ for $d \neq d'$ and $B(u_d, v_\Pi) = 0$ for all d. The inter-sector orthogonality is forced; the within-sector forms are free (normalization gauge).*

Proof. Construction. Choose any positive-definite form B_d on each sector M_d and B_Π on Π. Define $B(u, v) := \sum_d B_d(u_d, v_d) + B_\Pi(u_\Pi, v_\Pi)$ with zero cross-terms.

B is well-defined, symmetric, positive-definite by construction (direct sum of positive-definite forms).

K3 forces cross-term vanishing. For $u_d \in M_d$ and $v_{d'} \in M_{d'}$ with $d \neq d'$, K3 gives $\|u_d + v_{d'}\|_B^2 = \|u_d\|_B^2 + \|v_{d'}\|_B^2$ (disjoint supports). Expanding: $B(u_d, u_d) + B(v_{d'}, v_{d'}) + 2B(u_d, v_{d'}) = B(u_d, u_d) + B(v_{d'}, v_{d'})$, so $B(u_d, v_{d'}) = 0$. The same argument gives $B(u_d, v_\Pi) = 0$. □

Uses: T_{sep}, K3.

B.5 T_{adj}: Self-adjoint admissibility projections

Theorem B.5 (T_{adj})**.** *With respect to B from L_ω, each sector map E_d is a B-orthogonal projection: $E_d^\dagger = E_d$, $E_d^2 = E_d$.*

Proof. $\text{im}(E_d) = M_d$ and $\ker(E_d) = (\bigoplus_{d' \neq d} M_{d'}) \oplus \Pi$. By L_ω: $M_d \perp_B \ker(E_d)$. Therefore E_d is the B-orthogonal projection onto M_d: $B(E_d u, v) = B(u_d, v_d) = B(u, E_d v)$, so $E_d^\dagger = E_d$. □

B.6 L_{loc}: Locality (budget additivity)

Theorem B.6 (L_{loc})**.** *For independent interfaces Γ_1, Γ_2 with disjoint substrates $S_{\Gamma_1} \cap S_{\Gamma_2} = \emptyset$: $C(\Gamma_1 \cup \Gamma_2) = C(\Gamma_1) + C(\Gamma_2)$.*

Proof. Every distinction d has its anchor entirely within S_{Γ_1} or S_{Γ_2} (FD3: anchor-set locality). The cost sum partitions: $\sum_d \varepsilon(d) = \sum_{d \in \mathcal{D}(\Gamma_1)} \varepsilon(d) + \sum_{d \in \mathcal{D}(\Gamma_2)} \varepsilon(d) \leq C(\Gamma_1) + C(\Gamma_2)$. The bound is tight (each interface's budget can be independently saturated). □

Uses: A1, FD3. Does not use MD, L_{ε^*}, or any cost-functional structure.

B.7 L_{blk}: Block-diagonal defense failure

Theorem B.7 (L_{blk})**.** *Let $\Delta(d_1, d_2) > 0$. Then no idempotent with $\text{im}(P) \subseteq M_{d_1} \oplus M_{d_2}$ is an admissible joint defender.*

Proof. Such a P satisfies $P|_\Pi = 0$ (by idempotency and the direct-sum structure). It therefore has zero defense against Π-supported perturbations. But $\Delta > 0$ requires positive Π-defense (L_Δ, Step 3). Contradiction. □

B.8 T_{alg}: Noncommutativity

Theorem B.8 (T_{alg}). *If $\Delta(d_1, d_2) > 0$, the full admissibility algebra $\mathcal{A} := \mathrm{Alg}_{\mathbb{R}}(\{E_d\} \cup \{E_{\{d_1,d_2\}}\})$ is noncommutative.*

Proof. Step 1. The sector projections $\{E_d\}$ commute and generate $\mathcal{A}_{\text{diag}} \cong \mathbb{R}^{|\mathcal{B}|}$.

Step 2. By L_{blk}, the minimum-cost joint defender $E_{\{d_1,d_2\}} \notin \mathcal{A}_{\text{diag}}$. Define $F_\Pi := E_{\{d_1,d_2\}} - E_{d_1} - E_{d_2} \neq 0$.

Step 3. There exists $v \in M_{d_1}$ with $E_{\{d_1,d_2\}} v = v + \pi_v$, $\pi_v \in \Pi \setminus \{0\}$ (otherwise $E_{\{d_1,d_2\}}$ is block-diagonal, contradicting Step 2). Then: $[E_{d_1}, F_\Pi]\, v = E_{d_1}(\pi_v) - F_\Pi(v) = 0 - \pi_v = -\pi_v \neq 0$, using $E_{d_1}|_\Pi = 0$ (T_{sep}).

Step 4. $[E_{d_1}, F_\Pi] \neq 0$ in $\mathrm{End}(V)$. Any faithful $*$-homomorphism to a commutative algebra would map this to zero, contradicting faithfulness. Therefore $\mathcal{A}$ is noncommutative. □

Uses: T_{sep}, L_Δ, L_{blk}, FD5a. Pre-Hilbert.

B.9 L_{nc}: Non-closure (superadditivity at shared interfaces)

Lemma B.9 (L_{nc}). *At any shared interface Γ with $\Pi \neq \{0\}$ and co-located distinction sets G, R_Γ: $\varepsilon(G \cup R_\Gamma) > \varepsilon(G) + \varepsilon(R_\Gamma)$.*

Proof. This is L_Δ applied to the distinction sets G and R_Γ: the hypotheses of L_Δ (disjoint anchors, non-empty pool) are satisfied by the co-location conditions of the shared interface. □

B.10 L_{irr}: Irreversibility

Theorem B.10 (L_{irr}). *At any shared interface Γ with $\Pi \neq \{0\}$, there exist admissibility transitions that are locally irreversible: the capacity committed in the transition cannot be recovered by any operation confined to a single sector.*

Proof. Step 1. By L_Δ (§B.3), co-located admissibility at Γ commits a superadditive surplus $\Delta = \kappa_\Pi > 0$ in the pool sector Π.

Step 2. By L_{loc} (§B.6), admissibility at separate interfaces has independent budgets. A single-sector observer (operating within M_{d_1} alone) has access only to the capacity committed in M_{d_1}, not to the pool commitment κ_Π.

Step 3. The pool commitment κ_Π is a cross-sector correlation: it is committed in Π, which is shared between the sectors. No operation confined to M_{d_1} can access Π (by the direct-sum structure of T_{sep}: $M_{d_1} \cap \Pi = \{0\}$). Therefore the pool commitment is locally unrecoverable.

Step 4. A transition that increases κ_Π (e.g., by changing the gauge-sector distinctions at a shared interface) commits capacity that is locally unrecoverable. This is irreversibility: the forward transition costs more to reverse than the budget allows, because the reversal would require freeing pool capacity that no single-sector operation can access. □

Uses: L_Δ, L_{loc}, T_{sep}. The argument is pre-algebraic: no Hilbert space or GNS structure is invoked.

Summary. The ten results above (L_{ε^*}, T_{sep}, L_Δ, L_ω, T_{adj}, L_{loc}, L_{blk}, T_{alg}, L_{nc}, L_{irr}) constitute the complete set of results imported from the Paper 1 Supplement. Each is proved from A1, MD, K3, SP, and FD1–FD6 alone. With these proofs in hand, every theorem in this supplement can be verified end-to-end without consulting any external document.

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